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Information Dissemination and the Monetary Policy Uncertainty Premium: Evidence from China

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ABSTRACT

This paper proposes a novel monetary policy uncertainty (MPU) measure aggregating information from policy communication, policy operation and news coverage. We find that the aggregate MPU measure significantly commands a negative risk premium in the cross-section of the Chinese stock market. Stocks with higher (lower) MPU beta, favored (avoided) for hedging against MPU shocks, contribute to the premium through their underperformance (outperformance). Our results remain robust after controlling for economic uncertainty, economic policy uncertainty, and monetary policy surprises. We further show that the aggregate MPU measure outperforms its individual components in both portfolio- and stock-level asset pricing tests.

1. Introduction

Uncertainty regarding current and future monetary policies play an important role in both the economy and financial markets. Such uncertainty may stem from various sources in the policy information transmission, including ambiguous policy communications (Cieslak et al., 2021), intricate and unpredictable policy rules (Fernández-Villaverde et al., 2011; Bauer et al., 2022), and unclear interpretations in the media coverage (Husted et al., 2020; Ter Ellen *et al.*, 2021). Any single measure thereby cannot fully capture the overall level of monetary policy uncertainty (MPU). Moreover, the existing literature has studied the impact of monetary policy and its uncertainty shock on the temporal dynamics of stock market risk premium, exchange rates, yield curves, and crude oil prices (Bekaert et al., 2013; Mueller et al., 2017; Kurov and Stan, 2018; Tillmann, 2020), whereas the empirical evidence on the cross-section of equity returns is scant. In this paper, we propose a novel aggregate MPU measure and examine how stock returns react to MPU shocks.

Our investigation focuses primarily on China, where the complex monetary policy framework and the policy-driven financial market provide an ideal setting for studying the asset-pricing implications of MPU. First, the People's Bank of China (PBC) is a ministry-level agency under the State Council and, thereby, is constrained in decision-making and information provision (Huang et al., 2020). The PBC's policy considerations extend across a long list of objectives, and it employs various tools, including both quantity- and price-based instruments. This multi-objective and multi-instrument operating regime involves trade-offs between different objectives or policy tools. Second, the Chinese financial market is referred to as a "policy-driven" market, where the government employs intensive interventions to lean against short-term market fluctuations, and investors focus more on policies than fundamentals (Brunnermeier et al., 2021).[1] In particular, market participants are eager for any signal of monetary policy (McMahon et al., 2018). These features give rise to a considerable degree of MPU and elicit widespread attention in exploring its effects on such a financial market.

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[1] As the second largest equity market in the world, China is often regarded as a segmented market with many specificities. In addition to strong government intervention, the market remains centrally controlled, retail investor-dominated, and heavily relationship-driven (Amstad et al., 2020). However, over the past 15 years, China has undergone a significant internationalization process (Bekaert et al., 2022).

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In this paper, we begin our analyses by introducing a comprehensive MPU measure that aggregates information aligned with the dissemination process of policy information. We use three *individual* MPU measures: (1) a communication-based indicator constructed from counting keywords associated with uncertainty in PBC's communication texts, (2) an operation-based indicator constructed from the unexpected components of a set of monetary policy variables (tools), and (3) a news-based indicator constructed from uncertainty-related newspaper articles.[2] Following the standard methodology for variable combination (e.g. Baker and Wurgler, 2006), we use the first principal component of these three individual MPU indices as the *aggregate* measure. We then examine the economic determinants of our aggregate measure. We find that it is not only more consistent with the fluctuations in monetary policy but also captures the special episodes flagged by individual measures. The aggregate MPU measure is positively correlated with alternative uncertainty variables, such as economic uncertainty and economic policy uncertainty. Increases in MPU are typically associated with periods of monetary policy tightening and economic downturns.

Next, using the novel aggregate MPU measure, we show that MPU is significantly priced with a negative risk premium. The theoretical foundation for our study stems from Merton (1973)'s conditional intertemporal capital asset pricing model (ICAPM). Motivated by the well-established studies that connect macro-level uncertainty to systematic risks (e.g., Bloom, 2009; Jurado *et al.*, 2015) and our previous findings, we argue that MPU is a relevant state variable affecting future consumption and investment decisions. An increase in MPU represents an unfavourable shift in investment opportunities. Intertemporal hedging demand prompts investors to pay higher prices and accept lower returns for stocks with higher covariances with MPU shock. Consequently, the difference in returns between stocks with higher (positive) and lower (negative) covariance with MPU is negative.

To test this conjecture, we first estimate stock exposure to the aggregate MPU shock (ΔMPU), which we refer to as the *aggregate MPU beta* (β^{MPU}, or simply MPU beta). We then conduct univariate portfolio-level analysis and find that stocks in the highest MPU beta decile generate 4.188% lower excess return per annum than stocks in the lowest MPU beta decile. After controlling for the risk factors of Liu et al. (2019), the difference between the returns on the portfolios with the highest and lowest MPU beta remains negative and highly significant. Similarly, using bivariate portfolio sorts and Fama–MacBeth regressions, we show that the predictive power of the MPU beta is not subsumed by well-known cross-sectional return predictors. These results indicate that MPU commands a negative premium in the cross-section of expected returns. In addition, we find that the MPU premium is, to a great extent, persistent and robust in the next 2 to 3 months, and exhibits substantial increases during periods of heightened monetary policy uncertainty. By analyzing the economic determinants of the stock exposure to MPU shocks, we observe that the MPU betas in the finance and real estate sectors are more volatile compared to other industries. Stocks with high exposure to MPU shock are more likely to be growth, liquid, and have high coskewness and strong investment characteristics. We also construct a risk factor based on the MPU beta and provide evidence suggesting that it cannot be explained by established risk factors in the Chinese market.

To demonstrate the value of information aggregation, we first separately estimate stock exposure to each of the three individual MPU shocks, i.e., the *individual* MPU beta, and then replicate the cross-sectional asset-pricing tests using individual MPU betas as key explanatory variables. We find that the aggregate MPU measure outperforms all individual components by delivering higher and more significant return and alpha spreads. More importantly, none of the three individual MPU measures commands significant risk premium when we perform Fama–MacBeth regressions after controlling for other stock-level characteristics. Our results are consistent with recent literature highlighting the value of information aggregation in financial markets, e.g. Light et al. (2017) aggregate firm characteristics to predict expected stock returns, and Chen et al. (2022) aggregate investor sentiment proxies to predict the stock market risk premium.

Finally, we investigate the robustness of our findings. We first show that the return predictability of MPU cannot be explained by economic uncertainty (e.g., Bali *et al.*, 2017), economic policy uncertainty (e.g., Brogaard and Detzel, 2015), or monetary policy surprises (e.g., Ozdagli and Velikov, 2020). These results suggest that second-moment shocks to monetary policy have distinct asset-pricing implications that cannot be fully accounted for by alternative uncertainty variables and first-moment monetary shocks. Next, we estimate MPU betas from alternative factor models or MPU indices, and find that these betas remain significant predictors of future stock returns. To further validate the applicability of our study, we construct an aggregate MPU measure for the U.S. market and provide evidence of a negative premium, although this premium is weaker compared to what we have observed in China.

The contributions of this paper to the literature are threefold. First, we contribute to the literature on measuring monetary policy uncertainty. Previous studies have developed more specified proxies of MPU using interest rates volatility (Fernández-Villaverde et al., 2011), intraday derivatives (Kurov and Stan, 2018), professional forecast disagreement (Tillmann, 2020), and textual data (Husted et al., 2020). We complement these papers by conceptually exploring the sources of MPU across the dissemination of monetary policy information, and providing an aggregate measure. This comprehensive perspective is not only relevant for understanding the complexity of China's monetary policy framework but can also be applied to developed markets.

Second, we contribute to the longstanding literature that investigates the cross-sectional asset-pricing implications of macro-level risk or uncertainty, as in Ang et al. (2006), Ozoguz (2009), Bali et al. (2017), Farago and Tédongap (2018), etc. We extend these analyses by showing that MPU is an important determinant for predicting the cross-sectional dispersion in future stock returns. Our paper also connects to Ozdagli and Velikov (2020), who find that the exposure of a firm to monetary policy surprises command a negative risk premium in the cross-section of equity returns. Our analysis is distinct in demonstrating that the second-moment MPU shocks also matter.

Third, our paper contributes to the ongoing discussion on the efficiency of the Chinese financial market. The existing literature evaluates various frictions that give rise to market inefficiencies and anomalies in China, such as short-sale constraints (Chang et al., 2014), daily price limits (Chen et al., 2019), "T+1" trading mechanism (Qiao and Dam, 2020), direct government interventions (Dang et al., 2024), and institutional deficiencies in listing and delisting processes (Lee et al., 2023; Allen et al., 2024).[3] The MPU premium identified in this paper reflects another manifestation of market frictions. In a sense, our study extends this field by introducing a new perspective that emphasizes the role of monetary policy interventions in shaping cross-sectional return patterns.

[2] Numerous studies have utilized implied (or realized) volatility from financial derivatives to measure monetary policy uncertainty, e.g. Mueller et al. (2017), Kurov and Stan (2018) and Bauer et al. (2022). This type of measure is also pertinent to the monetary policy uncertainty perceived by the market, as financial derivatives reflect agents' expectations of future monetary policies. However, the financial derivatives market has been operating for only a short time with limited interest rate contracts in China. The derivative-based MPU proxies, such as implied volatility computed from interest rate option prices computed from intraday prices of interest rate futures, are not available in China.

[3] However, it is also worth noting that recent studies, such as Carpenter et al. (2021), have documented significant improvements in the price informativeness of China's stock market.

The remainder of this paper is organized as follows. Section 2 describes the institutional background and monetary policy uncertainty in China. Section 3 presents the asset-pricing results. Section 4 provides additional results and robustness tests. Section 5 concludes the paper.

2. Measuring China's monetary policy uncertainty

This section describes the institutional background on the unique features of monetary policies and the construction of the monetary policy uncertainty index in China.

2.1. Institutional background and the various sources of uncertainty

The most prominent feature that distinguishes Chinese monetary policymaking from that of developed countries is its complex policymaking mechanism.[4] First, China's monetary policies are a collective decision made by the State Council. The People's Bank of China (PBC) is one of 26 State Council ministries, which must formulate and implement monetary policies under the leadership of the State Council. In that sense, the PBC can provide advisory services and has limited operational independence at the monetary policy instrument level (McMahon et al., 2018). Therefore, China's monetary policymaking is much more complex than that of advanced economies, and the PBC's function is constrained by this collective decision mechanism.[5]

Second, China's monetary policy is characterized by a multi-objective and multi-instrument operating regime. The policy objectives of the PBC include price stability, economic growth, financial stability, and external account balance. It is difficult to achieve all these objectives simultaneously, leading to trade-offs in decision making. In particular, different objectives may take precedence at different times, which dampens the effectiveness of stabilizing expectations. However, China has never clarified its rule of monetary policymaking. In addition, the PBC has numerous policy instruments in its toolkit, including quantity-based (e.g., required reserve ratio, central bank bills) and price-based (e.g., benchmark deposit and lending rates, required reserve rate) tools. These policy instruments differ greatly in nature and do not move simultaneously, making it difficult to predict changes in monetary policy by monitoring various policy tools.

Third, the PBC prefers to ambiguously communicate with the market. According to Sun (2020), China's monetary policy stance is formulated in a vague and qualitative way without announcing any quantified operating target, making it difficult for the market to understand. For example, the PBC frequently uses "prudent" to describe its policy stance, whereas "prudent" refers to a narrow range around a neutral policy, which should be interpreted differently at different times.[6] Moreover, the PBC mainly conveys explanatory narratives but rarely uses forward guidance or releases official macroeconomic projections (Ma and He, 2020; Lin et al., 2023). This feature also creates confusion and obscurity in shaping well-guided market expectations.

Given the complexity of the Chinese monetary policy framework, various sources and facets of monetary policy uncertainty exist. From the perspective of monetary policymakers, the ambiguous communications and intricate policy rules stemming from the incorporation of multiple objectives and instruments are notable factors that drive policy uncertainty in China. More importantly, since the majority of financial participants in China are retail investors, which often lack economic and financial literacy, the complexity of China's monetary policy framework makes it more difficult for the public to form well-guided anticipations. The dissemination of policy information is essential in that sense.[7] Although the general public rarely reads monetary policy reports or monitors policy actions directly, they often digest policy information from the media (Blinder and Krueger, 2004; Munday and Brookes, 2021). Therefore, information dissemination plays a pivotal role in the process of monetary policymaking and implementation. This applies particularly to China, where the media serve as an important channel through which to transmit the voice of monetary policy.

2.2. An aggregate measure of monetary policy uncertainty

In this subsection, we provide an aggregate measure of monetary policy uncertainty from the perspective of information dissemination: the central bank (or policymaker) releases monetary policy information to the market via two channels, policy communications and policy operations. In addition, media coverage disseminates this information to market participants, influencing their perceptions of monetary policy and, consequently, their decision-making processes. Therefore, we consider three individual MPU measures, each corresponding to one of the three phases of monetary policy transmission, and then combine the information from these individual measures to construct our aggregate index. Before delving into the detailed discussion of the process, we must note that we focus on the sample from January 2003 to December 2020, which corresponds to the period when the PBC became the modern central bank for monetary policy implementation.

2.2.1. Communication-based individual measure

As documented above, the PBC often communicates ambiguous information to the market. Therefore, relevant information concerning monetary policy uncertainty in China is captured from the PBC's communication narratives. To do so, we first download all quarterly monetary policy executive reports (MPERs) from the PBC's official website, which regularly covers explanations about the most recent monetary policy decisions and outlooks on future monetary policy operations[8]. We adopt the procedure of Cieslak et al. (2021) to construct an individual MPU measure from policymakers' communications. Specifically, $\mathscr{M}$ denotes the list of phrases identified as pertaining to risk and uncertainty, and $\mathbf{w}_t = (w_{t,1}, \cdots, w_{t,N_t})$ is the list of phrases in a given MPER text t that has N_t terms in total. The most basic measure of the

[4] For comprehensive overviews of the monetary policy framework in China, please refer to Sun (2015, 2020), Chen et al. (2017) and Huang et al. (2020).

[5] A notable example to illustrate the drawbacks of dependence is the withdrawal of the stimulus policy in 2010, when the PBC started to tighten its monetary policy via its operations but still described its policy as "moderately easy" in its communications. This inconsistency arose because the PBC cannot alter its policy statements without approval from the State Council. The change on December 3, 2010 in the monetary policy stance from "moderately easy" to "prudent" was initially proposed in the Central Committee of the Communist Party of China and later reflected in the PBC's statement on December 27, 2010. The inconsistency between words and deeds during this period heightened monetary policy uncertainty, which we observed in the following.

[6] Sometimes, a "prudent" stance may be further clarified as "prudent with a tightening bias," "prudent neutral," or "prudent with an easing bias." Therefore, when the PBC describes its policy as "prudent," it could refer to "slightly tightening," "neural," or "slightly easing." For instance, the Monetary Policy Executive Reports (MPER) in 2010:Q4 illustrated that the prudent monetary policy stance was slightly expansionary from 1998 to 2002 and slightly contractionary from 2002 to 2007.

[7] As noted by the former PBC Governor Zhou, "Complicated models pose challenges to communication. … What the [PBC] does is to emphasize our communication with experts. We convey our messages as accurately as possible to those experts and let the messages trickle down to the public through experts, while also communicating in simplified forms directly to the public." Source: Michel Camdessus Central Banking Lecture at the IMF in Washington, DC on June 24, 2016. The full text can be accessed at https://www.imf.org/en/News/Articles/2016/07/06/17/45/SP062416-Xiaochuan-Zhou-Michel-Camdessus-Central-Banking-Lecture.

[8] Using MPER texts to analyze PBC communication is the most common practice in the literature, e.g., McMahon et al. (2018), Lin et al. (2023), etc. We do not consider oral communications since they are typically irregular and usually officially unpublished.

Table 1
Summary of the dataset related to monetary policy operations

Group	Number of Series	Group	Number of Series
Monetary Policy	14	Additional Instruments	6
Bond Market	8	Stock Market	6
Macro Economy	8	Price Level	5
Exchange Market	2	Government	2

Note: This table summarizes the dataset related to monetary policy operations, including 51 monthly monetary policy indicators. The original data are obtained from the National Bureau of Statistics of China and the China Economic Information (CEI) database. The sample spans from January 2003 to December 2020.

PBC's communication-based uncertainty aggregates the counts of terms in $\mathscr{M}$ in each MPER text and scales them by the number of total terms. More formally, the communication-based MPU index (MPU-MPC hereafter) is

$$\text{MPU}-\text{MPC}_t = \frac{\sum_{n=1}^{N_t} \mathbf{1}\left(w_{t,n} \in \mathscr{M}\right)}{N_t}, \tag{1}$$

where $\mathbf{1}(\cdot)$ represents the indicator function. As such, it depicts the intensity with which policymakers express uncertainty when discussing monetary policy.

The list of uncertainty phrases ($\mathscr{M}$) is derived from the keywords identified by Huang and Luk (2020)'s "uncertainty" criteria.[9] As the MPER is published quarterly, the communication-based MPU measure is calculated quarterly. To obtain a monthly series, we interpolate it accordingly.[10] This keyword counting method is akin to the approach employed by Baker et al. (2016) but on a word-level basis, and it has also been adopted by other relevant studies measuring political risk (Hassan et al., 2019).

2.2.2. Operation-based individual measure

In addition to uncertainty arising from communication, monetary policymakers could also introduce uncertainty through actual operational activities. Owing to the multi-objective and multi-instrument operating regimes in China, it may not be sufficient to simply measure operation-based MPU via second-moment variations of a particular policy tool. Following the approach of Jurado et al. (2015), we construct an operation-based individual MPU index (MPU-MPO hereafter) based on the unforecastable components of a broad set of monetary policy variables, such as intermediate targets and policy instruments. The first step is to estimate the uncertainty in an individual variable, and the second step is to aggregate individual uncertainty via a specific weighting scheme. To formalize this notion, we define h-period-ahead uncertainty in the policy variable $m_{j,t} \in \mathbf{M}_t = \left(m_{1,t}, \cdots, m_{N_m,t}\right)^\top$, denoted by $\mathscr{U}^m_{j,t}(h)$, as the conditional volatility of the purely unforecastable component of the future value of the series. Specifically,

$$\mathscr{U}^m_{j,t}(h) = \sqrt{E\left[\left(m_{j,t+h} - E\left[m_{j,t+h}|\mathscr{I}_t\right]\right)^2|\mathscr{I}_t\right]}, \tag{2}$$

where the expectation $E[\cdot|\mathscr{I}_t]$ is taken with respect to the information $\mathscr{I}_t$ available to economic agents at time t. Let $\widehat{\mathscr{U}}^m_{j,t}(h)$ be the estimated value of $\mathscr{U}^m_{j,t}(h)$. Next, the operation-based MPU measure can be constructed by aggregating the individual uncertainty variables at each date via aggregation weights w_j:

$$\text{MPU}-\text{MPO}_t = \sum_{j=1}^{N_m} w_j \widehat{\mathscr{U}}^m_{j,t}(h)\ . \tag{3}$$

We adopt the same indicators related to monetary policy operation as Wang et al. (2019) and Han and Wang (2023). As summarized in Table 1, this dataset includes 14 indicators associated with monetary policy, categorized under the "Monetary Policy" group, and 31 additional indicators utilized as latent common factors, classified under groups such as "Macro Economy" and "Bond Market." In addition, following the implementation of interest rate liberalization in 2015, we include six additional novel price-based policy tools, referred to as "additional instruments". These novel instruments include the standing lending facility (SLF), medium-term lending facility (MLF), and reverse repo rates, as outlined by the former PBC Governor Yi (2021). To obtain the operation-based MPU measure, we use $N_m = 14 + 6 = 20$ variables in the "Monetary Policy" and "Additional Instruments" groups. This approach allows us to extract the conditional volatility related to monetary policy (i.e., monetary policy uncertainty) while simultaneously controlling for the impact of other relevant economic variables. In addition, we estimate the monthly MPU-MPO series via the forecast horizon of $h = 1$ and equal weights, i.e., $w_j = 1/N_m,\ \forall j$.[11]

2.2.3. News-based individual measure

The media, which serve as a crucial intermediary in disseminating policy information, is also important for understanding monetary policy uncertainty. In the spirit of previous studies that have utilized newspapers to measure policy uncertainty in the U.S., we adopt a news-based proxy for the individual MPU measure (MPU-NEWS hereafter) proposed by Huang and Luk (2020). There are also several reasons for using a news-based measure, given the Chinese context. On the one hand, this measure avoids the limitation of focusing solely on the volatility of a single policy variable but captures the degree of MPU that the general public perceives in the PBC's policy actions and their consequences. On the other hand, this measure is more representative than other measures because it reflects a wider audience reading newspapers. Since Chinese financial markets are dominated by individual retail investors, the news-based MPU index is valuable for measuring individuals' perceptions of monetary policy uncertainty.

The compilation strategy of the MPU index follows Baker et al. (2016) and involves counting the number of occurrences of articles

[9] It is important to clarify that our text analysis is conducted in the original language of MPER, which is Chinese. Therefore, uncertainty terms are based on the Chinese vocabulary provided by Huang and Luk (2020) under the "uncertainty" criteria. In addition, we have supplemented this list with two phrases, "risk" (风险) and "unstable" (不稳定), to align with the common expressions in the PBC communication texts.

[10] Notably, the release of MPER typically exhibits a lag, often published in the second month of the next quarter. For instance, the MPER for 2008:Q2 was actually released on August 15. To avoid look-ahead bias, when calculating the MPU-MPC index for a given month, we use the most recently published MPER text.

[11] Appendix A.2 provides a brief description of the estimation procedure developed by Jurado et al. (2015) and details about the dataset. We use the equally weighted average since the measure of average uncertainty does not impose any structure on the individual uncertainties and beyond the assumptions on the latent volatility process (Jurado et al., 2015). The uncertainty series constructed from different forecast horizons and weighting schemes are highly correlated and show similar time-series trends. These different setups do not change our main empirical results.

discussing MPU in Chinese-language newspapers.[12] To our knowledge, Chen and Tillmann (2021) also constructed a news-based MPU measure for China. In contrast to Huang and Luk (2020), they used two English-speaking international newspapers and found that the overall evolution of the MPU was similar for these two indices, with a correlation of 0.700 from 2000–2018. Since we focus on the MPU premium in China's domestic stock market, we use the Huang and Luk (2020) MPU index constructed from local newspapers in mainland China.[13]

2.2.4. An aggregate monetary policy uncertainty measure

Prior to introducing the aggregate MPU measure, we delve into the temporal dynamics of the three individual measures. As shown in Fig. 1, these individual MPU indices generally exhibit similar movements and peaks. Following the global financial crisis in September 2008, these measures first surged and then gradually declined after large-scale stimulus policies and economic recovery began in 2009. As China's monetary policy became more prudent from 2012 onwards, these indices fluctuated less. However, their evolution diverges at some episodes, indicating that individual indicators may not comprehensively capture the overall MPU perceived by the market. For example, the communication-based index remained subdued in contrast to the peak of the MPU amid the withdrawal of expansionary policy in 2010–2011. In contrast, during the policy tightening in 2017, the remaining two indices either exhibited a lagged response or showed no significant change.[14]

Now, we propose a comprehensive monetary policy uncertainty index by aggregating information across individual measures. Our aggregate MPU measure is constructed using principal component analysis (PCA), which extracts the first principal component of the three individual MPU indices as the aggregate index. On the one hand, this aggregation technique is simple, efficient and widely used in the finance literature, e.g., Baker and Wurgler (2006), Huang et al. (2021), etc.[15] On the other hand, this approach could eliminate non-MPU noise and exploit the common component among these individual indices. In addition, as an unsupervised learning technique for dimensionality reduction, the advantage of PCA lies in its ability to aggregate information without needing to predetermine supervised targets.[16] As presented in Table 2, all three individual MPU measures demonstrate a highly positive correlation with the aggregate index. Among them, the news-based MPU index is the most informative and has the highest factor loading and correlation with the aggregate measure.[17]

Fig. 2 illustrates the time series variation in our aggregate MPU measure that reveals commonalities among the individual measures. We observe highly correlated dynamics of this index with changes in monetary policy and economic conditions in China. The most striking episodes of heightened MPU coincide with various unpredictable monetary policy shocks in China, such as the global spread of the financial crisis and aggressive policy stimulus in 2008, the subsequent withdrawal in 2010–2011, the cash crunch in 2013, the transition to the "new normal" periods in 2014–2016, the prudent and moderate policy adjustments in 2017–2018, and the COVID-19 pandemic and policy easing in 2020.[18] During these periods, the actions of monetary policy and its potential outcomes are difficult to predict for the market, leading to an increase in MPU. In general, the peaks of the aggregate index are more reasonable than those of the other indices and capture informative episodes indicated by certain individual measures in a rational manner.

2.2.5. Economic determinants of aggregate monetary policy uncertainty

We examine the time-series movements of our aggregate MPU measure by regressing it on a variety of variables related to economic uncertainty (EU), economic policy uncertainty (EPU), monetary policy stance, and economic state. Specifically, the EU index in China is constructed via the method of Jurado et al. (2015) using the macroeconomic dataset of Lin et al. (2023). We consider two news-based indices of EPU: one is proposed by Huang and Luk (2020) (EPU^{HL}) based on Chinese newspapers, and the other is from Baker et al. (2016) (EPU^{BBD}) based on English newspapers.[19] In terms of variables related to monetary policy, a comprehensive monetary policy index (MPI) is used to measure the PBC's monetary policy stance. Notably, an increase in the comprehensive monetary policy index indicates a contractionary monetary policy. The economic state variables include the output gap (OG), business cycle index (BCI), recession dummy (REC), and relative bond rate (RREL). The former two indicators are procyclical, whereas the last two are countercyclical.[20]

The first three columns in Table 3 present the univariate regression results of our aggregate MPU measure on other uncertainty measures. We find that the slope coefficient of EU is positive but statistically insignificant and that the adjusted goodness-of-fit (Adj. R^2) is quite small. This result indicates that our MPU index captures the uncertainty perceived by the market pertaining to monetary policy, as opposed to overall economic uncertainty. In column 2, we use Huang and Luk

[12] Huang and Luk (2020) first searched for articles containing at least one keyword in each of the four criteria, namely, (1) economy, (2) uncertainty, (3) policy, and (4) monetary policy, from 114 newspapers in mainland China. Then, they scaled the number of articles in each month by the number of articles that met the criterion (1) for the same month. Next, they standardized the series to have a standard deviation of unity from January 2000 to December 2011. Finally, they computed the simple average of the monthly series across all newspapers and normalized it to have an average value of 100 from January 2000 to December 2011. It is available at https://economicpolicyuncertaintyinchina.weebly.com/. One may be concerned that the Chinese government's stringent censorship on domestic media could affect the validity of a news-based measure. Huang and Luk (2020) conducted a comprehensive examination and discussion of the impact of media censorship on their index construction and found that there was no significant media bias. Therefore, we consider the news-based MPU to be effective.

[13] The MPU index developed by Chen and Tillmann (2021) is not yet publicly available.

[14] Appendix A.1 describes the summary statistics of the individual MPU components.

[15] Our aggregate MPU measure is robust for the aggregation method. In robustness tests, we consider an alternative aggregate index constructed by simply equal-weighting the three (standardized) individual MPU components. The intuition is that if each individual measure is unbiased, equal-weighting efficiently reduces the idiosyncratic errors.

[16] Given that our primary objective is to construct a comprehensive MPU index rather than solely to predict any specific target, we prefer the use of unsupervised techniques for information aggregation.

[17] Given the generality of information transmission in monetary policymaking and execution, our method is also applicable to the U.S. context. In this regard, we construct a measure of aggregate monetary policy uncertainty for the U.S. and examine its economic determinants. We also investigate its cross-sectional asset-pricing implications and find that this aggregate measure commands a negative risk premium in the U.S. market. However, the significance of the premium is somewhat weaker compared to that in China, indicating that the pricing power of MPU is stronger in a "policy-driven" environment. These results are provided in Appendix C. We appreciate the referee for this suggestion.

[18] For the timing of the 2008 financial crisis, instead of using the interval set by the NBER, we adopt the bankruptcy of Lehman Brothers in September 2008 as the starting point and PBC's increase in the reserve requirement rate (RRR) in January 2010 as the end marker for the policy shift.

[19] Appendix A.2 provides a detailed description of the methodology and data used in constructing the EU index. Huang and Luk (2020)'s EPU index is publicly available on their website. It is important to stress that their EPU index is constructed based on the same news article dataset and similar procedure as their MPU index, leading to a relatively high correlation between these two indices. Baker et al. (2016) developed an EPU index for China based on the South China Morning Post, which can be downloaded from https://www.policyuncertainty.com/china_epu.html. Furthermore, Davis et al. (2019) constructed an EPU index (EPUDLS) based on mainland newspapers, which is highly correlated with Baker et al. (2016)'s index. Hence, the main results are similar if we use EPUBBD and EPUDLS.

[20] Appendix A.2 and A.3 provides more details about the variables used here.

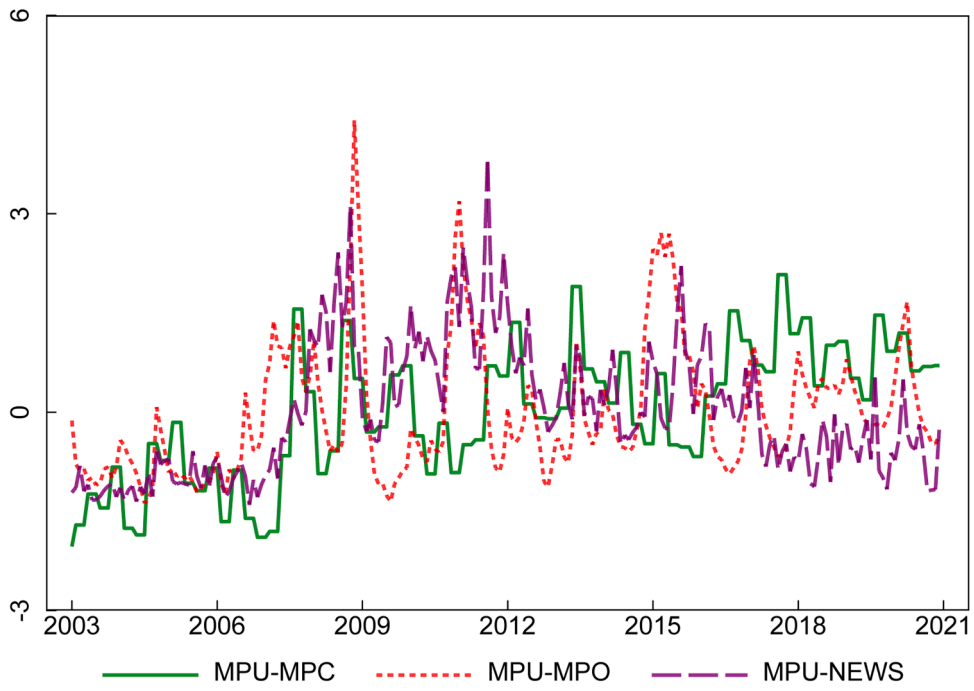


Fig. 1. Time series of the individual monetary policy uncertainty measures.
Note: This figure shows the time series of three individual monetary policy uncertainty measures, namely, communication-based (MPU-MPC, the green solid line), operation-based (MPU-MPO, the red short-dashed line) and news-based (MPU-NEWS, the purple long-dashed line) uncertainty. To make the measures comparable, we standardize each MPU index by its full-sample mean and standard deviation. The sample spans from January 2003 to December 2020.

Table 2
Loadings and correlations of the individual monetary policy uncertainty measures

	Loading	Correlation
MPU-MPC	0.474	0.590
MPU-MPO	0.605	0.752
MPU-NEWS	0.639	0.795

Note: This table reports the factor loadings of the aggregate monetary policy uncertainty measure on individual measures and the correlations between them. The sample spans from January 2003 to December 2020.

(2020)'s EPU index as the explanatory variable, the coefficient of which is positively estimated and statistically significant. Given that our MPU index aggregates information from their news-based individual measure, the adjusted R^2 is relatively high. However, when we consider Baker et al. (2016)'s EPU measure in column 3, the adjusted R^2 becomes small, whereas the coefficient remains positive and significant. Overall, these results suggest that although our aggregate MPU measure is positively correlated with extant uncertainty variables, it has distinctive information that cannot be fully explained by them.

Moreover, column 4 shows that the slope coefficient of the MPI is positive and highly significant. The underlying economic implication is that MPU tends to increase as monetary policy tightens. As presented in the last column, we include all the economic state variables simultaneously, and the multivariate regression results indicate that the aggregate MPU index exhibits a countercyclical pattern to some extent. For instance, the business cycle index loads significantly and negatively on our MPU measure, which suggests that MPU is greater during periods of economic distress.

In addition, we analyze the relationship between our novel aggregate MPU measure and the yield curve, noting that the yield curve itself can significantly impact stock returns.[21] As presented in Appendix D, we find that the aggregate MPU measure negatively and significantly affects the changes in short-end bond yields (with maturities of 3 to 12 months). This negative relationship is also consistent with the U.S. results concerning policy uncertainty and bond yields (Tillmann, 2020; De Pooter *et al.*, 2021; Leippold and Matthys, 2022), indicating flight-to-quality behaviour characterized by an increasing demand for safe government bonds in response to heightened monetary policy uncertainty. We also observe that yield volatility increases with increasing uncertainty and that the slope factor is most responsive to MPU shocks. Comparing different individual measures, we find that the operation-based MPU index has a greater impact on yield changes, while the news-based index more significantly influences yield volatility. These results suggest that the three components of the aggregate measure may affect different parts of the yield curve. Therefore, using an information aggregation technique is valuable for assessing the overall level of China's monetary policy uncertainty.

3. The asset-pricing implications of monetary policy uncertainty

In this section, we investigate the cross-sectional asset-pricing implications of the aggregate MPU measure by showing that it is priced with a negative premium.

3.1. Theoretical foundation

Our theoretical foundation for the cross-sectional asset-pricing tests is built on the framework of Merton (1973)'s conditional ICAPM, which implies the following equilibrium relation between risk and return:

$$E_t\left(R_{i,t+1}-R_{f,t+1}\right)=\gamma_m\cdot Cov_t\left(R_{i,t+1},R_{m,t+1}\right)+\gamma_x\cdot Cov_t\left(R_{i,t+1},\Delta z_{t+1}\right), \quad (4)$$

where $R_{i,t+1}$ is the ex-post return on asset i, $R_{f,t+1}$ is the risk-free rate, $R_{m,t+1}$ is the ex-post market return, and z_t is a vector of state variables that shift the investment opportunity set. The term γ_m is the relative risk aversion of market investors, and the term γ_x represents the covariance price of risk for shifts in the state vector that governs the stochastic

[21] We are grateful for this constructive suggestion from the referee.

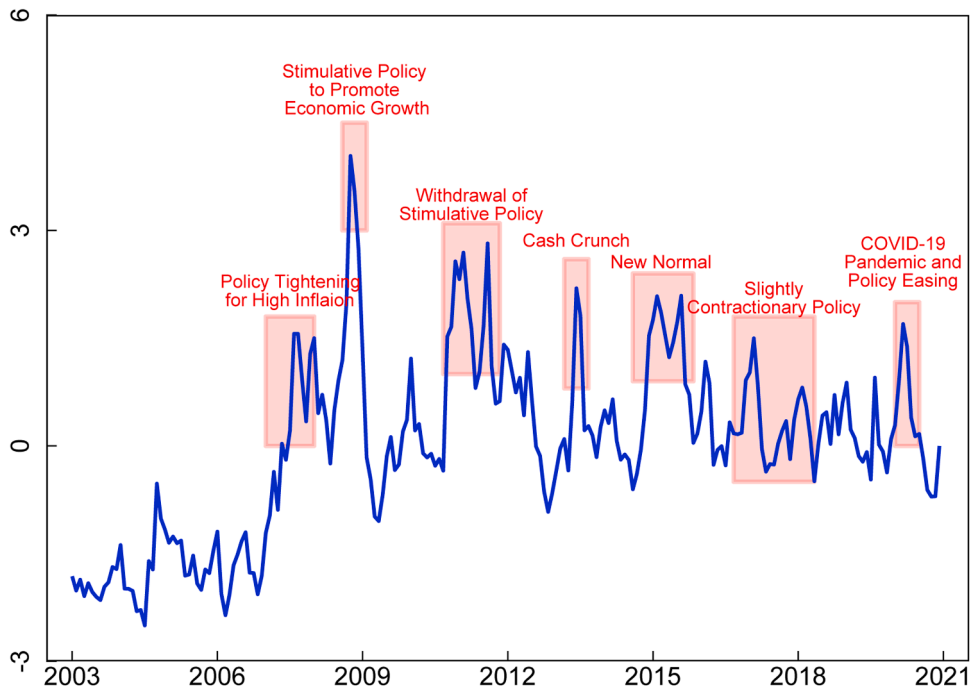


Fig. 2. Time series of the aggregate monetary policy uncertainty measure.
Note: This figure shows the time series of the aggregate monetary policy uncertainty measure (MPU, the blue line). The red shaded area represents the most striking episodes of heightened monetary policy uncertainty. The sample spans from January 2003 to December 2020.

Table 3
Economic determinants of the aggregate monetary policy uncertainty measure

	Uncertainty			Monetary policy	Economic state
	(1)	(2)	(3)	(4)	(5)
EU	6.286				
	(1.356)				
EPU^{HL}		0.026			
		(11.452)			
EPU^{BBD}			0.001		
			(2.186)		
MPI				1.042	
				(2.131)	
OG					-0.019
					(-0.403)
BCI					-0.069
					(-1.890)
REC					0.724
					(0.977)
RREL					0.258
					(0.844)
Intercept	-2.324	-3.292	-0.258	0.315	6.840
	(-1.416)	(-11.136)	(-0.994)	(1.718)	(1.901)
Obs.	216	216	216	216	216
Adj. R^2 (%)	3.954	58.073	3.574	6.815	3.388

Note: This table reports the estimates from univariate and multivariate regressions of the aggregate monetary policy index on a set of variables related to uncertainty, monetary policy, and economic state. Alternative uncertainty indices used are the economic uncertainty index (EU) and the economic policy uncertainty indices of Huang and Luk (2020) (EPU^{HL}) and Baker et al. (2016) (EPU^{BBD}). The monetary policy variable is the comprehensive monetary policy index (MPI), which measures the PBC's policy stance. The economic state variables used are the output gap (OG), business cycle index (BCI), recession dummy (REC) and relative bond rate (RREL). The sample spans from January 2003 to December 2020. Newey–West adjusted *t*-statistics are provided in parentheses using four lags.

investment opportunity set. Equation (4) states that in equilibrium, investors are compensated with higher expected returns for bearing systematic market risk and for bearing the risk of unfavourable shifts in the investment opportunity set.

There is an extensive body of literature providing both theoretical and empirical evidence showing that uncertainty shocks capture additional sources of risk and have a significant effect on asset prices (Anderson et al., 2009; Bloom, 2009; Drechsler, 2013; Jurado et al., 2015; Bekaert and Engstrom, 2017; Ludvigson et al., 2021). Furthermore, Pástor and Veronesi (2012, 2013) have presented models suggesting that government policy uncertainty raises the equity risk premium and, in particular, serves as a state variable affecting consumption and investment opportunities. Specifically, as monetary policies play a critical role in shaping interest rates and liquidity conditions, any uncertainty regarding its future course becomes an integral factor in an investor's consumption and investment decisions.

Drawing on these insights, we argue that MPU serves as a state variable, and that a positive MPU shock can lead to adverse changes in the investment opportunity set. This is because when MPU increases, investors become more concerned about discount rates and thereby the future value of investment, leading to a reduction in current investment demand. In the cross-section, to hedge against such an unfavourable shift, investors prefer holding stocks that have higher covariances (beta) with MPU shocks. This is because an increase in MPU will increase the returns on these stocks due to positive intertemporal correlation. Consequently, through intertemporal hedging demand, investors have greater demand for stocks with higher covariances with MPU shocks, bidding up their price and driving down their expected returns.

In addition to the conditional ICAPM framework, the negative monetary policy uncertainty premium can be motivated by limited participation in the stock market under heightened uncertainty. If investors' preferences about MPU are sufficiently dispersed and MPU is sufficiently high, investors with relatively high aversion against MPU

may cease or reduce their participation in a stock.[22] As a result of this limited participation, stocks with a high MPU beta are held only by investors with a sufficiently optimistic view of MPU or low aversion to it. Therefore, stocks with high MPU betas require low premiums.[23]

3.2. Cross-sectional tests

In this subsection, we examine the cross-sectional relationship between MPU shock and future stock returns. First, we estimate the exposure of individual stocks to the innovation in aggregate MPU measure. Second, we conduct nonparametric portfolio-level analyses. Third, we present the univariate and multivariate Fama–MacBeth cross-sectional regressions. Finally, we construct a risk factor based on the MPU beta.

3.2.1. Estimating exposures to monetary policy uncertainty

Our paper includes all common stocks traded on China's domestic stock market, i.e., the A-share market on the Shanghai (SSE) and Shenzhen (SZSE) stock exchanges from January 2003 to December 2020. For each stock and month in our sample, the exposures of individual stocks to MPU shocks are obtained from the monthly rolling regressions of excess stock returns (R^e) on the first-order difference (or the monthly change) in the aggregate MPU measure ($\Delta MPU_t = MPU_t - MPU_{t-1}$) with controlling for the Chinese risk factors of Liu et al. (2019):

$$R^e_{i,t} = \alpha_{i,t} + \beta^{MPU}_{i,t}\cdot\Delta MPU_t + \beta^{MKT}_{i,t}\cdot MKT_t + \beta^{SMB}_{i,t}\cdot SMB_t + \beta^{VMG}_{i,t}\cdot VMG_t + \beta^{PMO}_{i,t}\cdot PMO_t + \varepsilon_{i,t}, \quad (5)$$

where MKT, SMB, VMG and PMO are the market, size, value and turnover factors, respectively, in the Chinese four-factor (CH4) model. For each stock i in month t, the coefficient of ΔMPU represents its exposure to MPU shocks, which we refer to as the MPU beta, i.e., β^{MPU}. The use of innovations in MPU is also consistent with the ICAPM of Merton (1973).[24] That is, stock returns respond only to the unexpected component of MPU. We perform a 60-month fixed window estimation and require at least 48 available months for each stock.[25] Therefore, the first estimates of the MPU beta for individual stocks start in December 2007 and are obtained via the sample from January 2003 to December 2007.

3.2.2. Univariate portfolio-level analysis

We conduct a portfolio-level analysis by taking advantage of being nonparametric without any functional form restrictions on the relationship between the MPU beta and future stock returns. We start with univariate portfolio analysis. For each month t, we form both equal- and value-weighted decile portfolios by sorting individual stocks. The breakpoints are calculated as the 10th to 90th percentiles according to the past-month MPU beta, where the decile 1 (*Low*) portfolio contains stocks with the lowest MPU beta during the previous month, and the decile 10 (*High*) portfolio contains stocks with the highest MPU beta. Intuitively, this approach involves building a portfolio at the end of the previous month and holding it until the end of the subsequent month. We compute the average values of excess returns (RET−RF) and risk-adjusted returns (alphas) of the CAPM (α_{CAPM}) and the Chinese four-factor model (α_{CH4}).

Table 4 reports the one-month-ahead univariate portfolio results. The first column shows significant cross-sectional variations in the average values of the post-ranking MPU beta, in which the *Low* and *High* MPU betas are -4.624 and 5.350, respectively, with an absolute difference of 9.974. For the equal-weighted portfolio, column 2 shows that the overall next-month average excess stock return decreases from 0.946% to 0.597% per month, moving from decile 1 to decile 10. Moreover, the spread of the average excess return between *High* and *Low* portfolios is also economically large and statistically significant, with a value of -0.349% and a Newey–West adjusted *t*-statistic of -2.000. This result indicates that stocks in the lowest MPU beta decile generate 4.188% higher annual excess returns compared to stocks in the highest MPU beta decile. In addition, the next two columns present the magnitude and statistical significance of the risk-adjusted returns (alphas) from the CAPM and CH4 factor models. For example, the difference in CH4 alphas between *High* and *Low* portfolios is -0.446% per month, with a Newey–West *t*-statistic of -2.639.

The value-weighted portfolio results are consistent with the above findings and exhibit an even higher return spreads. Similar to previous studies, the value-weighted excess return and alphas for each decile portfolio are lower than those of the equal-weighted portfolios. However, the value-weighted average excess return between *High* and *Low* portfolios is greater, indicating that the premium of MPU is positively correlated with the size factor. As presented in the last three columns, the average excess returns are 0.691% per month among the lowest MPU beta stocks and -0.190% per month among the highest MPU beta stocks. The corresponding spread is negative and significant (-0.882% with a *t*-stat. = -3.108), indicating that the lowest MPU beta decile generates 10% more annual returns than do those in the highest MPU beta decile. The next two columns present the alpha results from the factor models. We find that the spreads between *High* and *Low* portfolios are larger in magnitude and statistically significant. For instance, the CH4 alpha spread is -1.132%, and the Newey–West *t*-statistic is -4.453. This result suggests that the cross-sectional spreads of MPU betas are sizable and significant. In summary, we provide portfolio-level evidence that MPU is priced in the Chinese market with a negative premium.

Notably, the MPU betas documented in Table 4 are the month over which we measure average portfolio returns. As we expected, the post-ranking MPU beta increases monotonically when moving from the lowest to the highest MPU beta decile. Therefore, investors have paid high prices for stocks that have exhibited high MPU betas in the past because of the expectation that this behaviour will be repeated in the future; however, a natural question is whether these expectations are rational. To address this question, we examine the persistence of the MPU beta by running stock-level cross-sectional regressions of the MPU beta on lagged the MPU beta and lagged stock-level characteristics. These stock-level variables are well-known cross-sectional return predictors, including the market beta (β^{MKT}), log of market capitalization (SIZE), earnings-price ratio (EP), operating profitability (ROE), investment (I/A), net operating assets (NOA), short-term reversal (REV), coskewness (COSKEW), turnover (TOVER) and illiquidity (ILLIQ).[26]

The first row in Table 5 reports the average coefficients on the MPU beta from the univariate and multivariate cross-sectional regressions. According to the univariate regression of the 1-year-ahead MPU beta on the lagged MPU beta, the coefficient is largely positive (0.726) and extremely statistically significant (*t*-stat. = -36.482). In other words, stocks with high MPU betas also tend to exhibit similar features in the following 12 months. When the stock-level control variables are added

[22] Studies such as Easley and O'Hara (2009) and Epstein and Schneider (2010) predict that stock market participation should be lower in the presence of uncertainty. Gábor-Tóth and Georgarakos (2019) provide empirical evidence documenting that households with more exposure to policy uncertainty news are less likely to invest in stocks directly or through mutual funds.

[23] Earlier studies find that limited participation in a stock can lead to a lower equity premium (Cao et al., 2005; Bossaerts et al., 2010).

[24] We establish the robustness of using alternative measures of innovations in MPU in Section 4.3.

[25] Unless otherwise stated, when constructing the MPU beta, we retain stocks labelled with ST or falling in the bottom 30% in terms of size. This approach provides a more precise estimation of the MPU beta. When conducting the cross-sectional pricing tests, we exclude those stocks from our main empirical analysis, in line with Liu et al. (2019).

[26] Appendix A.5 provides details of the definitions and construction of the stock-level variables.

Table 4
Univariate portfolios of stock returns sorted by the monetary policy uncertainty beta

		Equal-weighted			Value-weighted		
Decile	Post β^{MPU}	RET−RF	α_{CAPM}	α_{CH4}	RET−RF	α_{CAPM}	α_{CH4}
Low	-4.624	0.946	0.812	0.746	0.691	0.555	0.739
		(1.194)	(2.763)	(4.795)	(0.853)	(2.569)	(3.099)
2	-2.344	1.004	0.872	0.533	0.507	0.382	0.191
		(1.240)	(3.206)	(4.433)	(0.640)	(2.016)	(0.891)
3	-1.428	1.064	0.936	0.621	0.711	0.584	0.310
		(1.364)	(3.193)	(3.790)	(0.935)	(2.484)	(1.389)
4	-0.751	1.024	0.898	0.518	0.521	0.399	0.302
		(1.292)	(3.224)	(3.446)	(0.685)	(1.901)	(1.192)
5	-0.109	1.034	0.907	0.528	0.236	0.121	-0.172
		(1.293)	(3.315)	(4.269)	(0.304)	(0.477)	(-0.739)
6	0.497	0.810	0.681	0.271	0.282	0.169	-0.263
		(1.020)	(2.533)	(2.242)	(0.407)	(0.725)	(-1.148)
7	1.115	0.873	0.746	0.460	0.435	0.315	0.231
		(1.120)	(2.981)	(5.739)	(0.624)	(1.613)	(1.180)
8	1.823	0.798	0.671	0.368	0.382	0.255	0.091
		(1.016)	(2.294)	(3.280)	(0.521)	(1.342)	(0.432)
9	2.746	0.773	0.643	0.428	0.114	-0.016	-0.040
		(0.968)	(2.331)	(3.576)	(0.148)	(-0.095)	(-0.244)
High	5.350	0.597	0.464	0.300	-0.190	-0.321	-0.393
		(0.733)	(1.594)	(2.136)	(-0.241)	(-1.457)	(-2.002)
High−Low		-0.349	-0.348	-0.446	-0.882	-0.876	-1.132
		(-2.000)	(-1.983)	(-2.639)	(-3.108)	(-3.095)	(-4.453)

Note: This table reports the univariate portfolio results to analyze the cross-sectional relationship between the monetary policy uncertainty beta (β^{MPU}) and excess stock returns for the equal- and value-weighted portfolios. For each month, the equal- and value-weighted decile portfolios are formed by sorting individual stocks based on their MPU betas during the previous month, where decile 1 (*Low*) contains the stocks with the lowest MPU beta and decile 10 (*High*) contains the stocks with the highest MPU beta. The first column reports the average post-ranking MPU betas of the individual stocks in each decile, and the remaining columns report the average excess returns (RET–RF) and alphas, where α_{CAPM} denotes the alpha of the CAPM model containing the market (MKT) factor and α_{CH4} denotes the alpha of the CH4 model containing the market (MKT), size (SMB), value (VMG) and turnover (PMO) factors. The last row reports the differences in the average excess returns and alphas between decile 1 (*Low*) and decile 10 (*High*). The MPU beta is obtained from Equation (5). The average excess returns and alphas are defined in monthly percentage terms. The sample spans from December 2007 to December 2020. Newey–West adjusted *t*-statistics are given in parentheses using four lags.

Table 5
Persistence of the monetary policy uncertainty beta

h-year-ahead β^{MPU}	Univariate regressions	Controlling for stock characteristics
$h = 1$	0.726	0.679
	(36.482)	(31.040)
$h = 2$	0.542	0.480
	(22.803)	(21.832)
$h = 3$	0.354	0.298
	(14.858)	(15.416)
$h = 4$	0.156	0.126
	(6.175)	(7.312)
$h = 5$	-0.005	0.008
	(-0.360)	(0.569)

Note: This table examines the persistence of the monetary policy uncertainty beta (β^{MPU}) by running stock-level cross-sectional regressions of the MPU beta on the lagged MPU beta and lagged stock-level characteristics. The first column reports the average slope coefficients on the lagged MPU beta from the univariate regressions with the one-year- to five-year-ahead MPU beta as the dependent variable. The last column reports the average slope coefficients on the lagged MPU beta from the multivariate regressions after controlling for lagged stock-level characteristics. For each stock, the MPU beta is estimated via the 60-month rolling regression of Equation (5). The stock-level characteristics include the market beta (β^{MKT}), log of market capitalization (SIZE), earnings-price ratio (EP), operating profitability (ROE), investment (I/A), net operating assets (NOA), short-term reversal (REV), coskewness (COSKEW), turnover (TOVER) and illiquidity (ILLIQ). The sample spans from December 2007 to December 2020. Newey–West adjusted *t*-statistics are given in parentheses using four lags.

to the regression, the coefficient remains large and significant (0.679 with a *t*-stat. = -31.040). We also investigate the persistence of the MPU beta for two, three, four, and five years ahead. The last four rows in Table 5 show that the MPU beta remains highly persistent for up to five years into the future. These results indicate that the estimated historical MPU betas successfully predict future MPU betas and hence are good proxies for the true conditional betas, which is important for interpretations of the results in terms of an equilibrium model.

3.2.3. *Bivariate portfolio-level analysis*

To examine the incremental predictive power of the MPU beta for future stock returns after controlling for the well-known cross-sectional return predictors, we perform conditional bivariate portfolio sorts on the MPU beta and stock-level predictors, i.e., the market beta (β^{MKT}), market capitalization (SIZE), etc. For example, we control for the influence of the market beta by first forming decile portfolios ranked based on the market beta during the previous month. Then, in each market beta decile, we sort individual stocks into decile portfolios ranked via the MPU beta such that decile 1 (10) contains the stocks with the lowest (highest) MPU beta values. In each MPU beta decile, we average excess (or risk-adjusted) portfolio returns across the ten market beta deciles to produce decile portfolios with dispersion in the MPU beta but that contain all of the stocks' market beta. This procedure creates a set of MPU beta portfolios with very similar levels for the market beta; hence, these MPU beta portfolios control for variations in the market beta.

Table 6 presents the value-weighted bivariate portfolio-level analysis, where the one-month-ahead Chinese four-factor (CH4) alphas in each decile portfolio are reported. The first column shows that after controlling for the market beta, the MPU beta still produces a difference of -0.914% between the *High* and *Low* equal-weighted portfolios, with a Newey–West *t*-statistic of -3.153 after controlling for the market beta. Similar results are obtained after controlling for other stock-level characteristics, where the CH4 alpha spreads between the *High* and *Low* deciles are in the range of -0.672% and -0.998%, respectively. In summary, these findings indicate that well-known cross-sectional predictors cannot explain the negative and significant MPU premium in the Chinese stock market.

3.2.4. *Stock-level Fama–MacBeth regressions*

The portfolio-level results provide modest evidence that monetary policy uncertainty earns a significantly negative risk premium; however,

Table 6
Value-weighted bivariate portfolio sorts

Decile	β^{MKT}	SIZE	EP	ROE	I/A	NOA	REV	COSKEW	TOVER	ILLIQ
Low	0.777	0.736	0.459	0.712	0.530	0.411	0.718	0.555	0.502	0.673
	(3.497)	(2.800)	(2.355)	(2.941)	(2.434)	(2.051)	(3.166)	(2.597)	(2.253)	(2.755)
2	0.137	0.190	0.256	0.074	0.435	0.352	0.211	0.575	0.015	0.134
	(0.737)	(0.778)	(1.204)	(0.351)	(2.046)	(1.667)	(1.128)	(2.737)	(0.069)	(0.598)
3	0.187	0.020	0.210	0.374	0.089	0.244	0.082	-0.024	0.331	0.377
	(0.958)	(0.105)	(0.927)	(1.634)	(0.343)	(1.132)	(0.425)	(-0.117)	(1.673)	(1.727)
4	0.098	0.528	-0.153	-0.049	0.045	0.009	0.041	0.207	0.307	0.321
	(0.442)	(2.109)	(-0.948)	(-0.229)	(0.231)	(0.042)	(0.195)	(0.955)	(1.610)	(1.375)
5	-0.016	-0.473	0.152	-0.032	-0.179	0.000	0.052	-0.024	-0.201	-0.536
	(-0.071)	(-2.057)	(0.624)	(-0.134)	(-0.695)	(-0.002)	(0.213)	(-0.104)	(-0.855)	(-2.671)
6	-0.037	-0.064	0.141	-0.024	0.002	0.059	0.018	-0.348	0.004	-0.069
	(-0.166)	(-0.254)	(0.704)	(-0.111)	(0.009)	(0.244)	(0.077)	(-1.536)	(0.019)	(-0.266)
7	0.018	0.162	0.148	0.309	0.217	-0.120	0.191	0.146	0.096	0.301
	(0.082)	(0.758)	(0.667)	(1.301)	(0.919)	(-0.574)	(0.878)	(0.563)	(0.470)	(1.325)
8	-0.012	0.118	-0.142	-0.373	-0.044	0.047	-0.132	-0.051	-0.047	0.260
	(-0.064)	(0.623)	(-0.704)	(-2.067)	(-0.318)	(0.223)	(-0.687)	(-0.238)	(-0.215)	(1.597)
9	-0.001	-0.153	-0.109	0.037	-0.179	-0.056	-0.097	-0.221	-0.201	-0.250
	(-0.005)	(-0.796)	(-0.666)	(0.223)	(-0.992)	(-0.366)	(-0.541)	(-1.215)	(-1.084)	(-1.630)
High	-0.432	-0.332	-0.398	-0.387	-0.275	-0.373	-0.478	-0.254	-0.309	-0.394
	(-2.063)	(-1.742)	(-2.253)	(-1.970)	(-1.423)	(-1.953)	(-2.643)	(-1.383)	(-1.633)	(-2.020)
High−Low	-1.209	-1.068	-0.857	-1.099	-0.805	-0.784	-1.196	-0.809	-0.810	-1.067
	(-4.855)	(-4.028)	(-3.187)	(-3.907)	(-3.243)	(-3.204)	(-4.620)	(-2.978)	(-3.188)	(-4.247)

Note: This table reports the conditional bivariate portfolio results to analyze the cross-sectional relationship between the monetary policy uncertainty beta (β^{MPU}) and excess stock returns after controlling for stock-level characteristics. This table reports the value-weighted CH4 alphas (in percentages) relative to the market (MKT), size (SMB), value (VMG) and turnover (PMO) factors for each MPU beta decile, averaged across the ten control groups. For each month, the value-weighted decile portfolios are formed by first sorting the individual stocks based on one control variable and then sorting the stocks in each control variable decile into deciles based on their MPU betas during the previous month, where decile 1 (*Low*) contains the stocks with the lowest MPU beta and decile 10 (*High*) contains the stocks with the highest MPU beta. The MPU beta is obtained from Equation (5). The stock-level characteristics include the market beta (β^{MKT}), log of market capitalization (SIZE), earnings-price ratio (EP), operating profitability (ROE), investment (I/A), net operating assets (NOA), short-term reversal (REV), coskewness (COSKEW), turnover (TOVER) and illiquidity (ILLIQ). The last row reports the differences in alphas between decile 1 (Low) and decile 10 (High). The sample spans from December 2007 to December 2020. Newey–West adjusted *t*-statistics are given in parentheses using four lags.

this method makes it difficult to simultaneously control for multiple effects and overlooks a large amount of stock-level variation. We thus investigate the role of MPU in asset pricing at the stock level via the cross-sectional regressions of Fama and MacBeth (1973). Specifically, we use the MPU beta (β^{MPU}) from the 60-month rolling regressions based on Equation (5) and then conduct a series of monthly Fama–MacBeth regressions of one-month-ahead excess stock returns on the MPU beta with and without control variables. These regressions are run for the following specification and nested versions thereof:

$$R^{e}_{i,t+1} = \alpha_t + \lambda^{MPU}_t \cdot \beta^{MPU}_{i,t} + \mathbf{X}'_{i,t} \cdot \boldsymbol{\lambda}^{X}_t + \varepsilon_{i,t+1}, \quad (6)$$

where $R^{e}_{i,t+1}$ is the realized excess return on stock i in month $t+1$, $\beta^{MPU}_{i,t}$ represents the MPU beta for stock i in month t, and $\mathbf{X}_{i,t}$ denotes a collection of stock-level control variables such as market beta (β^{MKT}), SIZE, etc. We aim to test the time series averages of the slope coefficients on the MPU beta, i.e., $\bar{\lambda}^{MPU} = T^{-1}\sum_{t=1}^{T}\lambda^{MPU}_t$. The cross-sectional regressions are run at a monthly frequency over T=156 samples spanning from January 2008 to December 2020.

Table 7 summarizes the time series averages of the slope coefficients, as well as the Newey–West adjusted and Shanken-corrected *t*-statistics.[27] Panel A presents the results for one-month-ahead cross-sectional regressions, and Panel B reports the results for the regressions using long horizons of 2, 3 and 6 months. Starting from Panel A, the univariate regression reported in the first column shows that the average slope of the MPU beta is -0.033, with a Newey–West *t*-statistic of -2.048 and a Shanken-corrected *t*-statistic of -2.013, indicating a negative and statistically significant cross-sectional relation between the MPU beta and future stock returns. We find that the results obtained via the Shanken correction are similar to those obtained via the Newey–West adjustment, with the former appropriately reducing the *t* values. To determine the economic significance of this average slope coefficient, we use the average values of the MPU betas in the decile portfolios. Table 4 shows that the difference in β^{MPU} values between average stocks in the first and tenth deciles is 9.974 [= 5.350 - (-4.624)]. The average slope coefficient of -0.033 on β^{MPU} in Panel A of Table 7 represents an economically significant decrease of 0.329% per month, [-0.033 × 9.974 = -0.329%], or 3.950% per annum, in the average stock's expected return for moving from the first to the tenth decile of MPU beta.

As presented in column 2, the average slope of the MPU beta is negative and highly significant when controlling for stock-level characteristics, whereas the average coefficients on market beta, EP, I/A and NOA are statistically insignificant, the average slope of SIZE is highly significant and negative, and that of ROE is positive and weakly significant. The third column reflects the simultaneous control of all stock-level variables, whereas the average slope of the MPU beta remains negative (-0.036) and statistically significant (Newey–West *t*-statistic = -2.107). In the last three columns of Table 7, we control for industry effects. For each month, we assign each stock to one of the six major industries and replicate our cross-sectional regressions. Generally, controlling for potential industry effects is shown to have little influence on the value and significance of the cross-sectional relationship between the MPU beta and future excess returns, where the average slopes from the monthly regressions of realized returns on the MPU beta remain negative, in the range of -0.031 and -0.034, and statistically significant, with Newey–West *t*-statistics ranging from -2.078 to -2.176. Overall, these results verify that the MPU beta earns a negative and significant risk premium in all specifications, even when controlling for stock-level characteristics and industry effects.

[27] Since the MPU beta is estimated from the monthly rolling regressions of excess stock returns on the changes in MPU, a generated regressor problem, which is also known as an error-in-variable (EIV) problem, arises in the FMB regression. Shanken (1992) proposed a widely used correction method. We apply this method to adjust the estimates of standard errors for independent variables, whereas only the MPU beta (βMPU) and the market beta (βMKT) are subject to the EIV problem in our specification. We appreciate the referees' suggestion to check the robustness of the generated regressor issue.

Table 7
Fama–MacBeth cross-sectional regressions for monetary policy uncertainty beta

Panel A: One-month-ahead predictability of MPU beta						
	Without controlling for industry effects			With controlling for industry effects		
	(1)	(2)	(3)	(4)	(5)	(6)
β^{MPU}	-0.033	-0.034	-0.036	-0.031	-0.033	-0.034
	(-2.048)	(-2.116)	(-2.107)	(-2.176)	(-2.132)	(-2.078)
	[-2.013]	[-2.080]	[-2.074]	[-2.129]	[-2.093]	[-2.044]
β^{MKT}		0.025	0.065		-0.069	0.013
		(0.074)	(0.201)		(-0.216)	(0.042)
		[0.074]	[0.198]		[-0.213]	[0.041]
SIZE		-0.563	-0.465		-0.582	-0.498
		(-3.554)	(-3.149)		(-3.552)	(-3.267)
		[-3.547]	[-3.142]		[-3.546]	[-3.261]
EP		3.109	1.443		2.299	0.985
		(0.571)	(0.231)		(0.423)	(0.162)
		[0.570]	[0.230]		[0.422]	[0.161]
ROE		3.157	2.769		3.373	2.891
		(1.537)	(1.217)		(1.685)	(1.350)
		[1.534]	[1.214]		[1.682]	[1.348]
I/A		-0.052	-0.004		-0.047	-0.007
		(-1.022)	(-0.054)		(-1.058)	(-0.115)
		[-1.020]	[-0.054]		[-1.056]	[-0.115]
NOA		-0.237	-0.358		-0.135	-0.249
		(-0.864)	(-1.326)		(-0.531)	(-0.989)
		[-0.862]	[-1.323]		[-0.530]	[-0.987]
REV			0.004			0.005
			(0.544)			(0.648)
			[0.543]			[0.647]
COSKEW			-6.458			-7.412
			(-2.194)			(-2.521)
			[-2.189]			[-2.516]
TOVER			-0.279			-0.297
			(-5.556)			(-6.062)
			[-5.545]			[-6.051]
ILLIQ			0.170			0.154
			(2.772)			(2.552)
			[2.766]			[2.548]
Intercept	0.900	9.397	7.828	0.686	9.397	8.007
	(1.141)	(3.301)	(2.939)	(0.880)	(3.165)	(2.876)
	[1.139]	[3.295]	[2.933]	[0.878]	[3.160]	[2.871]
Obs.	187,472	126,920	122,701	187,472	126,920	122,701
Adj. R^2 (%)	0.202	5.225	7.926	2.194	6.912	9.473
Panel B: Long-term predictability of MPU beta						
	Without controlling for industry effects			With controlling for industry effects		
	$h = 2$	$h = 3$	$h = 6$	$h = 2$	$h = 3$	$h = 6$
β^{MPU}	-0.032	-0.041	-0.036	-0.029	-0.040	-0.034
	(-1.727)	(-2.075)	(-1.928)	(-1.752)	(-2.123)	(-1.867)
	[-1.703]	[-2.049]	[-1.902]	[-1.725]	[-2.094]	[-1.840]
Obs.	116,576	114,143	108,052	116,576	114,143	108,052
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2 (%)	8.731	8.265	7.093	10.653	10.141	8.940

Note: This table reports the time series averages of the slope coefficients from Equation (6) by regressing the one-month-ahead excess stock returns on the monetary policy uncertainty beta (β^{MPU}) and a collection of lagged stock-level characteristics as control variables via the Fama and MacBeth (1973) methodology. The entries in columns 1 to 3 under "Without controlling for industry effects" present the results without controlling for industry effects, and the entries in columns 4 to 6 under "With controlling for industry effects" present the same set of regression results controlling for industry effects. For each stock, the MPU beta is estimated via the 60-month rolling regression of Equation (5). The stock-level characteristics include the market beta (β^{MKT}), log of market capitalization (SIZE), earnings-price ratio (EP), operating profitability (ROE), investment (I/A), net operating assets (NOA), short-term reversal (REV), coskewness (COSKEW), turnover (TOVER) and illiquidity (ILLIQ). The sample spans from December 2007 to December 2020. Newey–West adjusted and Shanken-corrected *t*-statistics are given in parentheses (using four lags) and square brackets, respectively.

We also investigate the persistence of the negative relation between the MPU beta and future stock returns by conducting long-term, cross-sectional regressions with horizons h of 2, 3 and 6 months. Panel B in Table 7 presents the corresponding results from controlling for all stock-level characteristics, where the dependent variable is the h-month-ahead monthly realized returns. Column 1 shows that the average slope coefficient on MPU beta remains negative (-0.032) and statistically significant (t-stat. = -1.727). As the forecast horizon h increases, the negative relation between the MPU beta and future monthly returns remains statistically significant. These results document that the MPU beta is a valuable and robust predictor of future stock returns, even in the long run. In summary, the Fama–MacBeth regressions provide stock-level evidence that MPU commands a negative and significant risk premium, which is consistent with the portfolio-level results.

3.2.5. *Time series variation in the monetary policy uncertainty premium*

As discussed above, the Chinese economy has undergone significant structural changes over the past three decades, as has the monetary policy. For example, the PBC has primarily relied on quantity-based instruments, using price-based instruments less intensively. Since

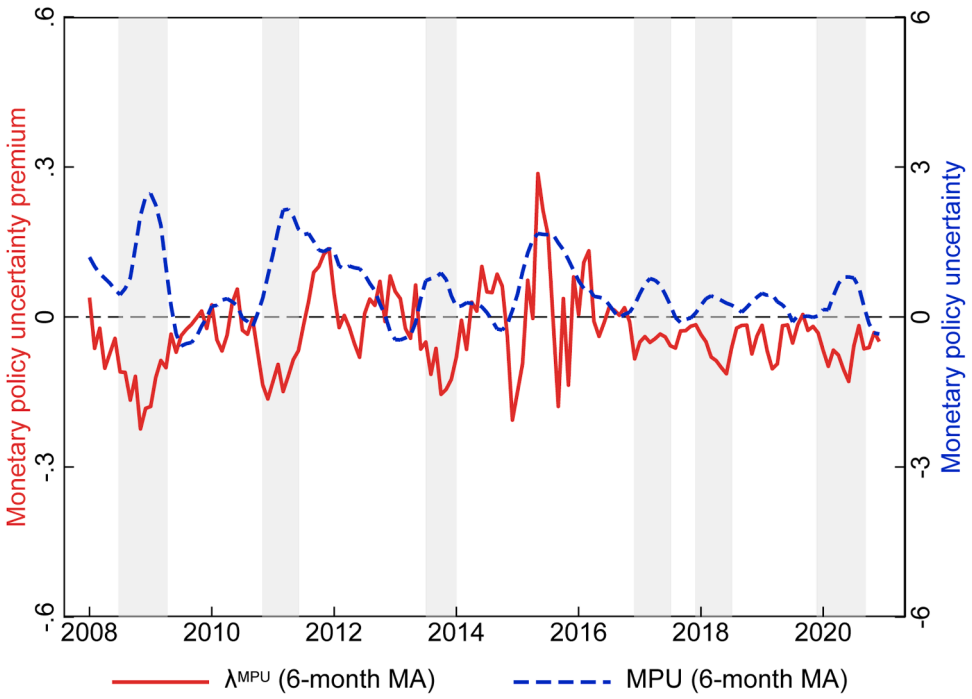


Fig. 3. Monetary policy uncertainty premium and monetary policy uncertainty.
Note: This figure shows the monthly series of monetary policy uncertainty premiums and monetary policy uncertainty. The red solid line depicts the MPU premium (λ^{MPU}) series smoothed by its 6-month moving average (MA) value, and the blue dashed line depicts the aggregate MPU series smoothed by its 6-month moving average value. The shaded areas highlight periods during which there is a substantial gap between these two series. The sample spans from January 2008 to December 2020.

2013, the PBC has made significant improvements in building a modern central banking system, e.g., by guiding market expectations with regular public communications and experimenting with new monetary policy tools such as the SLF and MLF. In line with the fact that there are structural changes in China's monetary policy practices, we examine whether the premium in response to MPU is state dependent and changes over time.

Specifically, we test whether the MPU premium is higher during periods of high MPU. The MPU premium (λ^{MPU}) is computed as the slope coefficient on the MPU beta in the Fama–MacBeth regression and is estimated by controlling for all stock-level characteristics and industry effects in each month (column 6 in Panel A of Table 7).[28] Since the original series of the aggregate MPU measure and its premium both fluctuate widely, we smooth them with their 6-month moving average values and plot the time series in Fig. 3. We make several observations. First, the MPU premium indeed exhibits high volatility over time. Second, the MPU premium becomes negative and has a large absolute magnitude when the MPU series increases. The full-sample correlation of these two series is -0.130. Noteworthy occurrences (highlighted by shaded areas) include the implementation of a stimulative monetary policy at the end of 2008, followed by its subsequent withdrawal during 2010–2011, a rapid economic slowdown and the entry of "new normal" periods after 2014, the transition towards contractionary policy in 2017–2018, and the economic downwards pressure facing the COVID-19 pandemic in 2020. A notable feature of those periods is that MPU was obviously high, accompanied by a substantial decline in the MPU premium.

To investigate economic significance, we first determine high vs. low MPU periods via the median MPU index, and we find that the average slope coefficient of the MPU beta is -0.003 (*t*-stat. = -0.142) during low MPU periods (MPU ≤ Median), whereas the average slope coefficient is much greater, at -0.051 (*t*-stat. = -2.148) during high MPU periods (MPU > Median). Similar results are obtained when an alternative classification method is applied: we define high (low) MPU months as those when the MPU index in the current month exceeds (falls below) its 6-month moving average (MA) value, which is calculated on the basis of the preceding 6 months on a rolling basis each month.[29] In this context, we find that the average slope coefficient on the MPU beta is -0.015 (*t*-stat. = -0.707) during low MPU periods (MPU ≤ 6-month MA), whereas the average coefficient is -0.058 (*t*-stat. = -2.351) during high MPU periods (MPU > 6-month MA). Overall, these results indicate that the MPU premium is greater during high MPU periods than during other periods and validate the nonlinear time-varying nature of the MPU premium.

3.2.6. *Economic determinants of the monetary policy uncertainty beta*

In this subsection, we investigate the cross-industry and stock characteristics of the MPU beta. First, Fig. 4 illustrates the industry averages of the MPU beta over time.[30] The industry averages for the finance and real estate sectors exhibit significant time variability, whereas those of the other three sectors show minimal fluctuations over time. For instance, the financial and real estate industries experienced pronounced changes in the MPU beta during periods of monetary policy uncertainty, such as the global financial crisis in 2008 and the real estate boom since 2015. During the global financial crisis, the financial

[28] Importantly, this estimation procedure is the most rigorous, as it accounts for a comprehensive range of other factors. Our results are robust if we use different estimates of the MPU premium.

[29] The use of detrended series is also consistent with the use of innovations in asset-pricing tests.

[30] We appreciate the referees' suggestion to discuss the economic determinants of MPU beta. Firms are classified into six industries, including manufacturing, business, utilities, finance, real estate, and others. Due to the diverse industry components included in the "others" category (such as agriculture, construction, and other enterprises not classified into the above five industries), we do not report the results for this category.

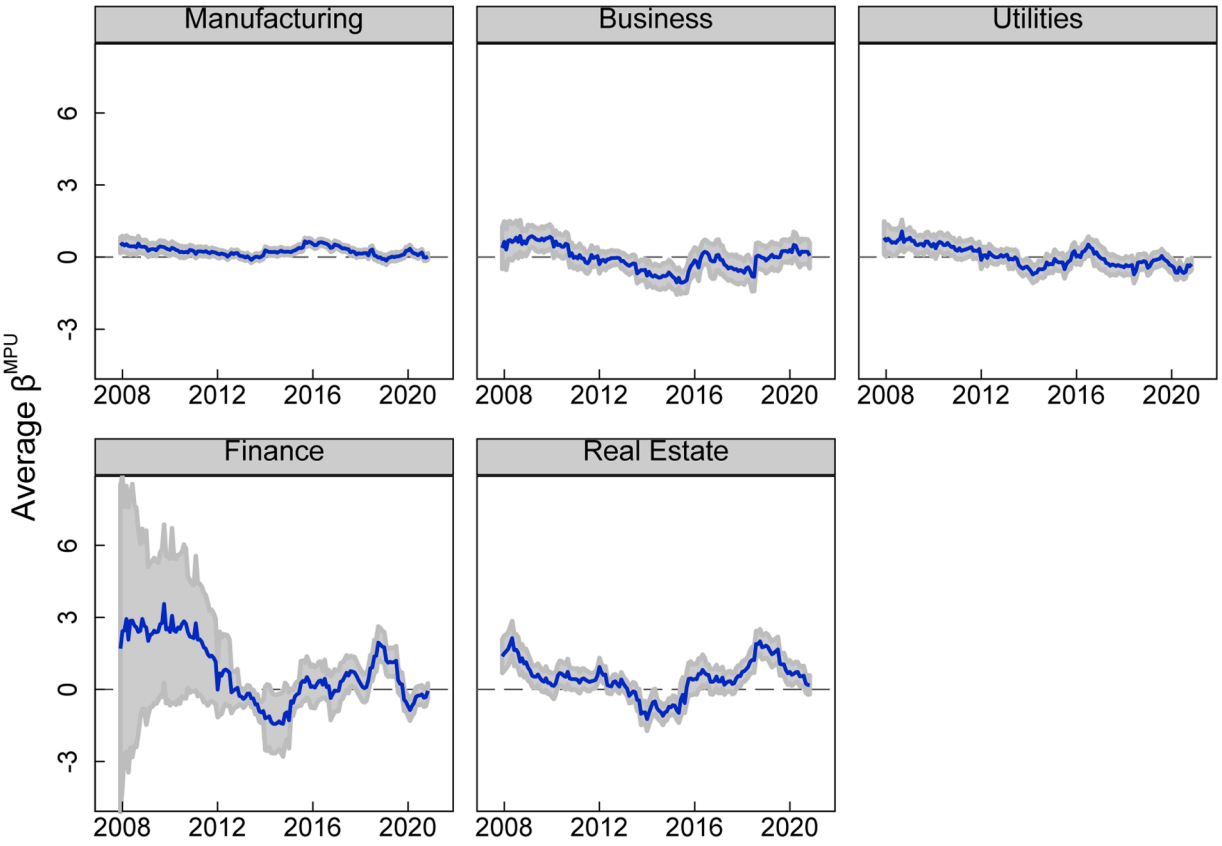


Fig. 4. Industry averages of the monetary policy uncertainty beta over time.
Note: This figure presents the time series of the average MPU beta (β^{MPU}) for five different industries, namely, manufacturing, business, utilities, finance, and real estate. Each panel represents the average beta for an industry over time. The grey areas represent the 90% confidence intervals. The sample spans from December 2007 to December 2020.

industry tended to have higher MPU betas, indicating that investors preferred holding these assets because of their better performance with expansionary policies. This suggests that the exposures to MPU of firms in finance and real estate are highly sensitive to monetary policy and fluctuated since the performance of firms in these two industries is strongly influenced by financial liquidity. In contrast, firms in manufacturing, business, and utilities exhibit relatively less volatile MPU betas because these industries are less sensitive to monetary policy changes.[31]

Second, we examine the average characteristics of stocks with low versus high MPU betas through Fama and MacBeth (1973) cross-sectional regressions:

$$\beta_{i,t}^{MPU} = \varphi_{0,t} + \mathbf{X}'_{i,t}\cdot\boldsymbol{\varphi}_{1,t} + \varepsilon_{i,t}, \tag{7}$$

where the dependent variable $\beta_{i,t}^{MPU}$ is the MPU beta of stock i in month t, and the independent variables $\mathbf{X}_{i,t}$ denote a collection of stock-level controls, including the market beta (β^{MKT}), SIZE, etc. The cross-sectional regressions are run at a monthly frequency from December 2007 to December 2020.

Table 8 presents the time series averages of the slope coefficients from the regressions of the MPU beta on the stock-level characteristics. The results show that the stocks with a high MPU beta are small and growth firms with better profitability, which are likely to have greater sensitivity to monetary policy changes. Column 1 shows that the average slope coefficient on β^{MKT} is positive and significant, which is consistent with Frazzini and Pedersen (2014) that stocks with high β^{MKT} have low future returns. Column 2 reports the average slope coefficients for stock-level accounting characteristics. The coefficient of EP is significantly negative; i.e., stocks with low MPU betas are value stocks. ROE and I/A are both significantly positive, indicating that stocks with high MPU betas have high profitability and growth of book assets (Fama and French, 2015). In column 3, we present the results including all the variables simultaneously. Stocks with a high MPU beta have high coskewness, lower turnover, and greater liquidity, which is consistent with the finding that these assets generate lower future returns and alphas (Harvey and Siddique, 2000; Amihud, 2002; Dey, 2005). Columns 4 to 6 present the robustness of our results controlling for industry effects. Overall, these results align with findings in the U.S. market. For example, Bali et al. (2017) also find that stocks with high exposure to uncertainty are more likely to be growth and liquid, along with high coskewness and strong investment characteristics.

3.2.7. *A risk factor based on the monetary policy uncertainty beta*

Finally, we construct a high-minus-low portfolio based on the MPU beta as an additional risk factor and examine its price of risk, as well as its relation to the extant risk factors.[32] Specifically, we apply the factor-forming technique pioneered by Fama and French (1993), which was also used by Bali et al. (2017) in a context similar to ours. At the end of each month, we sort all stocks into two groups based on the market

[31] However, firms within the finance sector tend to have more dispersed MPU betas especially before 2012, resulting in the insignificance of the industry average. This can be interpreted as a reflection of the high level of heterogeneity. This diversity in volatile betas also indicates that, to some extent, the finance sector features high sensitivity to MPU.

[32] We appreciate the referee's insightful comments on this issue.

Table 8
Monetary policy uncertainty beta and firm characteristics

	Without controlling for industry effects			With controlling for industry effects		
	(1)	(2)	(3)	(4)	(5)	(6)
β^{MKT}	0.688	0.661	0.578	0.645	0.656	0.575
	(1.885)	(1.867)	(1.496)	(1.759)	(1.814)	(1.459)
SIZE		-0.020	-0.243		-0.020	-0.247
		(-0.602)	(-6.369)		(-0.554)	(-6.632)
EP		-3.721	-3.378		-4.426	-4.038
		(-2.565)	(-2.629)		(-3.149)	(-2.996)
ROE		1.651	1.370		1.726	1.456
		(2.011)	(1.819)		(2.275)	(1.999)
I/A		0.110	0.107		0.107	0.104
		(2.124)	(2.113)		(2.137)	(2.106)
NOA		-0.242	-0.222		-0.311	-0.264
		(-1.132)	(-1.091)		(-1.626)	(-1.436)
REV			-0.000			-0.001
			(-0.162)			(-0.199)
COSKEW			14.064			14.218
			(3.922)			(3.962)
TOVER			-0.100			-0.100
			(-5.333)			(-5.083)
ILLIQ			-0.145			-0.146
			(-4.219)			(-4.423)
Intercept	-0.366	0.087	4.273	-0.406	0.030	4.295
	(-0.995)	(0.182)	(6.655)	(-1.355)	(0.064)	(7.449)
Obs.	122,701	122,701	122,701	122,701	122,701	122,701
Adj. R^2 (%)	4.167	6.075	11.867	5.324	6.913	12.612

Note: This table reports the time series averages of the slope coefficients from Equation (7) by regressing the monetary policy uncertainty beta (β^{MPU}) on a collection of stock-level characteristics as control variables via the Fama and MacBeth (1973) methodology. The entries in columns 1 to 3 under "Without controlling for industry effects" present the results without controlling for industry effects, and the entries in columns 4 to 6 under "With controlling for industry effects" present the same set of regression results controlling for industry effects. For each stock, the MPU beta is estimated via 60-month rolling regression. The stock-level characteristics include the market beta (β^{MKT}), log of market capitalization (SIZE), earnings-price ratio (EP), operating profitability (ROE), investment (I/A), net operating assets (NOA), short-term reversal (REV), coskewness (COSKEW), turnover (TOVER), and illiquidity (ILLIQ). The sample spans from December 2007 to December 2020. Newey–West adjusted *t*-statistics are given in parentheses using four lags.

capitalization (SIZE), with the breakpoint dividing the two groups being the median market capitalization of stocks traded on the A-share markets. We independently sort all stocks into three groups based on an ascending sort of the MPU beta (β^{MPU}) using the 30th and 70th percentile values. The intersections of the two size groups and the three β^{MPU} groups generate six portfolios. The MPU beta factor is taken to be the average return of the two value-weighted high-β^{MPU} portfolios minus the average return of the two value-weighted low-β^{MPU} portfolios. This design ensures that the MPU beta factor captures returns associated with the uncertainty premium while remaining neutral to market capitalization.

Table 9 presents the price of the benchmark MPU beta factor (F_{MPU}) in the first row, with the average return spread in column 1, followed by the risk-adjusted alpha spread from the CAPM and CH4 factor models in the subsequent columns. We observe that F_{MPU} is economically large and statistically significant, with an average monthly return of -0.318% and a Newey–West adjusted *t*-statistic of -2.089. The alphas remain negative, in the range of -0.317% and -0.361% per month, and statistically significant, with *t*-statistics ranging from -2.065 to -2.340. This result provides preliminary evidence that the MPU beta factor is not explained by well-known factors. We further consider different sorting breakpoints by using the quintile or decile values in the second and third rows of Table 9, resulting in 10 (2 × 5) or 20 (2 × 10) portfolios, respectively. Our results are robust with a larger return and spreads when we use decile values in the portfolio construction.

Table 9
Average monthly returns and alphas of the monetary policy uncertainty beta factors

	Avg. return	α_{CAPM}	α_{CH4}
F_{MPU} (2 × 3, Benchmark)	-0.318	-0.317	-0.361
	(-2.089)	(-2.065)	(-2.340)
F_{MPU} (2 × 5)	-0.378	-0.375	-0.375
	(-2.246)	(-2.224)	(-2.229)
F_{MPU} (2 × 10)	-0.636	-0.631	-0.865
	(-2.667)	(-2.630)	(-3.666)

Note: This table reports the average monthly returns and alphas of the monetary policy uncertainty (MPU) beta factors. Each month, value-weighted portfolios are formed by sorting the individual stocks based on their market capitalization in the most recent June and their MPU betas (β^{MPU}) from the previous month. We first sort all the stocks into two groups with the median market capitalization as the breakpoint. In the first row, we independently sort all the stocks into three groups based on an ascending sort of the MPU beta using the 30th and 70th percentile values. In the second and third rows, we independently sort all stocks into five or ten groups based on the quintile or decile values of the β^{MPU}. The MPU beta factor is taken to be the average return of the two value-weighted high-β^{MPU} portfolios minus the average return of the two value-weighted low-β^{MPU} portfolios. The average returns and alphas are defined in monthly percentage terms, where α_{CAPM} denotes the alpha of the CAPM model containing the market (MKT) factor, and α_{CH4} denotes the alpha of the CH4 model containing the market (MKT), size (SMB), value (VMG), and turnover (PMO) factors. The sample spans from December 2007 to December 2020. Newey–West adjusted *t*-statistics are given in parentheses using four lags.

Table 10
Long-short portfolios based on the MPU beta strategy and risk factors

	Avg. Return (%)	Std. Dev. (%)	SR	Drawdown (%)
MPU	0.318	2.333	0.473	7.200
MKT	0.119	7.622	0.054	14.851
SMB	0.607	4.492	0.468	16.729
VMG	0.841	3.569	0.816	13.924
PMO	0.684	3.660	0.647	11.823

Note: This table reports the high-minus-low portfolio performance of the monetary policy uncertainty beta (β^{MPU}), market (MKT), size (SMB), value (VMG), and turnover (PMO) factors. The MPU beta portfolios are formed by going long stocks in the top 30th percentile of high-performing characteristics and short stocks within the bottom 30th percentile of high-performing characteristics. All portfolios are value weighted and orthogonalized to market capitalization with independent sorting. The factor portfolios are the returns of each factor. "Avg. Return" and "Std. Dev." denote the mean and standard deviation of the monthly long-short portfolio return, respectively. "SR" denotes the annualized Sharpe ratio. "Drawdown" denotes the maximum drawdown. The sample spans from December 2007 to December 2020.

In addition, we construct a value-weighted zero-cost factor-timing strategy based on the MPU beta, finding that it has relatively low variation and a relatively high annualized Sharpe ratio. We consider long stocks in the top 30th percentile of high-performing characteristics (low β^{MPU}) and short stocks within the bottom 30th percentile of low-performing characteristics (high β^{MPU}). The portfolios are neutral to market capitalization with independent sorting, which is similar to the risk factor construction procedure above. In Table 10, we present the monthly average return, standard deviation, annualized Sharpe ratio, and maximum drawdown of the MPU beta strategy. We also include the Chinese four-factor (CH4) portfolios for comparison. Although the average excess return of the long-short portfolio based on the MPU beta is not very large, it has relatively low variation and a higher annualized Sharpe ratio. More importantly, the MPU beta strategy results in the lowest drawdown compared with the CH4 factors. In summary, these analysis provide some evidence demonstrating the value of monetary policy uncertainty risk in the Chinese equity market.

Table 11
Cross-sectional predictive power of the monetary policy uncertainty betas: Horse race

Panel A: Univariate portfolio sorts

	High−Low return spreads		
MPU beta	RET−RF	α_{CAPM}	α_{CH4}
β^{MPU}	-0.882	-0.876	-1.132
	(-3.108)	(-3.095)	(-4.453)
$\beta^{MPU\text{-}MPC}$	-0.270	-0.274	-0.375
	(-0.941)	(-0.970)	(-1.374)
$\beta^{MPU\text{-}MPO}$	-0.474	-0.474	-0.493
	(-1.778)	(-1.778)	(-1.920)
$\beta^{MPU\text{-}NEWS}$	-0.722	-0.708	-0.917
	(-2.682)	(-2.837)	(-4.089)

Panel B: Fama–MacBeth regressions

	Without controlling for industry effects				With controlling for industry effects			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
β^{MPU}	-0.036				-0.034			
	(-2.107) [-2.074]				(-2.078) [-2.044]			
$\beta^{MPU\text{-}MPC}$		-0.001				-0.001		
		(-0.074)				(-0.044)		
		[-0.074]				[-0.044]		
$\beta^{MPU\text{-}MPO}$			-0.027				-0.023	
			(-1.735)				(-1.535)	
			[-1.170]				[-1.130]	
$\beta^{MPU\text{-}NEWS}$				-0.028				-0.027
				(-1.541)				(-1.544)
				[-1.529]				[-1.533]
Obs.	122,701	122,701	122,701	122,701	122,701	122,701	122,701	122,701
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2 (%)	7.926	7.848	7.922	7.924	9.473	9.395	9.474	9.466

Note: This table reports the performance of aggregate and individual monetary policy uncertainty betas in predicting the cross-sectional dispersion in future stock returns. Panel A presents the univariate portfolio results. For each month, the value-weighted decile portfolios are formed by sorting the individual stocks based on their MPU betas during the previous month, where decile 1 (*Low*) contains the stocks with the lowest MPU beta and decile 10 (*High*) contains the stocks with the highest MPU beta. Each column reports the differences in average excess returns (RET−RF) and alphas, where α_{CAPM} denotes the alpha of the CAPM model containing the market (MKT) factor and α_{CH4} denotes the alpha of the CH4 model containing the market (MKT), size (SMB), value (VMG) and turnover (PMO) factors. The average excess returns and alphas are defined in monthly percentage terms. Panel B presents the time series average of the slope coefficients for the monetary policy uncertainty beta by regressing the one-month-ahead excess stock returns on the aggregate or individual MPU beta and a collection of lagged stock-level characteristics as control variables via the Fama and MacBeth (1973) methodology. The entries in columns 1 to 4 under "Without controlling for industry effects" present results without controlling for industry effects, and the entries in columns 5 to 8 under "With controlling for industry effects" present the same set of regression results controlling for industry effects. The MPU betas are obtained from the specification of Equation (5), where β^{MPU}, $\beta^{MPU\text{-}MPC}$, $\beta^{MPU\text{-}MPO}$ and $\beta^{MPU\text{-}NEWS}$ correspond to the aggregate, communication-, operation- and news-based MPU betas, respectively. The stock-level characteristics include the market beta (β^{MKT}), log of market capitalization (SIZE), earnings-price ratio (EP), operating profitability (ROE), investment (I/A), net operating assets (NOA), short-term reversal (REV), coskewness (COSKEW), turnover (TOVER) and illiquidity (ILLIQ). The sample spans from December 2007 to December 2020. Newey–West adjusted and Shanken-corrected *t*-statistics are given in parentheses (using four lags) and square brackets, respectively.

4. Additional results and robustness tests

We have thus far provided portfolio- and stock-level evidence that the aggregate measure of monetary policy uncertainty is priced in the cross-section of the Chinese stock market. In this section, we investigate the role of information aggregation and check for confounding effects. In addition, we perform a series of robustness tests.

4.1. The value of information aggregation

To investigate the role of information aggregation, we conjecture that the aggregate MPU measure outperforms its three individual measures in cross-sectional asset-pricing tests. First, we separately estimate a stock's exposure to each individual MPU measure, namely, the individual MPU beta, using the same specification of Equation (5). For example, for each stock and month in our sample, we estimate the news-based MPU beta ($\beta^{MPU\text{-}NEWS}$) from the 60-month rolling regressions of excess stock returns on the change in the news-based MPU index (ΔMPU-NEWS) while controlling for the four Chinese factors. Similarly, the estimates of these individual MPU betas start in December 2007 and end in December 2020. Second, we directly compare the performance of different MPU betas in predicting cross-sectional dispersion through a horse-race approach. We start by conducting portfolio-level analysis. For each MPU beta, we form decile portfolios by sorting individual stocks based on their MPU betas during the previous month and compute average return (or alpha) spreads between decile 10 (*High*) and decile 1 (*Low*).

Panel A of Table 11 presents the value-weighted portfolio results using the three individual MPU beta measures. For the sake of comparison, the first row shows the results of the aggregate MPU beta (β^{MPU}), noting that this is merely a replication of the last row in Table 4. The remaining rows report the results for the communication-based MPU beta ($\beta^{MPU\text{-}MPC}$), operation-based MPU beta ($\beta^{MPU\text{-}MPO}$) and news-based MPU beta ($\beta^{MPU\text{-}NEWS}$) separately. We first analyze the portfolio-level results for each individual MPU beta. As shown in the second row, for the communication-based MPU beta, the difference in average excess returns between *High* and *Low* portfolios is -0.270% per month. However, this result is statistically insignificant, with a Newey–West adjusted *t*-statistic of -0.941. In addition, the differences in alphas are also negative and insignificant: the CAPM alpha spread is -0.274% per month (*t*-stat. = -0.970), and the CH4 alpha spread is -0.375% per month (*t*-stat. = -1.374).

Regarding the operation-based MPU beta (row 3), we find that the spreads of the average return and CAPM alpha are both negative and significant at the 10% level. After controlling for the well-known market, size, value and turnover factors, the alpha spreads tend to be larger;

e.g., the CH4 alpha spread is -0.493% per month and is more significant (*t*-stat. = -1.920). Compared with the aggregate measure, the differences in the return spread of the operation-based component are much smaller. For example, the CH4 alpha of the aggregate measure (-1.132%) is twice that of the operation-based measure (-0.493%).

The last row shows that the average return and alpha spreads based on the news-based MPU beta are negative and highly significant. For instance, the difference in CH4 alphas between *High* and *Low* portfolios is -0.917% per month (*t*-stat. = -4.089). In summary, these results indicate that only the news-based MPU beta earns a robust long-short portfolio premium. Next, when comparing the average return and alpha spreads of individual MPU betas with the aggregate MPU beta, we find that the aggregate MPU beta outperforms the individual betas across all cases by commanding higher return spreads and greater statistical significance. For instance, although the average excess return and alpha spreads between the high-β^{MPU} and low-β^{MPU} exposure stocks are similar for both the aggregate and news-based measures, the annualized differences range from 1.920% to 2.580%, which are substantial from an economic perspective.[33] In Appendix B.1, we perform conditional bivariate portfolio sorts on aggregate and individual MPU betas. We find that after controlling for aggregate MPU measure, individual MPU does not exhibit a significant risk premium. When we reverse the order of sorting betas, the aggregate MPU measure remains a significant predictor for subsequent stock returns. These results provide us with portfolio-level evidence to demonstrate the role of information aggregation in the cross-sectional asset-pricing implications of MPU.

In addition to the portfolio-level analysis, we conduct stock-level Fama–MacBeth cross-sectional regressions to provide more comprehensive evidence. For each stock and each month, we replace the aggregate MPU beta with the individual MPU beta in Equation (6). Panel B of Table 11 presents the time series averages of the slope coefficients from the regressions of one-month-ahead stock returns on the MPU beta controlling for all stock-level characteristics (market beta, SIZE, EP, etc.). For the sake of comparison, we also include the average slope coefficient on the aggregate MPU beta in the first column, which aligns with the results in column 3 in Panel A of Table 7. Starting with the average slope coefficient of the communication-based MPU beta (column 2), we find that it is relatively small and statistically insignificant after controlling for stock-level characteristics. With respect to the operation-based MPU beta, column 3 shows that its average slope is negatively estimated with weak statistical significance, which is consistent with the univariate portfolio results. Column 4 presents the average slope of the news-based MPU beta, which is also negative but insignificant. As presented in the last four columns, the Fama–MacBeth regression results are analogous if we control for the industry effect. Overall, these results indicate that individual MPU betas are not robust predictors in predicting the cross-section of stock returns. In contrast to these individual measures, the aggregate MPU measure earns a significant risk premium even when controlling for well-known cross-sectional return predictors. Therefore, these stock-level results provide us with stronger evidence demonstrating the importance of aggregating information.[34]

4.2. Alternative explanations and confounding effects

We now examine whether other measures of uncertainty, such as overall economic uncertainty (Bali et al., 2017), economic policy uncertainty (Brogaard and Detzel, 2015), and first-order monetary policy shocks (Ozdagli and Velikov, 2020), affect the asset-pricing implications of MPU. First, a reasonable concern over the significant MPU premium is that the MPU is not a separate source of uncertainty risk, which could also confound the effects of economic uncertainty (EU) or economic policy uncertainty (EPU). To address this concern, we estimate the innovation in the aggregate MPU measure by orthogonalizing MPU separately with respect to EU and EPU:

$$\mathrm{MPU}_t = \varphi_0 + \varphi_1 \cdot \mathrm{MPU}_{t-1} + \varphi_2 \cdot \mathrm{UNC}_t + \nu_t, \tag{8}$$

where UNC_t represents the alternative uncertainty index and the estimated residual (ν_t) of Equation (8) is the orthogonalized MPU measure that isolates the effects of other uncertainty variables. We reestimate stock exposure to the orthogonalized MPU index, which we refer to as the orthogonalized MPU beta ($\beta^{MPU\perp}$), and replicate the Fama–MacBeth cross-sectional regressions controlling for stock-level characteristics and industry effects.

We present the results in Table 12. In column 1, the orthogonalized MPU beta isolates the confounding effect of EU, and its average slope coefficient remains negative and statistically significant. In the next two columns, we orthogonalize the impact of EPU. In the case of Huang and Luk (2020)'s index, the average slope on the orthogonalized MPU beta is negative with weak significance. However, for Baker et al. (2016)'s index, the average slope remains negative and highly significant. In summary, the asset-pricing implications of MPU are not influenced by alternative uncertainty variables, confirming that MPU is priced in the Chinese stock market.

Second, the PBC adopts aggressively expansionary or contractionary monetary policy during our sample periods (e.g., stimulus policy following the 2008 global financial crisis), leading to the peak of

Table 12
Fama–MacBeth regressions for the orthogonalized monetary policy uncertainty beta

Orthogonal to	EU	EPU		MPS	
		HL	BBD	McCallum Rule	Taylor Rule
$\beta^{MPU\perp}$	-0.037	-0.032	-0.039	-0.038	-0.038
	(-2.266)	(-1.828)	(-2.344)	(-2.292)	(-2.360)
	[-2.221]	[-1.791]	[-2.301]	[-2.251]	[-2.317]
Obs.	122,701	122,701	122,701	122,701	122,701
Controls	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes
Adj. R^2 (%)	9.468	9.470	9.467	9.473	9.454

Note: This table reports the results obtained by regressing the one-month-ahead excess stock returns on the orthogonalized monetary policy uncertainty beta and a collection of lagged stock-level characteristics as control variables via the Fama and MacBeth (1973) methodology. The aggregate MPU index is orthogonalized to alternative uncertainty variables or monetary policy surprises via the specification of Equation (8), and the orthogonalized MPU beta is then obtained via Equation (5) by replacing the innovation of the MPU with the orthogonalized MPU index. Column 1 under "EU" shows the result that orthogonalizes the effect of economic uncertainty (EU). The entries in columns 2 and 3 under "EPU" present the results that orthogonalize the effect of economic policy uncertainty (EPU): "HL" denotes Huang and Luk (2020)'s index, and "BBD" denotes Baker et al. (2016)'s index. The entries in columns 4 and 5 under "MPS" present results that orthogonalize the effect of the monetary policy surprise (MPS): "McCallum Rule" denotes the quantity rule-based index, and "Taylor Rule" denotes the interest rate rule-based index. The stock-level characteristics include the market beta (β^{MKT}), log of market capitalization (SIZE), earnings-price ratio (EP), operating profitability (ROE), investment (I/A), net operating assets (NOA), short-term reversal (REV), coskewness (COSKEW), turnover (TOVER) and illiquidity (ILLIQ). The sample spans from December 2007 to December 2020. Newey–West adjusted and Shanken-corrected *t*-statistics are given in parentheses (using four lags) and square brackets, respectively.

[33] We thank the referee's comment of documenting the annualized difference from an economic perspective.

[34] Appendix B.1 provides further evidence to demonstrate the comparative advantages of the aggregate MPU measure. For instance, we perform a factor-timing strategy to compare the high-minus-low portfolio performance for different MPU measures and conduct a series of orthogonalized beta tests to validate the significant premium for the aggregate MPU measure in the Chinese stock market.

monetary policy surprise (MPS). Since both first- and second-order shocks to monetary policy are essential to asset prices, we attempt to distinguish monetary policy uncertainty from monetary policy surprises. In doing so, we construct a rule-based MPS index following the methodology of Chen et al. (2018). This approach proposes a China-specific quantity-based (or McCallum) monetary policy rule and uses the residual as the monetary policy surprise. In addition, we construct an interest rate rule (Taylor)-based MPS index to align with the increased importance of price-based monetary policy tools during recent periods.[35] Following a similar specification to Equation (8), we orthogonalize the aggregate MPU index with respect to MPS and then perform Fama–MacBeth cross-sectional regressions using the orthogonalized MPU beta. As presented in the last two columns in Table 12, for each rule-based MPS index, the average slope coefficient on the orthogonalized MPU beta is negatively estimated and statistically significant. Hence, our aggregate MPU measure remains a robust return predictor even when controlling for the first-order shock of monetary policy.

4.3. Additional robustness tests

We provide a battery of robustness checks in this subsection.[36] We start by testing whether alternative measures of the MPU beta are priced in the cross-section. As discussed in Equation (5), we estimate the MPU beta based on the first-order difference (innovation) in the aggregate MPU series while simultaneously controlling for the market, size, value and turnover factors (CH4 model). To check the robustness, we generate two alternative measures. The first specification follows Bali et al. (2017), in which we estimate the MPU beta using the level of the aggregate MPU measure:

$$R^{e}_{i,t} = \alpha_{i,t} + \beta^{\mathrm{MPU}}_{i,t}\cdot \mathrm{MPU}_t + \beta^{\mathrm{MKT}}_{i,t}\cdot \mathrm{MKT}_t + \beta^{\mathrm{SMB}}_{i,t}\cdot \mathrm{SMB}_t + \beta^{\mathrm{VMG}}_{i,t}\cdot \mathrm{VMG}_t + \beta^{\mathrm{PMO}}_{i,t}\cdot \mathrm{PMO}_t + \varepsilon_{i,t}. \quad (9)$$

Hence, this version of the MPU beta evaluates an individual stock's exposure to the current level of MPU rather than its change. Column 1 in Table 13 reports the average slope coefficients from the Fama–MacBeth cross-sectional regression on the MPU beta estimated via Equation (9). The MPU premium is still economically and statistically significant, with an average slope of -0.108 and a *t*-statistic of -2.388 when controlling for all stock-level characteristics and industry effects.

In the second specification, we extract the innovation in the aggregate MPU measure by conducting an AR(p) regression in the spirit of Brogaard and Detzel (2015):

$$\mathrm{MPU}_t = \varphi_0 + \varphi_1\cdot \mathrm{MPU}_{t-1} + \epsilon^{\mathrm{MPU}}_t, \quad (10)$$

where the best lag length is selected as $p = 1$ to minimize the Bayesian information criterion (BIC; up to 12 lags). We estimate this regression via ordinary least squares and then use the estimated residual (ϵ^{MPU}) as an alternative measure of innovation in the aggregate MPU measure. Following a similar beta estimation procedure, we compute the MPU beta based on the estimated residual of Equation (10). As presented in column 2, we find that the average slope coefficient on the MPU beta has a magnitude very similar to that reported in column 6 in Panel A of Table 7, which equals -0.039 with Newey–West adjusted *t*-statistics of -2.327. Overall, these results indicate that the use of level or AR(1) residual in estimating MPU beta do not alter our main conclusion.

Next, we examine whether alternative specifications of the factor model used in Equation (5) could alter our primary results. As shown in columns 3 and 4 in Table 13, we estimate the MPU beta based on the 5-factor model (including CH4 factors and momentum factor) and the 7-factor model (including CH4 factors, momentum, profitability, and investment factors), respectively. The average slope coefficient of the MPU beta based on the 5-factor model is negatively estimated (-0.036) and statistically significant (*t*-stat. = -2.135). Although the average coefficient of the MPU beta based on the 7-factor model decreases slightly, it is negative but statistically weak. Hence, alternative measures of the MPU beta remain significant predictors of future stock returns.

Finally, we examine the significance of the MPU premium via an alternative method for information aggregation. Following Huang et al. (2021), an alternative aggregate MPU measure is constructed by equally weighting the three (standardized) individual MPU measures. Intuitively, if each individual measure is unbiased, equal weighting (EW) efficiently reduces idiosyncratic errors. The last column in Table 13 presents the results of the Fama–MacBeth regressions for the equal-weighted MPU index. The average slope coefficient of the MPU beta is also negative and highly significant, suggesting that alternative methods for aggregating individual MPU measures command similar MPU premiums in the Chinese stock market.

Table 13
Robustness checks of the Fama–MacBeth cross-sectional regressions for the monetary policy uncertainty beta

	Innovation measures		Factor models		Weighting methods
	Level	AR(1) residual	5-factor	7-factor	EW
β^{MPU}	-0.108	-0.039	-0.036	-0.025	-0.018
	(-2.388)	(-2.327)	(-2.135)	(-1.757)	(-2.011)
	[-2.277]	[-2.282]	[-2.101]	[-1.721]	[-1.977]
Obs.	122,701	122,701	122,701	122,701	122,701
Controls	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes
Adj. R^2 (%)	9.522	9.467	9.470	9.406	9.470

Note: This table reports various robustness checks by regressing the one-month-ahead excess stock returns on the monetary policy uncertainty beta and a collection of lagged stock-level characteristics as control variables via the Fama and MacBeth (1973) methodology. The MPU betas are obtained from alternative specifications of Equation (5): The entries in columns 1 and 2 under "Innovation measures" present results using the level and estimated AR (1) residual of the MPU series. The entries in columns 3 and 4 under "Factor models" present the results obtained via the 5- and 7-factor models. The entry in column 5 under "Weighting methods" presents the result obtained via equal weighting for aggregation. The stock-level characteristics include the market beta (β^{MKT}), log of market capitalization (SIZE), earnings-price ratio (EP), operating profitability (ROE), investment (I/A), net operating assets (NOA), short-term reversal (REV), coskewness (COSKEW), turnover (TOVER) and illiquidity (ILLIQ). The sample spans from December 2007 to December 2020. Newey–West adjusted and Shanken-corrected *t*-statistics are given in parentheses (using four lags) and square brackets, respectively.

5. Conclusion

In this paper, we propose an aggregate monetary policy uncertainty measure from the perspective of information dissemination, and examine its asset-pricing implications in the cross-section of the Chinese stock market. We show that the aggregate MPU measure is closely associated with economic states, monetary policy stance and yield curve dynamics. In accordance with the ICAPM theory, we estimate stock exposure to the innovation in the aggregate MPU measure, and then thoroughly conduct portfolio- and stock-level analyses. Our findings reveal that the MPU beta significantly predicts cross-sectional dispersion in future stock returns. Furthermore, we delve into the role of information aggregation, and present comprehensive evidence illustrating that the aggregate MPU measure surpasses its individual components in cross-sectional asset pricing tests. Notably, the MPU premium cannot be attributed to alternative uncertainty variables or to first-order monetary policy surprises. Finally, we bolster our findings with a battery of robustness checks.

[35] See Appendix A.4 for more details about the construction of the monetary policy surprise index.
[36] Additional robustness tests are provided in Appendix B.2.

Overall, we demonstrate that the MPU premium in a policy-driven stock market is economically and statistically important. Our work introduces a novel perspective on understanding the information dissemination of monetary policy and extends the literature on asset pricing with uncertainty. In future work, it would be interesting to examine other economic consequences of the aggregate MPU measure, or to investigate the asset-pricing implications of other specific policy uncertainties, such as regulatory policy uncertainty.

CRediT authorship contribution statement

Jianhao Lin: Writing – review & editing, Writing – original draft. **Jiacheng Fan:** Writing – review & editing, Writing – original draft, Data curation. **Yifan Zhang:** Writing – review & editing, Writing – original draft, Data curation.

Declaration of competing interest

None.

Appendix A

Variable definition and construction

Summary statistics of individual MPU measures

Panel A of Table A.1 presents the descriptive statistics of the three individual MPU measures. It is apparent that the scales across MPU measures vary dramatically due to differences in construction methods and data sources. For instance, the mean of the news-based MPU index is 117.939, while the mean of the operation-based MPU index is 0.185. Thus, to make them comparable, we standardize each MPU measure by its full-sample mean and standard deviation in our empirical analyses. Moreover, all three MPU indices are strongly serially correlated, with first-order autocorrelations in the range of 0.827–0.889. Therefore, Panel B presents the pairwise correlations between individual MPU measures. All of the measures are positively correlated but not strongly correlated to some extent. The correlation matrix indicates that different MPU indices may reflect different dimensions or aspects of the aggregate MPU measure.

Table A.1
Summary statistics and pairwise correlations of individual monetary policy uncertainty measures.

Panel A: Summary statistics

	Avg	Std	Max	Min	Skew	Kurt	1st AC
MPU-MPC	0.002	0.001	0.003	0.001	-0.119	2.219	0.879
MPU-MPO	0.185	0.053	0.420	0.112	1.327	4.977	0.889
MPU-NEWS	117.939	62.505	357.999	30.406	0.823	3.393	0.827

Panel B: Pairwise correlations

	MPU-MPC	MPU-MPO	MPU-NEWS
MPU-MPC	1.000		
MPU-MPO	0.173	1.000	
MPU-NEWS	0.240	0.387	1.000

Note: This table reports the summary statistics and pairwise correlations of three individual monetary policy uncertainty measures in Panels A and B, respectively. The communication-based MPU index (MPU-MPC) is obtained by counting uncertainty-related keywords in PBC communication texts. The operation-based MPU index (MPU-MPO) is obtained from the unforecastable components of a broad set of monetary policy variables following the method of Jurado et al. (2015). The news-based MPU index (MPU-NEWS) is obtained from Huang and Luk (2020). The sample spans from January 2003 to December 2020.

Uncertainties based on unforecastable components of a large set of variables

Jurado et al. (2015) develop a factor-based estimate of (macro)economic uncertainty using a rich set of time series that represent broad categories of macroeconomic activities. They first estimate the conditional volatility of the unpredictable component of the future value of each series and then aggregate individual conditional volatilities into a measure of overall economic uncertainty. In our paper, we adopt this method and construct two uncertainty measures: (1) the operation-based monetary policy uncertainty (MPU-MPO) index, which is derived from the unforecastable components of monetary policy variables, and (2) the economic uncertainty (EU) index, which is derived from the unforecastable components of a wide range of macroeconomic variables.

We start by briefly introducing the method of Jurado et al. (2015). Let $\mathbf{X}_t = \left(X_{1,t}, \cdots, X_{N,t}\right)^\top$ represent the set of variables in period t with N. We first establish an approximate factor structure of variable X_{it} that takes the form:

$$X_{i,t} = \Lambda_i^F \mathbf{F}_t + e_{i,t}^X, \tag{A.1}$$

where $\mathbf{F}_t$ is a $r_F \times 1$ vector of latent common factors with a loading vector of Λ_i^F and $e_{i,t}^X$ is the idiosyncratic error. The dynamic structure of $\mathbf{F}_t$ is formed as

$$\mathbf{F}_t = \Phi^F \mathbf{F}_{t-1} + \mathbf{v}_t^F, \tag{A.2}$$

where Φ^F is the autoregressive coefficient matrix and $\mathbf{v}_t^F$ is a $r_F \times 1$ vector of the error term. We drop the subscript that indices the factor for simplicity, and Equation (A.2) imposes a univariate AR(1) process for each element of $\mathbf{F}_t$. Its error is given as $v_t^F = \sigma_t^F \varepsilon_t^F, \varepsilon_t^F \sim \mathcal{N}(0,1)$.

Let $y_{j,t}$ generically denote a series for which we wish to compute uncertainty in and its value in period $h \geq 1$.[37] Assume that we estimate its one-period-ahead forecast using a factor-augmented predicting model:

[37] In general, $y_{j,t}$ is a variable in $\mathbf{X}_t$.

$$y_{j,t+1} = \phi_j^y(L) y_{j,t} + \gamma_j^F(L)\widehat{\mathbf{F}}_t + v_{j,t+1}^y, \tag{A.3}$$

where $\widehat{\mathbf{F}}_t$ is the consistent estimate of $\mathbf{F}_t$ and $\phi_j^y(L)$ and $\gamma_j^F(L)$ are finite-order polynomials in the lag operator L of order p_y and p_F, respectively. The disturbance term is $v_{j,t+1}^y = \sigma_{j,t+1}^y \epsilon_{j,t+1}^y$ with $\epsilon_{j,t+1}^y \sim \mathcal{N}(0,1)$. Then, we use an autoregressive structure to introduce the time-varying stochastic volatility of σ_t^F and $\sigma_{j,t+1}^y$:

$$\log\left(\sigma_t^F\right)^2 = \alpha^F + \beta^F \log\left(\sigma_{t-1}^F\right)^2 + \tau^F \eta_t^F, \quad \eta_t^F \sim \mathcal{N}(0,1), \tag{A.4}$$

$$\log\left(\sigma_{j,t+1}^y\right)^2 = \alpha_j^y + \beta_j^y \log\left(\sigma_{j,t}^y\right)^2 + \tau_j^y \eta_{j,t+1}, \quad \eta_{j,t+1} \sim \mathcal{N}(0,1). \tag{A.5}$$

Following Jurado et al. (2015), we write a more compact representation of the system above using factor augmented vector autoregression (FAVAR). Let $\mathbf{Z}_t \equiv \widehat{\mathbf{F}}_t$ be a $r = r_F$ vector that collects r_F estimated factors and defines $\mathscr{Z}_t \equiv \left(\mathbf{Z}_t^\top, \cdots, \mathbf{Z}_{t-q+1}^\top\right)^\top$. Similarly, let $Y_{j,t+h} = \left(y_{j,t}, \cdots, y_{j,t-q+1}\right)^\top$. Then, the h-step-ahead forecast can be obtained from the FAVAR system, stacked in first-order companion form:

$$\begin{pmatrix} \mathscr{Z}_t \\ Y_{j,t} \end{pmatrix} = \begin{pmatrix} \Phi^{\mathscr{Z}} & 0 \\ \Lambda_j^\top & \Phi_j^Y \end{pmatrix} \begin{pmatrix} \mathscr{Z}_{t-1} \\ Y_{j,t-1} \end{pmatrix} + \begin{pmatrix} \mathscr{V}_t^{\mathscr{Z}} \\ \mathscr{V}_{j,t}^{Y} \end{pmatrix}, \tag{A.6}$$

or

$$\mathscr{Y}_{j,t} = \Phi_j^{\mathscr{Y}} \mathscr{Y}_{j,t-1} + \mathscr{V}_{j,t}^{\mathscr{Y}}.$$

Given quadratic loss, the h-step-ahead forecast error variance at time t is

$$\Omega_{j,t}^{\mathscr{Y}}(h) \equiv E_t\left[\left(\mathscr{Y}_{j,t+h} - E_t\left[\mathscr{Y}_{j,t+h}\right]\right)\left(\mathscr{Y}_{j,t+h} - E_t\left[\mathscr{Y}_{j,t+h}\right]\right)^\top\right], \tag{A.7}$$

where $E_t[\cdot] \equiv E[\cdot|\mathscr{I}_t]$ is the conditional forecast based on information $\mathscr{I}_t$. Since the optimal h-step-ahead forecast is the conditional mean, i.e., $E_t\left[\mathscr{Y}_{j,t+h}\right] = \left(\Phi_j^{\mathscr{Y}}\right)^h \mathscr{Y}_{j,t}$, the forecast error variance evolves according to

$$\Omega_{j,t}^{\mathscr{Y}}(h) = \Phi_j^{\mathscr{Y}} \Omega_{jt}^{\mathscr{Y}}(h-1) \Phi_j^{\mathscr{Y}^\top} + E_t\left[\mathscr{V}_{j,t+h}^{\mathscr{Y}} \mathscr{V}_{j,t+h}^{\mathscr{Y}^\top}\right]. \tag{A.8}$$

The expected forecast uncertainty of the scalar series $y_{j,t+h}$ given information at time t, denoted as $\mathscr{U}_{j,t}^y(h)$, is

$$\mathscr{U}_{j,t}^y(h) = \sqrt{\mathbf{1}_j^\top \Omega_{j,t}^{\mathscr{Y}}(h) \mathbf{1}_j}, \tag{A.9}$$

where $\mathbf{1}_j$ is a selection vector.

We use Jurado et al. (2015)'s toolkit to conduct the estimation procedure above. For the specific subset of variables in $\mathbf{X}_t$ that we are interested in, defined as $\mathbf{y}_t = \left(y_{1,t}, \cdots, y_{N_y,t}\right)^\top$, we form weighted averages of estimates of individual uncertainty to obtain the estimate of aggregate uncertainty:

$$\overline{\mathscr{U}}_t^y(h) = \sum_{j=1}^{N_y} w_j \widehat{\mathscr{U}}_{j,t}^y(h), \tag{A.10}$$

where $\widehat{\mathscr{U}}_{j,t}^y(h)$ indicates the estimated value of $\mathscr{U}_{j,t}^y(h)$, w_j represents the aggregate weights, and N_y is the number of variables of interest. We use a simple weighting scheme that gives every series the equal weight of $w_j = 1/N_y$, which does not impose any structure on the individual uncertainties or beyond the assumptions of the latent volatility process (Jurado et al., 2015).[38] Unless otherwise specified, the forecast horizon is set as $h = 1$ in our analyses.[39]

In this paper, we constructed two uncertainty indices using the method proposed by Jurado et al. (2015). One is the operation-based monetary policy uncertainty (MPU-MPO) index. The other is the economic uncertainty (EU) index.

We first describe the monetary policy indicators used in constructing the MPU-MPO index. This is presented in Table A.2. The majority of the variables, totaling 45, are sourced from the dataset provided by Wang et al. (2019). Variables related to monetary policy, output, price, and financial markets are included. Additionally, we introduce an "Additional Instruments" group containing six variables to incorporate more recently developed monetary policy tools. Therefore, the total number of monetary policy indicators that we used is 45 + 6 = 51, which forms the FAVAR system. Importantly, when estimating the MPU-MPO index, we use $N_y = 14 + 6 = 20$ monetary policy variables in the "Monetary Policy" and "Additional Instruments" groups. This approach allows us to extract the conditional volatility related to monetary policy (i.e., monetary policy uncertainty) while simultaneously controlling for the impact of other relevant economic variables. Fig. 1 of the main text illustrates the MPU-MPO series, highlighting notable peaks during the 2008 financial crisis and the monetary policy tightening in 2010.

[38] As suggested by Jurado et al. (2015), an alternative measure is to average individual uncertainties by their loadings on the first principal component (PCA) of the covariance matrix of individual uncertainties. Our empirical results remain robust when using this weighting method.

[39] While the level of uncertainty increases with forecast horizon (on average), the variability of uncertainty decreases because the forecast tends to the unconditional mean as the forecast horizon tends to infinity. However, the uncertainty series constructed from different forecast horizons are highly correlated and show similar time-series trends. Our main results are also robust to this minor setup.

Table A.2
Description of monetary policy indicators.

	Indicator name	Calculation method
Monetary Policy	Money supply (M0)	Year on year growth
	Money supply (M1)	Year on year growth
	Money supply (M2)	Year on year growth
	Required Reserve Ratio	Level value
	Rediscount Ratio	Level value
	Nominal lending rate: within 1 year	Level value
	Nominal lending rate: 1-5 year	Level value
	Nominal lending rate: over 5 years	Level value
	Household savings deposits rate: 1-year	Level value
	Household savings deposits rate: 2-year	Level value
	Household savings deposits rate: 3-year	Level value
	Interbank offered rate: 7-day	Level value
	Interbank offered rate: 1-month	Level value
	Interbank offered rate: 3-month	Level value
Additional Instruments	Standing lending facility: overnight	Level value
	Standing lending facility: 7-day	Level value
	Standing lending facility: 1-month	Level value
	Medium-term lending facility: current month	Level value
	Medium-term lending facility: 1-year	Level value
	Reverse repo rate: 7-day	Level value
Bond Market	Treasury bond spread: 1-year	Minus 3-month Treasury bond yield
	Treasury bond spread: 3-year	Minus 3-month Treasury bond yield
	Treasury bond spread: 5-year	Minus 3-month Treasury bond yield
	Treasury bond spread: 10-year	Minus 3-month Treasury bond yield
	3A corporate bond spread: 1-year	Minus 3-month Treasury bond yield
	3A corporate bond spread: 3-year	Minus 3-month Treasury bond yield
	3A corporate bond spread: 5-year	Minus 3-month Treasury bond yield
	3A corporate bond spread: 10-year	Minus 3-month Treasury bond yield
Stock Market	Shanghai stock exchange return	Return of SSE composite index
	Shenzhen stock exchange return	Return of SZSE composite index
	Shanghai stock exchange volatility	GARCH(1,1) Conditional Std.
	Shenzhen stock exchange volatility	GARCH(1,1) Conditional Std.
	Shanghai stock exchange turnover	Level value
	Shenzhen stock exchange turnover	Level value
Macro Economy	Value added of industry	Year on year growth
	Business cycle leading index	Year on year growth
	Business cycle coincident index	Year on year growth
	Business cycle lagging index	Year on year growth
	Business cycle early-warning index	Year on year growth
	Purchasing managers' index: manufacture	Year on year growth
	Retail sales of consumer goods	Year on year growth
	Fixed asset investment	Year on year growth
Price Level	Consumer price index	Year on year growth
	Retail price index	Year on year growth
	Corporate good price	Year on year growth
	Producer price index	Year on year growth
	Producer price index: Agricultural input	Year on year growth
Exchange Market	RMB to USD	Year on year growth
	RMB exchange rate index: BIS	Level value
Government	Government revenue	Year on year growth
	Government expenditure	Year on year growth

Note: This table summarizes the dataset related to monetary policy operation, including 51 monthly monetary policy indicators. The original data are obtained from the National Bureau of Statistics of China and the China Economic Information (CEI) database. The sample spans from January 2003 to December 2020.

The economic uncertainty (EU) index is constructed using the dataset of Lin et al. (2023), which comprises a total of 151 macroeconomic and financial indicators in $\mathbf{X}_t$. We provide a summary of the dataset in Table A.3. There are 15 groups such as real output, consumption, financial markets, price index, and international economy. More details about the definition and data preprocessing procedure can be found in their appendix. The data originate from the National Bureau of Statistics of China and the China Economic Information (CEI) database from January 2003 to December 2020. All variables are suitably transformed to render the series stationary and are standardized before estimation. To obtain the EU index, we calculate the aggregate uncertainty based on the stochastic volatility components of all 13 groups related to China's economy ($N_y = 138$), while excluding 13 series associated with the U.S. economy and international commodity prices.

Table A.3
Description of the dataset used to construct the economic uncertainty index.

Group	Number of Series	Group	Number of Series
Real Output	12	Public Finance	3
Real Investment	4	Money and Credit	9
Real Estate	9	Price Index	28
Consumption	4	Interest Rate	23
Import and Export	9	Treasury Bond Yield	20
Futures Market	9	U.S. Economy	7
Stock Market	2	Commodity	6
Exchange Rate	6	Total	151

Note: This table summarizes the dataset related to monetary policy operation, including 51 monthly monetary policy indicators. The original data are obtained from the National Bureau of Statistics of China and the China Economic Information (CEI) database. The sample spans from January 2003 to December 2020.

Fig. A.1 illustrates the time series of economic uncertainty from 2003 to 2020. It shows a notable spike in uncertainty around the 2008 financial crisis. After 2010, the index exhibits a downward trend and fluctuates with some minor peaks until 2015. This stability during the period is largely due to minimal fluctuations in key economic variables, which contributed to a more robust economic environment. Afterward, this index gradually increases again towards 2020. It reaches a peak during the COVID-19 pandemic, reflecting heightened economic uncertainty during this period. Overall, the series captures significant variations in economic uncertainty corresponding to major economic events.

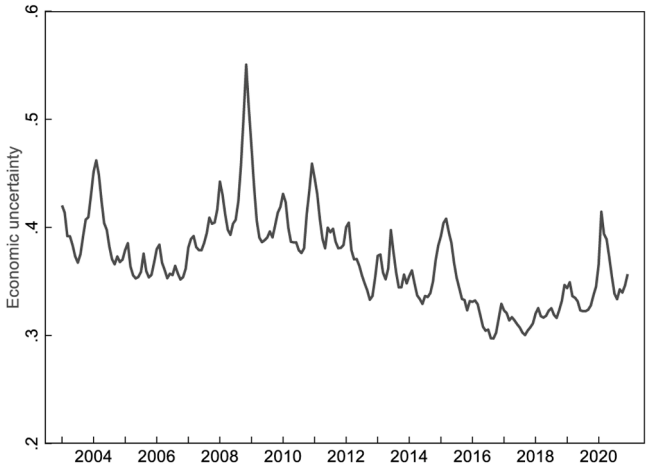


Fig. A.1. Time series of economic uncertainty measure.
Note: This figure shows the time series of economic uncertainty measure. The sample spans from January 2003 to December 2020.

Monetary policy and economic state indicators

Following Xiong (2012), we construct a comprehensive monetary policy index to measure the PBC's stance based on multiple instruments, denoted as the MPI:

$$\mathrm{MPI}_t = \frac{1}{N_t}\sum_{i=1}^{N_t}\frac{\Delta \mathrm{Tool}_{i,t}}{\Delta \mathrm{Max}_{i,t}}, \tag{A.11}$$

where the numerator ($\Delta \mathrm{Tool}_{i,t}$) represents the change in monetary policy tool i in month t, the denominator ($\Delta \mathrm{Max}_{i,t}$) is the maximum change in tool $\underline{i}$ up to month t, and N_t denotes the number of monetary policy tools used in month t. Therefore, we scale each tool index by its historical maximum change and use equal weights for aggregation. The following two price- and quantity-based monetary policy instruments are used: (1) the reserve requirement ratio (RRR), (2) the benchmark deposit rate, (3) the benchmark lending rate, (4) the medium-term lending facility (MLF) rate, (5) the standing lending facility (SLF) rate, and (6) the M2 growth rate.[40] We adjust the direction of M2 growth such that a positive value in the comprehensive index indicates a contractionary monetary policy, and a negative value indicates an expansionary monetary policy.

We also create various economic indicators, such as the output gap (OG), recession dummy (REC), business cycle index (BCI) and relative bond rate (RREL). To construct the output gap series, we use the monthly data of the month-on-month growth rate of real gross domestic product (GDP) provided by the Federal Reserve Bank of Atlanta (Chang et al., 2016; Higgins et al., 2016).[41] Following Chen et al. (2018), the gap between GDP growth and the

[40] The PBC uses the reserve requirement ratio as an effective tool to directly regulate the liquidity of the banking system, especially during the period of high inflation of 2007–2008. The benchmark interest rates for deposits and loans adjust the capital costs in the credit and money markets to change market liquidity. Considering the interest rate liberalization reform and the introduction of new monetary policy tools, we introduce the 1-year MLF rate and the 7-day SLF rate into the toolkit. These two products have the highest issuance frequency and the largest issuance scale. In 2015, the PBC proposed to "explore the SLF rate as the upper limit of the interest rate corridor," which enhanced its importance. The quantity-based instrument, M2 growth rate, serves as the intermediate target for China's monetary policy. Since M2 growth exhibits structural change in our sample, we detrend the original series by its 12-month rolling average.

[41] See https://www.atlantafed.org/cqer/research/china-macroeconomy/.

GDP growth target is defined as the output gap.[42] Considering the negative shocks to the Chinese economy due to the 2008 financial crisis, we take the recession dummy equal to one during these periods: September 2008 to December 2009.[43] For the remaining periods, the recession dummy is assigned a value of zero. The business cycle index data are compiled and released by the National Bureau of Statistics of China and obtained from the China Economic Information (CEI) database. Following previous studies (Fama and French, 1989; Brogaard and Detzel, 2015), we also create a variable from bond returns. The Chinese bond return data are available from Wind Information, Inc. (WIND). The relative bond rate is equal to the 3-month Treasury bond yield minus its 12-month rolling average.

Monetary policy surprises

We construct a rule-based MPS index following the methodology of Chen et al. (2018). This approach proposes a China-specific quantity-based (or McCallum) monetary policy rule with the specification of

$$g_{m,t} = \gamma_0 + \gamma_m g_{m,t-1} + \gamma_\pi(\pi_{t-1} - \pi^*) + \gamma_{x,t}\left(g_{x,t-1} - g^*_{x,t-1}\right) + \varsigma_{m,t}, \tag{A.12}$$

where $g_{m,t}$ is the broad money (M2) supply growth rate in month t, π_{t-1} represents the consumer price index (CPI) inflation rate, $g_{x,t-1}$ denotes the gross domestic product (GDP) growth rate in the previous month, π^* represents the inflation target, and $g^*_{x,t-1}$ is the GDP growth target.[44] The residual ($\varsigma_{m,t}$) represents the monetary policy surprise, which is a serially independent random shock that has a normal distribution with a mean of zero and a time-varying standard deviation. In line with Chen et al. (2018), we adopt the asymmetric setup for the output coefficient and heteroskedasticity:

$$\gamma_{x,t} = \begin{cases} \gamma_{x,a}, & \text{if } g_{x,t-1} - g^*_{x,t-1} \geq 0 \\ \gamma_{x,b}, & \text{if } g_{x,t-1} - g^*_{x,t-1} < 0 \end{cases}, \quad \sigma_{x,t} = \begin{cases} \sigma_{x,a}, & \text{if } g_{x,t-1} - g^*_{x,t-1} \geq 0 \\ \sigma_{x,b}, & \text{if } g_{x,t-1} - g^*_{x,t-1} < 0 \end{cases}. \tag{A.13}$$

In addition to the quantity rule-based MPS index, we construct an interest rate rule (Taylor)-based index by replacing M2 growth in Equation (A.12) with 7-day SHIBOR, a kind of market interest rate in China.[45] The underlying reason is that although China's current monetary policy framework is still mainly quantity based, the importance of price-based tools has increased recently, particularly with the promotion of interest rate liberalization. This increased importance is also in line with the fact that both quantity- and price-based instruments play important but different roles in China.

Chinese risk factors and stock-level characteristics

The data we use, which include data on stock returns, risk factors, and firm characteristics, are from the WIND and China Stock Market and Accounting Research (CSMAR) data libraries, both of which are prominent financial data providers in China.

Instead of simply replicating a general asset pricing model in China, we use the market factor (MKT), size factor (small minus big, SMB), value factor (value minus growth, VMG) and turnover factor (pessimistic minus optimistic, PMO) constructed by Liu et al. (2019). Their Chinese-specified versions of size and value factors are intended to avoid overapplying research paradigms developed for U.S. settings and consider the specificities of China's financial system, such as excluding the effects of shell firms in reverse mergers. For instance, the size factor excludes the smallest 30% of firms, which are referred to as potential shells in reverse mergers when IPO constraints in China are tight. The value factor is calculated using the earnings-price ratio (EP), which is different from the conventional BM-based value factor of Fama and French (1992). To capture the sentiment effects, the turnover factor is constructed by considering long low-turnover stocks and short high-turnover stocks for which investors are pessimistic and optimistic, respectively. All risk factors from the Chinese factor model can be downloaded from their website.

Several stock-level characteristics are included in our empirical discussion. For each stock and for each month in our sample, we estimate the market beta from the 60-month rolling regressions of excess stock returns on four Chinese risk factors:

$$R^e_{i,t} = \alpha_{i,t} + \beta^{\mathrm{MKT}}_{i,t}\mathrm{MKT}_t + \beta^{\mathrm{SMB}}_{i,t}\mathrm{SMB}_t + \beta^{\mathrm{VMG}}_{i,t}\mathrm{VMG}_t + \beta^{\mathrm{PMO}}_{i,t}\mathrm{PMO}_t + \varepsilon_{i,t}, \tag{A.14}$$

where $R^e_{i,t}$ denotes the excess return of stock i in month t and MKT, SMB, VMG and PMO are the market, size, value and turnover factors, respectively, in the Chinese four-factor (CH4) model. The slope coefficient on market factor is the estimate of the market beta (β^{MKT}). Following general settings (Fama and French, 2015; Bali et al., 2017; Liu et al., 2019), the stock's market capitalization (SIZE) is computed as the natural logarithm of the product of the price per share during the previous month and the number of total A shares outstanding, including nontradable shares. As suggested by Liu et al. (2019), the earnings-price ratio (EP) best captures the cross-sectional variation in average stock returns to explain the value effect in China; this variable is computed as the ratio of earnings to market capitalization. The quarterly operating profitability (ROE) equals the ratio of a firm's earnings to book equity. A firm's investment is measured using its annual asset growth rate (I/A). Following Hirshleifer et al. (2004), we construct firm-level net operating assets (NOA) by calculating the gap between operating assets and operating liability and dividing by total assets in the prior period. Following Jegadeesh (1990), short-term reversal (REV) is obtained using the stock return over the prior month and is computed as the cumulative return over the past 20 trading days. Following Harvey and Siddique (2000), coskewness (COSKEW) is defined as

[42] The results for using alternative measures such as the Hodrick and Prescott (1997) filtered series are similar.

[43] The data show that in April 2020, industrial value-added growth turned from negative to positive, and the monthly MoM GDP growth data also exhibited a similar trend. Therefore, we define the end point of the short-term negative shock of the COVID-19 pandemic as March 2020.

[44] To obtain the monthly MPS series, we construct a monthly version of Chen et al. (2018)'s monetary policy rule. Notably, Chen et al. (2018)'s monetary policy rule is conducted using month-on-month (MoM) data. Therefore, we use a preprocessed dataset containing MoM time-series of the Chinese macroeconomy from the Federal Reserve Bank of Atlanta. The dataset is widely used in macroeconomic research on China (Chen et al., 2016; Higgins et al., 2016). In line with Chen et al. (2018), the monthly inflation target is set to 0.292%, which corresponds to an annual rate of 3.500%. The GDP growth target is set annually by the Chinese State Council.

[45] As documented by Huang et al. (2020), Shanghai Interbank Offered Rate (SHIBOR) has behaved as an important indicator of the market's liquidity condition, which may serve as an intermediate target.

$$COSKEW_{i,t} = \frac{E\left[\varepsilon_{i,t} R_{m,t}^2\right]}{\sqrt{E\left[\varepsilon_{i,t}^2\right] E\left[R_{m,t}^2\right]}}, \tag{A.15}$$

where $\varepsilon_{i,t}$ is the residual from the regression of the excess individual stock monthly return against the contemporaneous excess market return $R_{m,t}$ using observations over the prior 60 months with at least 24 monthly return observations available (Mitton and Vorkink, 2007). We measure monthly turnover (TOVER) as the average daily turnover over the prior month, where daily turnover is calculated as trading volume divided by total shares (Liu et al., 2019). Following Amihud (2002), we measure illiquidity (ILLIQ) using the ratio of daily absolute stock return to daily trading volume averaged over the month.[46] Finally, we also consider a series of industry dummies to control for industry effects based on the industrial classification code from the CSMAR data library.

We also impose several filters in the spirit of Liu et al. (2019). First, given the small free float capital of newly listed stocks and the large number of restricted shares to be released after one year, we require stocks to be listed for more than one year. Second, we exclude stocks with less than 15 days of trading in the most recent month to prevent our results from being influenced by returns that follow long trading suspensions. Third, we drop special treatment (ST) stocks with 5% daily price-change limits that have been specially treated by the Shanghai or Shenzhen exchanges and even face the risk of delisting. Finally, the bottom 30% of stocks, often known as "shell stocks," are valued significantly as potential shells in reverse mergers but are less related to operating fundamentals. Thus, we eliminate them when conducting our empirical analysis.

Appendix B. Additional results

Further results of the value of information aggregation

B.1.1 Bivariate portfolio sorts

We perform conditional bivariate portfolio sorts on aggregate and individual MPU betas. As presented in Table B.1, Panel A show that after controlling for aggregate MPU measure, individual MPU measure does not exhibit a significant risk premium. When we reverse the order of sorting betas (Panel B), the aggregate MPU measure remains a significant predictor for subsequent stock returns.

Table B.1
Bivariate portfolio sorts for aggregate and individual monetary policy uncertainty betas.

Panel A: Controlling for aggregate MPU beta			
	Individual MPU beta		
High−Low	$\beta^{\text{MPU-MPC}}$	$\beta^{\text{MPU-MPO}}$	$\beta^{\text{MPU-NEWS}}$
RET−RF	0.135	-0.187	-0.064
	(0.484)	(-0.705)	(-0.235)
α_{CAPM}	0.133	-0.195	-0.047
	(0.473)	(-0.752)	(-0.179)
α_{CH4}	0.117	0.059	-0.221
	(0.401)	(0.228)	(-0.813)
Panel B: Controlling for individual MPU beta			
	Control variable		
High−Low	$\beta^{\text{MPU-MPC}}$	$\beta^{\text{MPU-MPO}}$	$\beta^{\text{MPU-NEWS}}$
RET−RF	-1.154	-0.877	-0.571
	(-4.258)	(-2.776)	(-2.123)
α_{CAPM}	-1.144	-0.868	-0.581
	(-4.364)	(-2.841)	(-2.100)
α_{CH4}	-1.359	-1.077	-0.353
	(-5.549)	(-3.830)	(-1.334)

Note: This table reports the results of conditional bivariate portfolio to analyze the cross-sectional relationship between the aggregate and individual monetary policy uncertainty betas and excess stock returns based on one monetary policy uncertainty beta as the control variable. Panel A controls for the aggregate MPU beta, and Panel B controls for the individual MPU beta. For each month, the value-weighted decile portfolios are formed by first sorting individual stocks based on the control variable and then sorting stocks in each control variable decile into deciles based on the other MPU beta during the previous month, where decile 1 (*Low*) contains stocks with the lowest MPU beta and decile 10 (*High*) contains stocks with the highest MPU beta. Each row reports the differences in average excess returns (RET−RF) and alphas, where α_{CAPM} denotes the alpha of the CAPM model containing the market (MKT) factor and α_{CH4} denotes the alpha of the CH4 model containing the market (MKT), size (SMB), value (VMG) and turnover (PMO) factors. The average excess returns and alphas are defined in monthly percentage terms. The MPU betas are obtained from the specification of Equation (5) by replacing the aggregate MPU measure with the individual MPU indices, where $\beta^{\text{MPU-MPC}}$, $\beta^{\text{MPU-MPO}}$ and $\beta^{\text{MPU-NEWS}}$ correspond to the communication-, operation- and news-based MPU betas, respectively. The sample spans from December 2007 to December 2020. Newey−West adjusted *t*-statistics are given in parentheses.

B.1.2 A factor-timing strategy

[46] Amihud's illiquidity measure is scaled as 108, which is from CSMAR data library.

We construct a value-weighted zero-cost factor-timing strategy based on the MPU beta (β^{MPU}) to investigate the differences between the aggregate and the individual MPU measures, where we long stocks in the top 30th percentile of high-performing characteristics (low β^{MPU}) and short stocks within the bottom 30th percentile of low-performing characteristics (high β^{MPU}). The high-minus-low portfolios are neutral to market capitalization with independent sorting. We apply the factor-forming technique pioneered by Fama and French (1993), which was also used by Bali et al. (2017) in a similar context to ours.[47]

Fig. B.1 presents the cumulative excess returns of the factor-timing strategy for each different MPU measure. On the one hand, the aggregate MPU beta portfolio demonstrates superior excess returns compared with other individual MPU measures for the majority of the observed period. On the other hand, no single individual index is always better than the others, indicating the necessity of aggregating information. For example, the portfolios based on the communication- and operation-based indices exhibit higher returns before 2015, whereas the portfolios based on the news-based index outperform the other two afterward.

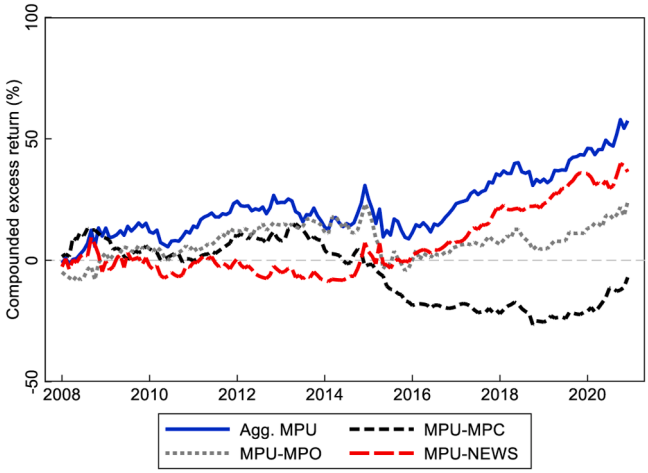


Fig. B.1. Cumulative excess returns of monetary policy uncertainty factors.
Note: This figure reports the cumulative excess returns of high-minus-low portfolios on the monetary policy uncertainty (MPU) beta strategies, where "Agg. MPU", "MPU-MPC", "MPU-MPO", and "MPU-NEWS" denote the aggregate, communication-, operation-, and news-based MPU measures, respectively. The compounded excess return for month t is calculated as 100 times the cumulative product of one plus the monthly return up to and including the given month. We normalize the returns so that the initial return is zero. The sample is from December 2007 to December 2020.

In addition, Table B.2 provides the average monthly return, standard deviation, annualized Sharpe ratio, and maximum drawdown of different MPU beta strategies. These results suggest that the aggregate MPU measure provides a more stable and better strategy performance return than the individual measures do. For instance, the aggregate MPU beta portfolio yields an average monthly return of 0.318%, with an annualized Sharpe ratio of 0.473 and a maximum drawdown of 7.200%. However, the news-based MPU beta portfolio, which is the best among the individual measures, only commands an average return of 0.226%, with an annualized Sharpe ratio of 0.337. Compared with the extant Chinese four factors in Panel B, our aggregate MPU beta strategy remains valuable and outperforms the market and size factors

Table B.2
Long-short portfolios based on the MPU beta strategy and Chinese four factors.

	Avg. Return (%)	Std. Dev. (%)	SR	Drawdown (%)
Panel A: MPU beta strategy				
Agg. MPU	0.318	2.333	0.473	7.200
MPU-MPC	-0.030	1.840	0.056	7.781
MPU-MPO	0.163	2.231	0.253	7.421
MPU-NEWS	0.226	2.320	0.337	8.396
Panel B: Chinese four factors				
MKT	0.119	7.622	0.054	14.851
SMB	0.607	4.492	0.468	16.729
VMG	0.841	3.569	0.816	13.924
PMO	0.684	3.660	0.647	11.823

Note: This table reports the high-minus-low portfolio performance of the monetary policy uncertainty beta (MPU) in Panel A and Chinese four factors (Liu et al., 2019) in Panel B. "Avg. Return" and "Std. Dev." denote the mean and standard deviation of the monthly long-short portfolio return, respectively. "SR" denotes the annualized Sharpe ratio. "Drawdown" denotes the maximum drawdown. The Chinese four factors are the market (MKT), size (SMB), value (VMG), and turnover (PMO). The sample is from December 2007 to December 2020.

B.1.3 Orthogonalized beta test

In the spirit of Lee et al. (2019), we further show the information gains from aggregation by orthogonalizing the individual MPU betas with respect

[47] Specifically, at the end of each month, we sort all stocks into two groups based on market capitalization, with the breakpoint dividing the two groups being the median market capitalization of stocks traded on the A-share markets. We independently sort all stocks into three groups based on an ascending sort of the MPU beta using the 30th and 70th percentile values. The intersections of the two size groups and the three βMPU groups generate six portfolios. The return of the βMPU strategy is calculated as the average return of the two value-weighted high-βMPU portfolios minus the average return of the two value-weighted low-βMPU portfolios.

to the aggregate MPU beta. Specifically, for each month, we run a cross-sectional regression of individual MPU betas on the intercept and our aggregate MPU beta. The intercept plus residual is computed as the orthogonalized individual MPU beta. Next, we run Fama–MacBeth cross-sectional regressions of one-month-ahead stock returns on the aggregate MPU beta, the orthogonalized individual MPU beta, and other stock-level variables.

Table B.3 presents the time series averages of the slope coefficients on these MPU betas. The results show that the aggregate MPU measure is priced by commanding a significant negative risk premium, whereas the orthogonalized individual uncertainty is not. This finding indicates that individual MPU measures may have idiosyncratic errors that are unrelated to cross-sectional stock returns. As a result, information aggregation provides a better proxy for measuring the overall level of MPU in China by extracting the common component from diverse sources of individual measures. More importantly, it is a common component that has real asset pricing implications in the Chinese stock market.

Table B.3
Fama–MacBeth cross-sectional regressions for orthogonalized beta tests.

	Without controlling for industry effect			With controlling for industry effect		
	(1)	(2)	(3)	(4)	(5)	(6)
β^{MPU}	-0.034	-0.035	-0.035	-0.032	-0.034	-0.033
	(-2.028)	(-2.060)	(-2.002)	(-2.001)	(-2.053)	(-1.950)
$\beta^{MPU\text{-}MPC\perp}$	0.012			0.011		
	(0.661)			(0.612)		
$\beta^{MPU\text{-}MPO\perp}$		-0.012			-0.009	
		(-0.707)			(-0.552)	
$\beta^{MPU\text{-}NEWS\perp}$			-0.013			-0.017
			(-0.393)			(-0.560)
Obs.	122,701	122,701	122,701	122,701	122,701	122,701
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2 (%)	8.032	8.101	8.115	9.573	9.634	9.641

Note: This table reports the time-series average of the slope coefficients for the monetary policy uncertainty beta by regressing one-month-ahead excess stock returns on the aggregate MPU beta (β^{MPU}), orthogonalized individual MPU beta, and a collection of lagged stock-level characteristics as control variables via the Fama and MacBeth (1973) methodology. The orthogonalized procedure is conducted as follows: for each month, we run a cross-sectional regression of the individual MPU beta on the intercept and our aggregate MPU beta. The intercept plus residual is computed as the orthogonalized individual MPU beta (notated with subscript "⊥"). The individual MPU indices are communication-based (MPU-MPC), operation-based (MPU-MPO) and news-based (MPU-NEWS). The stock-level characteristics include the market beta (β^{MKT}), log of market capitalization (SIZE), earnings-price ratio (EP), operating profitability (ROE), investment (I/A), net operating assets (NOA), short-term reversal (REV), coskewness (COSKEW), turnover (TOVER), and illiquidity (ILLIQ). The sample spans from December 2007 to December 2020. Newey–West adjusted *t*-statistics are given in parentheses using four lags.

Additional robustness tests

Table B.4.

Table B.4
Bivariate portfolio sorts: Controlling for alternative uncertainty variables and monetary policy surprises

	β^{EU}	β^{EPU}		β^{MPS}	
Decile		HL	BBD	McCallum Rule	Taylor Rule
Low	0.653	0.612	0.815	0.571	0.623
	(2.443)	(2.590)	(3.539)	(2.668)	(2.673)
2	0.318	0.119	0.189	0.311	0.189
	(1.843)	(0.743)	(0.909)	(1.539)	(0.950)
3	0.013	0.396	0.199	-0.061	0.062
	(0.068)	(1.777)	(1.037)	(-0.300)	(0.326)
4	0.047	-0.056	-0.120	0.448	0.389
	(0.228)	(-0.326)	(-0.612)	(1.552)	(1.332)
5	-0.138	0.050	-0.100	-0.138	0.104
	(-0.656)	(0.214)	(-0.434)	(-0.671)	(0.472)
6	-0.111	-0.145	0.066	-0.392	-0.235
	(-0.487)	(-0.640)	(0.252)	(-1.607)	(-1.082)
7	-0.008	0.134	-0.058	0.245	0.027
	(-0.036)	(0.710)	(-0.227)	(1.202)	(0.171)
8	0.356	-0.060	0.172	-0.220	0.055
	(1.556)	(-0.348)	(0.813)	(-1.361)	(0.261)
9	-0.016	-0.145	0.006	0.083	-0.232
	(-0.096)	(-0.778)	(0.036)	(0.621)	(-1.235)
High	-0.338	-0.364	-0.480	-0.187	-0.269
	(-2.065)	(-1.595)	(-2.210)	(-1.024)	(-1.460)
High−Low	-0.991	-0.976	-1.296	-0.758	-0.892
	(-3.523)	(-3.584)	(-4.709)	(-2.891)	(-3.575)

Note: This table reports the results of conditional bivariate portfolio to analyze the cross-sectional relationship between the aggregate monetary policy uncertainty beta and excess stock returns using the alternative uncertainty beta or monetary policy surprise beta as control variables. For each month, the value-weighted decile portfolios are formed by first sorting individual stocks based on the alternative uncertainty beta or monetary policy surprise beta as the control variable and then sorting stocks in each control variable decile into deciles based on their aggregate MPU betas during the previous month, where decile 1 (*Low*) contains stocks with the lowest aggregate MPU beta and decile 10 (*High*) contains stocks with the highest aggregate MPU beta. The aggregate MPU beta is obtained from the specification of Equation (5). The alternative uncertainty beta and monetary policy surprise beta are also obtained from the specification of Equation (5) by replacing the aggregate MPU measure with the alternative uncertainty index or the monetary policy surprise index. Therefore, β^{EU}, β^{EPU} and β^{MPS} correspond to economic uncertainty (EU), economic policy uncertainty (EPU) and monetary policy surprise (MPS) betas, respectively. The entries in columns 2 and 3 under "EPU" present the results controlling for EPU betas: "HL" denotes Huang and Luk (2020)'s index, and "BBD" denotes Baker et al. (2016)'s index. The entries in columns 4 and 5 under "MPS" present the

results controlling for MPS betas: "McCallum Rule" denotes the quantity rule-based index, and "Taylor Rule" denotes the interest rate rule-based index. Each row reports the alpha of the CH4 model containing the market (MKT), size (SMB), value (VMG) and turnover (PMO) factors. The alphas are defined in monthly percentage terms. The sample spans from December 2007 to December 2020. Newey–West adjusted *t*-statistics are given in parentheses using four lags.

Table B.5.

Table B.5
Additional robustness tests.

	Alternative operation-based indices			Considering media censorship	
	PCA-weighted	Horizon = 3	Horizon = 12	10-year dummy	5-year dummy
β^{MPU}	-0.030	-0.036	-0.033	-0.046	-0.054
	(-2.148)	(-2.192)	(-2.110)	(-2.089)	(-2.063)
Obs.	122,701	122,701	122,701	122,701	122,701
Controls	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes
Adj. R^2 (%)	9.484	9.454	9.454	9.459	9.443

Note: This table reports various robustness checks by regressing one-month-ahead excess stock returns on the monetary policy uncertainty beta and a collection of lagged stock-level characteristics as control variables using the Fama and MacBeth (1973) methodology. The MPU betas are obtained from the specifications of Equation (5) by replacing the aggregate MPU measure with alternative measures. The entries in columns 1–3 under "alternative operation-based indices" present the results obtained using the alternative operation-based MPU measure when constructing the aggregate index. We use PCA-weighted and change the forecast horizon (3 or 12) to obtain alternative MPU-MPO indices. Columns 4 and 5 under "Considering media censorship" present the results obtained using an alternative news-based MPU measure when constructing the aggregate index. We include time dummy variables with durations of 10 years and 5 years to account for the impact of regime changes in the central government, mitigating potential variations in media censorship. The stock-level characteristics include the market beta (β^{MKT}), log of market capitalization (SIZE), earnings-price ratio (EP), operating profitability (ROE), investment (I/A), net operating assets (NOA), short-term reversal (REV), coskewness (COSKEW), turnover (TOVER) and illiquidity (ILLIQ). The sample spans from December 2007 to December 2020. Newey–West adjusted *t*-statistics are given in parentheses using four lags.

Appendix C. An aggregate monetary policy uncertainty measure for the U.S

Given the generality of information transmission in monetary policy-making and execution, our method is also applicable in the U.S. context. Therefore, we construct an empirical measure of aggregate monetary policy uncertainty for the U.S. and examine its economic determinants. More importantly, we compare the results with those from China to highlight the special features of Chinese monetary policy-making and its MPU index.

The individual MPU indices

To maintain consistency with our MPU index in China, we use individual measures based on monetary policy communications, operations, and news texts.[48] For the communication-based measure, we adopt the monetary policymaker's uncertainty (PMU) index of Cieslak et al. (2021).[49] Their study uses transcripts from Federal Open Market Committee (FOMC) meetings to measure uncertainty about various dimensions of the economic environment. Therefore, they construct a series of meeting-level, topic-specific PMU indices, including inflation, real economy, financial markets, models, and unclassified topics.[50] We average across five topics and obtain a single PMU index. It captures the overall perception of uncertainty from the view of the central bank, which is also conveyed to the market though its communications. The sample consists of 228 meetings from August 1987 through December 2015, which is also the period for the following U.S.-based analysis. To convert it from meeting-level to monthly data, we use the value from the most recent preceding meeting if there is a missing value for the current month. The operation-based MPU measure, referred to as MPU-JLN, is constructed following the framework of Jurado et al. (2015). It is derived from the unforecastable components based on a panel of monetary policy variables, similar to the method in our paper. Specifically, we use monthly macroeconomic data from the FRED-MD dataset.[51] This dataset consists of eight groups with 126 macroeconomic time series, including output and income, the labor market, housing, prices, etc. We use 20 series classified into the groups of "money and credit" and "interest and exchange rates", which are relevant monetary policies, to compute the measure of MPU.[52] The remaining variables serve as the conditional information to construct factor structures in the FAVAR system. For the news-based measure, we use the monetary policy uncertainty index proposed by Baker et al. (2016), denoted by MPU-BBD hereafter.[53] This index quantifies the public's concerns regarding MPU in the U.S., which corresponds to our argument about the media channel in monetary policy dissemination.

Table C.1 presents the descriptive statistics of the three individual measures. Similar to our findings in China, it is apparent that the scales across different MPU measures vary dramatically due to differences in construction methods and data sources. To make them comparable, we standardize each MPU measure by its full-sample mean and standard deviation. Moreover, all three MPU indices are strongly serially correlated, with first-order autocorrelations in the range of 0.630–0.973. In addition, Panel B presents the pairwise correlations between individual MPU measures. All of the measures are positively correlated but not strongly correlated to some extent. The correlation matrix indicates that different MPU indices may reflect different dimensions or aspects of the aggregate level of MPU. These U.S. primitive results also support our main motivation for measuring monetary policy uncertainty from the different perspectives of information dissemination.

[48] Notably, we use indices published in the literature as much as possible to ensure the authority and comparability with previous findings.
[49] The data is publicly available from the authors' website: https://www.feduncertainty.com/data.
[50] This is done by using a slightly more complicated keyword counting method, which analyzes the local co-occurrence of terms that denote risk and uncertainty with terms related to specific economic concepts like inflation, wages, and economic growth. Cieslak et al. (2021) distinguish the statements by the FOMC members and the Federal Reserve Board staff. Here, we use the index that includes both types of texts.
[51] Available at https://research.stlouisfed.org/econ/mccracken/fred-databases/. This dataset is described in detail in McCracken and Ng (2016), which was previously used in Jurado et al. (2015) and Ludvigson et al. (2021). We use April 2024 vintage.
[52] In the latest version of the FRED-MD appendix, the variables correspond to the IDs: 70–79, 84–90, and 132–134.
[53] Available at https://www.policyuncertainty.com/monetary.html. Details can be found in their paper.

Table C.1
Summary statistics and pairwise correlations of individual monetary policy uncertainty measures in the U.S.

Panel A: Summary statistics

	Avg	Std	Max	Min	Skew	Kurt	1st AC
PMU	0.003	0.001	0.005	0.001	0.965	4.415	0.630
MPU-JLN	0.647	0.135	1.244	0.381	0.861	5.142	0.973
MPU-BBD	92.819	59.120	407.941	16.575	1.851	7.872	0.607

Panel B: Pairwise correlations

	PMU	MPU-JLN	MPU-BBD
PMU	1.000		
MPU-JLN	0.002	1.000	
MPU-BBD	0.184	0.311	1.000

Note: This table reports the summary statistics and pairwise correlations of three individual monetary policy uncertainty measures in Panels A and B, respectively. The communication-based MPU index is Cieslak et al. (2021) policymakers' uncertainty (PMU). The operation-based MPU index (MPU-JLN) is obtained from the unforecastable components of a broad set of monetary policy variables following the method of Jurado et al. (2015). The news-based MPU index (MPU-BBD) is obtained from Baker et al. (2016). The sample spans from August 1987 to December 2015.

An aggregate measure

In line with our manuscript, the aggregate MPU measure is constructed via principal component analysis (PCA), which extracts the first principal component of the three individual indices as the aggregate index. As presented in Table C.2, all three individual MPU measures demonstrate a highly positive correlation with the aggregate index. Among them, the news-based MPU index is the most informative and has the highest factor loading and correlation with the aggregate measure. This observation is similar to our results in China. The underlying economic reason may be that the news media captures a broad and timely range of information, significantly influencing market perceptions and expectations.

Notably, the communication-based MPU index is relatively less important in the U.S. market, with a factor loading (correlation) of 0.362 (0.474) in the U.S. versus 0.474 (0.590) in China. This may be attributed to ambiguous communication with the PBC. Unlike advanced central banks, the PBC frequently uses ambiguous words and expressions, such as "prudent," in its communication. This ambiguity makes it more challenging for the market to understand monetary policy, leading to a greater correlation between communication-based MPU and aggregate MPU.

Table C.2
Loadings and correlations of individual monetary policy uncertainty measures in the U.S.

	U.S.		China	
	Loading	Correlation	Loading	Correlation
Communication-based MPU	0.362	0.423	0.474	0.590
Operation-based MPU	0.609	0.710	0.605	0.752
News-based MPU	0.706	0.824	0.639	0.795

Note: This table reports the factor loadings of the aggregate monetary policy uncertainty measure on individual measures and the correlations between them. The sample spans from August 1987 to December 2015 for the U.S. The loadings and correlations for China are copied from Table 2 in the manuscript.

Fig. C.1 illustrates the time series variation in the U.S. aggregate MPU measure. The results indicate that monetary policy uncertainty peaks before major policy changes. For instance, during a phase of unusually low interest rates from 2002 to 2003, uncertainty was high due to concerns about the onset of the tightening cycle, which eventually commenced in 2004. The level of monetary policy uncertainty surged sharply at the beginning of the 2008 financial crisis. Fig. C.1 also contains information on NBER-dated recessions. Unsurprisingly, during recessions, MPU tends to increase. This finding is also consistent with our results for China and the literature, e.g., Tillmann (2020) and others.

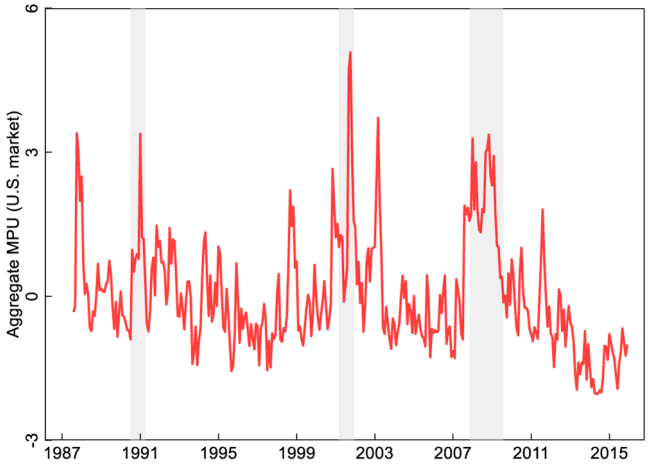


Fig. C.1. Time series of aggregate monetary policy uncertainty measure in the U.S.
Note: This figure shows the time series of the aggregate monetary policy uncertainty measure. The shaded gray area corresponds to NBER-dated recessions. The sample spans from August 1987 to December 2015.

Economic determinants of the U.S. aggregate measure

To examine the economic determinants of the U.S. aggregate MPU measure, we conduct time-series regressions of the MPU index on alternative uncertainty variables, monetary policy stance, and the economic state. For uncertainty measures, we use the economic uncertainty (EU) index of Jurado et al. (2015), the news-based economic policy uncertainty (EPU) index of Baker et al. (2016), and the financial uncertainty (FU) index constructed by Ludvigson et al. (2021). In terms of variables related to monetary policy, the text-based Hawk–Dove (HD) score of Cieslak et al. (2021) is used to measure the Fed's policy stance.[54] Notably, an increase in the HD score indicates a contractionary monetary policy. We also include the change in the effective federal funds rate (EFFR) target to measure the policy preference. Following Brogaard and Detzel (2015), the economic state variables are the output gap, NBER-dated recession, Chicago Fed National Activity Index, relative bond rate, term spread, and default spread.[55]

Similar to the results in China, we find that the U.S. aggregate MPU measure is positively correlated with broader uncertainties and exhibits a countercyclical pattern. For instance, as presented in the second column in Panel A of Table C.3, the slope coefficient of EPU is positive and statistically significant at the 1% level. However, the maximum adjusted goodness-of-fit (Adj. R^2) is 46.273%, indicating that more than 50% of the variation cannot be explained by these uncertainty variables. In the final panel, both univariate and multivariate regressions indicate that MPU increases with a decrease in output and real economic activity, as well as an increase in procyclical relative bond rates. Conversely, the countercyclical term spread and default spread load significantly and positively on MPU. Likewise, it also increases during recessions, which is analogous to our findings in China.

However, there indeed are differences: U.S. monetary policy uncertainty tends to decrease as monetary policy tightens, whereas the opposite is observed in China. The first column in Panel B shows that the slope coefficient of the HD score is negative and highly significant. We also observe a similar result when using the change in the effective FFR target. This interesting finding contrasts with our observations in China, where monetary policy uncertainty increases with contractionary policy. There is a possible reason for this difference.[56] In the U.S., an increase in MPU may reflect investors' concerns about a future path toward monetary policy tightening. As a result, monetary policy uncertainty falls when contractionary policies are implemented. However, in China, the overriding objective of the central government is to achieve output growth (Chen et al., 2018). The monetary authority thereby develops an accommodative policy environment to promote economic growth most of the time. Hence, the market can barely anticipate the action of the PBC during hawkish periods, exacerbating the uncertainty about monetary policy.

Table C.3
Economic determinants of aggregate monetary policy uncertainty measures in the U.S.

Panel A: Other uncertainty

	(1)	(2)	(3)	(4)
EU	0.523			0.293
	(7.100)			(3.443)
EPU		0.562		0.405
		(5.394)		(4.348)
FU			0.527	0.164
			(6.368)	(1.789)
Obs.	341	341	341	341
Adj. R^2 (%)	27.115	31.389	27.590	46.273

Panel B: Monetary policy

	(1)	(2)	(3)
HD	-0.453		-0.295
	(-6.787)		(-4.533)
ΔEFFR		-0.439	-0.257
		(-5.172)	(-2.720)
Obs.	341	341	341
Adj. R^2 (%)	20.300	19.010	24.201

Panel C: Economic state

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
OG	-0.059						0.121
	(-0.415)						(1.580)
REC		0.529					0.279
		(7.305)					(2.869)
CFNAI			-0.435				-0.052
			(-6.471)				(-0.623)
RREL				-0.467			-0.244
				(-4.679)			(-2.523)
TERM					0.107		0.012

(*continued on next page*)

[54] The HD score is the difference between the Hawk and Dove scores, and is calculated at the meeting level. Therefore, we apply the same conversion method to transform it into monthly data. Details about the construction could be found in Cieslak et al. (2021).

[55] The constructions of these variables are as follows. The output gap (OG) is the cyclical component of monthly industrial production, derived using the Hodrick and Prescott (1997) filter. The relative bond rate (RREL) is equal to the yield on the three-month Treasury bill minus its 12-month rolling average. The term spread (TERM) is the yield on the 10-year Treasury bond minus the yield on the three-month Treasury bill. The default spread (DEFA) is the Moody's BAA corporate bond yield minus the Moody's AAA corporate bond yield. The macro and bond data are obtained from the FRED-MD dataset, and the Chicago Fed National Activity Index is downloaded from https://www.chicagofed.org/research/data/cfnai/current-data.

[56] Our explanation is intuitive and informal. Generally, the relationship between monetary policy uncertainty and monetary policy stance an intriguing topic, especially given the divergence observed between the two largest markets, China and the United States. Further discussion on this issue may extend beyond the primary scope of this study, and we leave it for future research.

Table C.3 (*continued*)

Panel C: Economic state							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
					(1.236)		(0.135)
DEFA						0.416	0.188
						(5.946)	(2.018)
Obs.	341	341	341	341	341	341	341
Adj. R^2 (%)	0.055	27.742	18.641	21.610	0.844	17.024	32.879

Note: This table reports the estimates from univariate and multivariate regressions of the aggregate monetary policy index on a set of variables related to uncertainty, monetary policy, and economic state, which are conducted based on the U.S. data. Alternative uncertainty indices used are the economic uncertainty index (EU), the economic policy uncertainty index (EPU), and the financial uncertainty index. The monetary policy variables are the text-based Hawk–Dove score (HD) and the change in the effective federal funds rate (ΔEFFR). The economic state variables used are the output gap (OG), NBER-dated recession (REC), Chicago Fed National Activity Index (CFNAI), relative bond rate (RREL), term spread (TERM), and default spread (DEFA). The sample spans from August 1987 to December 2015. All the variables are standardized. Newey–West adjusted *t*-statistics are provided in parentheses using five lags according to the rule of thumb in Newey and West (1987).

C.4 The asset-pricing implications of the U.S. aggregate measure

Now, we examine the cross-sectional pricing performance of the U.S. aggregate MPU measure. For a brief overview, we find that, similar to China, MPU commands a negative risk premium in the U.S. However, the magnitude and significance of the premium are somewhat weaker compared to China, indicating that the pricing power of MPU is stronger in a "policy-driven" environment. We believe this finding underscores the distinctive nature of showing China's specialty. The details are provided below, with both the data and testing methods following standard practices from the literature.

C.4.1 Data and variables

Our stock sample includes all U.S.-based common stocks traded on the NYSE, AMEX, and NASDAQ exchanges. The sample period is from August 1987 through December 2015, which covers the time-series of the aggregate MPU measure. We eliminate stocks with a price per share less than $5 or more than $1,000 (Bali et al., 2017). The monthly stock return data are from the CRSP, and the risk-free security return data are taken from Ken French's data library. We follow Bali et al. (2016) to adjust stock returns for delisting.

The exposures of individual stocks to MPU are estimated following the methodology of Bali et al. (2017). Specifically, for each stock and for each month in our sample, we estimate the U.S. MPU beta ($\beta^{\text{MPU-US}}$) from the monthly rolling regressions of excess stock returns on the aggregate MPU measure over a 60-month fixed window while controlling for the market (MKT), size (SMB), book-to-market (HML), momentum (UMD), liquidity (LIQ), investment ($R_{I/A}$), and profitability (R_{ROE}) factors of Fama and French (1993), Carhart (1997), Pástor and Stambaugh (2003), and Hou et al. (2015). We require at least 24 monthly observations be available for estimating uncertainty betas.[57]

We obtain the firm-level accounting variables as well as the well-known cross-sectional return predictors from the dataset provided by Chen and Zimmermann (2022), and merge them with the CRSP returns data.[58] The stock-level return predictors include the market beta (β^{MKT}), log of market capitalization (SIZE), book-to-market ratio (BM), operating profitability (ROE), investment (I/A), net operating assets (NOA), short-term reversal (REV), coskewness (COSKEW), and illiquidity (ILLIQ).

C.4.2 Empirical results

The first set of U.S. MPU betas ($\beta^{\text{MPU-US}}$) are obtained using the sample from August 1987 to July 1992. Then, these monthly MPU betas are used to predict the cross-sectional stock returns in the following month (August 1992). This monthly rolling regression approach is used until the sample is exhausted in December 2015. The cross-sectional return predictability results are reported from August 1992 through December 2015.

We first conduct univariate portfolio-level analysis to examine the relation between the MPU beta and future stock returns in the U.S. market. The portfolio formation is similar as before. For each month, we form decile portfolios by sorting individual stocks based on their MPU betas ($\beta^{\text{MPU-US}}$), where decile 1 contains stocks with the lowest $\beta^{\text{MPU-US}}$ during the past month, and decile 10 contains stocks with the highest $\beta^{\text{MPU-US}}$ during the previous month. We compute the average values of excess returns (RET−RF) and risk-adjusted returns (alphas) of the Fama and French (2015)'s five factor model (α_{FF5}) and Hou et al. (2021)'s augmented *q*-factor model (α_{HMXZ}).

Table C.4 reports the one-month-ahead univariate portfolio results. The first column presents the average MPU betas for the decile portfolios. The next five columns show the average excess returns and the alphas on the equal-weighted portfolios, and the last five columns report the average excess returns and the alphas on the value-weighted portfolios. For the equal-weighted portfolio, column 2 shows that the overall next-month average excess stock return decreases from 1.686% to 1.347% per month, moving from decile 1 to decile 10. Since the returns for deciles 9 and 10 are relatively high, the monotonicity of the portfolio returns is not very clear to some extent. Nevertheless, the spread of the average excess return between *High* and *Low* portfolios is economically sizable and statistically significant, with a value of -0.340% and a Newey–West adjusted *t*-statistic of -2.135. As presented in the next two columns, the alpha spreads of the FF5 and *q*-factor models are also negative and highly significant.

However, the negative risk premium diminishes when using the value-weighted portfolios. For example, the average excess returns are 1.084% per month among the lowest MPU beta stocks and 0.887% per month among the highest MPU beta stocks. The corresponding spread is slightly negative but statistically insignificant (-0.197% with *t*-stat. = -0.767). The FF5 alpha spread exhibits a similarly weak result, whereas the *q*-factor alpha spread is significantly negative (-0.604% with *t*-stat. = -2.521). In summary, these results demonstrate that our measure of the aggregate MPU is priced in both the Chinese and the U.S. markets, although the cross-sectional spreads of MPU betas are weaker in the latter. This phenomenon can be attributed to the fact that China operates as a policy-driven market, where the influence of monetary policy signals is paramount (McMahon et al., 2018), and uncertainty surrounding policies rather than fundamentals is thereby a significant concern among the market participants. In contrast, the U.S. monetary policy is more developed and transparent, resulting in the market perceiving lower risk premia associated with uncertainty stemming from monetary policy.

[57] The excess market returns and the Fama-French factors (small-minus-big, high-minus-low, and winner-minus-losers) are from Kenneth French's data library. The liquidity factor (LIQ) is from Lubos Pástor's data library. The investment and profitability factors are from Hou et al. (2015)'s data library. The results using alternative measures of βMPU-US estimated with different combinations of these factors turn out to be very similar to those reported in our main tables.

[58] The website link is https://www.openassetpricing.com/.

Table C.4
Univariate portfolios of stock returns sorted by monetary policy uncertainty beta.

		Equal-weighted			Value-weighted		
Decile	Post $\beta^{MPU\text{-}US}$	RET−RF	α_{FF5}	α_{HMXZ}	RET−RF	α_{FF5}	α_{HMXZ}
Low	-3.677	1.686	1.015	1.143	1.084	0.755	0.894
		(4.101)	(7.842)	(7.614)	(2.306)	(3.478)	(3.872)
2	-1.634	1.474	0.635	0.700	0.902	0.254	0.243
		(4.356)	(6.115)	(5.259)	(2.602)	(1.851)	(1.624)
3	-0.961	1.192	0.342	0.433	0.766	0.050	0.060
		(3.947)	(4.005)	(3.944)	(2.313)	(0.518)	(0.640)
4	-0.505	1.194	0.360	0.422	0.657	-0.085	-0.053
		(4.176)	(4.204)	(3.610)	(2.227)	(-1.002)	(-0.584)
5	-0.141	1.098	0.285	0.344	0.750	0.056	0.033
		(3.987)	(3.751)	(3.525)	(2.985)	(0.815)	(0.374)
6	0.186	1.030	0.205	0.277	0.680	0.049	0.079
		(3.818)	(2.755)	(2.649)	(2.668)	(0.595)	(0.758)
7	0.527	0.998	0.189	0.233	0.397	-0.234	-0.279
		(3.822)	(3.008)	(2.358)	(1.459)	(-2.407)	(-2.676)
8	0.941	0.972	0.150	0.206	0.582	-0.158	-0.204
		(3.631)	(2.281)	(2.311)	(2.349)	(-1.709)	(-1.850)
9	1.572	1.131	0.379	0.395	0.694	0.123	0.080
		(3.731)	(4.442)	(4.093)	(2.441)	(1.021)	(0.608)
High	3.666	1.347	0.658	0.621	0.887	0.378	0.290
		(3.750)	(5.507)	(5.398)	(2.185)	(2.038)	(1.595)
High−Low		-0.340	-0.357	-0.522	-0.197	-0.376	-0.604
		(-2.135)	(-2.339)	(-3.677)	(-0.767)	(-1.531)	(-2.521)

Note: This table reports the univariate portfolio results to analyze the cross-sectional relationship between the monetary policy uncertainty beta ($\beta^{MPU\text{-}US}$) and excess stock returns for the equal- and value-weighted portfolios. For each month, the equal- and value-weighted decile portfolios are formed by sorting individual stocks based on their MPU betas during the previous month, where decile 1 (*Low*) contains stocks with the lowest MPU beta and decile 10 (*High*) contains stocks with the highest MPU beta. The first column reports the average post-ranking MPU betas of individual stocks in each decile, and the remaining columns report the average excess returns (RET–RF) and alphas, where α_{FF5} denotes the alpha of the Fama-French five factor model containing the market (MKT), size (SMB), value (HML), investment (CMA), and profitability (RMW) factors, and α_{HMXZ} denotes the alpha of the augmented *q*-factor model containing the market, size, value, investment ($R_{I/A}$), profitability (R_{ROE}), and expected growth (R_{EG}) factors. The last row reports the differences in the average excess returns and alphas between decile 1 (*Low*) and decile 10 (*High*). The average excess returns and alphas are defined in monthly percentage terms. The sample spans from August 1992 to December 2015. Newey–West adjusted *t*-statistics of five lags are given in parentheses.

We now examine the cross-sectional relation between the uncertainty beta and expected returns at the stock level using the Fama and MacBeth (1973) regressions. Table C.5 presents the time-series averages of the slope coefficients from the regressions of one-month-ahead stock returns on the MPU beta ($\beta^{MPU\text{-}US}$) with and without control variables. The univariate regression results reported in the first column indicate a negative and statistically significant relation between the MPU beta and the cross-section of future stock returns. The average slope from the monthly regressions of realized returns on $\beta^{MPU\text{-}US}$ alone is -0.062 with a Newey-West *t*-statistic of -2.311. To determine the economic significance of this average slope coefficient, we use the average values of the uncertainty betas in the decile portfolios. Table 9 shows that the difference in $\beta^{MPU\text{-}US}$ values between average stocks in the first and tenth deciles is 7.343[= 3.666 - (-3.677)]. Therefore, the average coefficient of −0062 represents an economically significant decrease of 0.46% per month, $[-0.062 \times 7.343 = -0.46\%]$, in the average stock's expected return for moving from the first to the tenth decile. As presented in column 2 and 3, the average slope coefficients on $\beta^{MPU\text{-}US}$ remain negative and highly significant when controlling for several stock-level characteristics such as the market beta, size, BM ratio, ROE, etc. However, when controlling for a large set of stock-level predictors simultaneously, the average slope of the MPU beta is estimated to be negative but statistically insignificant. In the last four columns, we control for industry effects by assigning each stock to one of the twelve industries based on its four-digit Standard Industrial Classification (SIC) code.[59] Generally, the coefficients when controlling for potential industry effects are similar with previous results but with a decrease in *t*-statistics. Overall, these results verify that the MPU beta commands a negative and significant risk premium in some specifications in the U.S., whereas the asset-pricing performance is weaker compared to that in the Chinese market.

[59] The industry definitions are obtained from the online data library of Kenneth French.

Table C.5
Fama–MacBeth cross-sectional regressions for monetary policy uncertainty beta.

	Without controlling for industry effect				With controlling for industry effect			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
β^{MPU-US}	-0.062	-0.066	-0.057	-0.042	-0.055	-0.058	-0.043	-0.030
	(-2.311)	(-2.618)	(-2.068)	(-1.490)	(-2.283)	(-2.426)	(-1.643)	(-1.154)
β^{MKT}		0.349	0.329	0.239		0.301	0.226	0.128
		(1.539)	(1.433)	(1.030)		(1.634)	(1.205)	(0.696)
SIZE			-0.205	-0.168			-0.201	-0.166
			(-5.707)	(-4.484)			(-5.852)	(-4.720)
BM			0.052	0.047			0.102	0.096
			(0.957)	(0.923)			(2.363)	(2.292)
ROE			-0.129	-0.087			-0.067	-0.025
			(-2.057)	(-1.343)			(-1.039)	(-0.373)
I/A			0.003	0.003			0.003	0.003
			(1.081)	(1.059)			(1.009)	(0.926)
NOA			0.385	0.354			0.493	0.446
			(2.830)	(2.760)			(5.053)	(4.992)
REV				0.028				-0.096
				(0.069)				(-0.262)
COSKEW				0.237				0.195
				(0.999)				(1.108)
ILLIQ				0.046				0.049
				(2.337)				(2.495)
Intercept	1.215	0.879	2.441	2.161	1.174	0.869	2.558	2.234
	(4.119)	(3.447)	(6.549)	(5.617)	(3.712)	(3.173)	(7.129)	(5.915)
Obs.	868,805	868,805	681,739	663,220	868,805	868,805	681,739	663,220
Adj. R^2 (%)	0.177	2.882	4.397	5.063	4.631	6.061	7.345	7.940

Note: This table reports the time-series averages of the slope coefficients by regressing one-month-ahead excess stock returns on the monetary policy uncertainty beta (β^{MPU-US}) and a collection of lagged stock-level characteristics as control variables using the Fama and MacBeth (1973) methodology. Entries in columns 1 to 3 under "Without controlling for industry effect" present results without controlling for the industry effect, and entries in columns 4 to 6 under "With controlling for industry effect" present the same set of regression results controlling for the industry effect. For each stock, the MPU beta is estimated via a 60-month rolling regression. The stock-level characteristics include the market beta (β^{MKT}), log of market capitalization (SIZE), book-to-market ratio (BM), operating profitability (ROE), investment (I/A), net operating assets (NOA), short-term reversal (REV), coskewness (COSKEW), and illiquidity (ILLIQ). The sample spans from August 1992 to December 2015. Newey–West adjusted *t*-statistics of five lags are given in parentheses.

Appendix D. Monetary policy uncertainty and the yield curve

Following the growing literature that studies monetary policy and its impact on the yield curve (Rudebusch and Wu, 2008; Born and Pfeifer, 2014; Tillmann, 2020; Boeck and Feldkircher, 2021), we study the Chinese government bonds of various maturities and the heterogeneous response of yield changes and volatility to MPU shocks for both individual and aggregate measures. In addition, we examine the impact of the MPU on the level, slope, and curvature factors of the yield curve. The remainder of this reply is organized as follows: data introduction, econometric setup, and empirical results.

Yield data and variables

We utilize a dataset on zero-coupon Chinese Treasury bonds from March 1, 2006 to December 31, 2020, which was compiled and provided by China Central Depository & Clearing Co., Ltd. (CCDC).[60] Consistent with previous studies (Lin and Niu, 2021; Umar et al., 2021), our sample comprises eleven available bond yields, with monthly maturities of 3, 6, 9, 12, 36, 60, 84, 120, 180, 240, and 360 months. Bonds with severe data missingness, such as those with medium-term maturities at 4 or 6 years, as well as longer terms such as 40 years and 50 years, are not included. Table D.1 provides key descriptive statistics for these yields.

Table D.1
Summary statistics for bond yields.

Maturity	3	6	9	12	36	60	84	120	180	240	360
Avg	2.475	2.555	2.598	2.641	2.976	3.192	3.392	3.502	3.800	3.939	4.068
Std	0.756	0.732	0.717	0.717	0.613	0.533	0.483	0.472	0.416	0.423	0.411
Max	4.611	4.174	4.165	4.219	4.416	4.458	4.582	4.552	4.788	5.011	5.114
Min	0.844	0.895	0.929	0.961	1.323	1.789	2.169	2.538	2.980	3.043	3.134
Skew	-0.093	-0.240	-0.273	-0.277	-0.160	0.145	0.257	0.383	0.329	0.280	0.299
Kurt	2.709	2.580	2.585	2.543	2.694	2.463	2.404	2.467	2.653	2.838	2.900
Obs.	178	178	178	178	178	178	178	178	178	178	178

Note: This table reports the summary statistics of eleven Chinese Treasury bond yields. The data are monthly, and the yields are expressed as percentages. Maturities are in months. The sample is from March 2006 to December 2020.

Following Leippold and Matthys (2022), we investigate the impact of monetary policy uncertainty (MPU) on yield changes, yield volatility, and yield curve factors. To ensure data frequency consistency, we use the yield from the last business day of each month as the monthly yield. For observed

[60] The China Central Depository & Clearing Co., Ltd. (CCDC) is a state-owned financial institution, approved and funded by the State Council of China. The CCDC offers comprehensive information services on China's bond market, including yield curves, bond valuation, and bond indices. As a key piece of national financial market infrastructure, CCDC's information products serve as crucial references for investors in market transactions, as well as for government authorities in market surveillance and policy implementation. The daily yield data can be freely downloaded from https://yield.chinabond.com.cn/cbweb-mn/yield_main?locale=en_US.

yield volatility, we employ realized variance aggregated on a monthly level from business day data. Adopting the method of Sihvonen (2024), we extract the first three principal components as yield curve factors, encompassing all eleven maturities between 3 months and 30 years. The factor loadings for the bond yields are presented in Table D.2. The principal components appear standard. The first principal component (pc^1) explains approximately 84.76% percent of the variation in yields, which is often called a *level* factor since it loads fairly evenly on all maturities. The second component (pc^2), which explains approximately 13.26% of the yield variation, has negative loadings on short-term yields and positive loadings on long-term yields; thus, it is referred to as the *slope* factor. The third component (pc^3), with positive loadings on short- and long-maturity yields and negative loadings on mid-maturity yields, is identified as the *curvature* factor. Collectively, these three components account for 99.11% of the variation in yields according to our data.

Table D.2
Factor loadings for bond yields.

Maturity	3	6	9	12	36	60	84	120	180	240	360
pc^1	0.290	0.295	0.296	0.300	0.318	0.323	0.320	0.304	0.292	0.287	0.291
pc^2	-0.356	-0.353	-0.346	-0.327	-0.148	0.031	0.124	0.290	0.363	0.376	0.357
pc^3	0.339	0.189	0.093	0.077	-0.380	-0.424	-0.412	-0.190	0.066	0.414	0.360

Note: This table reports the factor loadings of eleven Chinese Treasury bond yields, where pc^1, pc^2, and pc^3 represent the first, second, and third principal components of the yields, respectively. Maturities are in months. The sample is from March 2006 to December 2020.

Econometric setup

We start by analyzing the yield and volatility term structures in response to a monetary policy uncertainty shock. Specifically, we run the following monthly regression:

$$y_t^{(n)} = \alpha_0^{(n)} + \alpha_1^{(n)}\mathrm{MPU}_t + \mathbf{X}'_t\boldsymbol{\alpha}_2^{(n)} + \varepsilon_t^{(n)}, \tag{D.1}$$

where $y_t^{(n)}$ is either the change in the bond yield for a given time-to-maturity n or its realized volatility, MPU_t represents our aggregate or individual monetary policy uncertainty measure, and ε_t is the error term. Plotting the estimated $\alpha_1^{(n)}$ as a function of the maturity n gives the term structures of yield changes to the MPU shock at t. To account for the effects of the first-order moment of monetary policy and other policy uncertainties, we introduce a monetary policy surprise (MPS) index and Huang and Luk (2020)'s economic policy uncertainty (EPU) index in $\mathbf{X}_t$.[61] Nevertheless, we include additional control variables to capture business cycle influences on yields beyond the monetary policy shock. Thus, the vector $\mathbf{X}_t$ also includes current inflation and the GDP growth rate.

For yield curve factors, we use a similar econometric specification by substituting the dependent variable with principal components, that is,

$$pc_t^i = \varphi_0^i + \varphi_1^i\mathrm{MPU}_t + \mathbf{X}'_t\boldsymbol{\varphi}_2^i + \varepsilon_t^i, \tag{D.2}$$

where $i = 1$, 2, or 3 denotes the first (*level*), second (*slope*), or third (*curvature*) principal component, respectively. The control variables $\mathbf{X}_t$ are the same as those used above.

We estimate the model for the eleven maturities previously mentioned, using monthly data over the sample period from March 2006 to December 2020. We standardize the MPU series to make the coefficients comparable across different MPU indices. Since the dependent variable is dated h periods ahead, the error terms will exhibit serial correlation. We apply a Newey–West correction (Newey and West, 1987) to our estimation errors. The maximum lag for the correction is set to 4 according to the rule of thumb in Newey and West (1987). The Newey–West corrected errors are used to construct confidence bands around our estimates.

Empirical results

D.3.1 Response of yield changes

First, we analyze the relationship between yield changes and monetary policy uncertainty by plotting $\alpha_1^{(n)}$ as a function of maturity n for different MPU measures. The slope coefficients from the contemporaneous regression are given in Fig. D.1. We find that the aggregate MPU measure negatively and significantly affects the changes in short-end bond yields (with maturities of 3 to 12 months). This finding indicates that yields decrease when monetary policy uncertainty increases, which is also consistent with the U.S. results concerning policy uncertainty and bond yields (Tillmann, 2020; De Pooter *et al.*, 2021; Leippold and Matthys, 2022). This behavior is consistent with the flight-to-quality behavior documented in numerous theoretical and empirical studies (Bloom, 2009; Baker et al., 2016; Leippold and Matthys, 2022). In turbulent times, heightened uncertainty makes capital investments riskier, eventually inducing investors to seek shelter in safer assets such as government bonds. As a result, bond prices rise, driving their yields down. However, the negative contemporaneous relationship between monetary policy uncertainty and yields weakens as maturity increases. This could be attributed to the fact that China's monetary policy is still relatively underdeveloped compared with that of more advanced economies. In particular, the majority of the policy instruments in the PBC toolkit are quantitative, and the effectiveness of price-based tools needs to be improved (Koivu, 2009; Sun, 2015; Huang et al., 2020). The transmission of monetary policy to long-term interest rates may exhibit a lag, and as a result, such policy shocks struggle to exert an immediate and effective influence on these rates.

[61] We use an interest rate rule-based MPS index, constructed from the residual of the Taylor rule as a measure of monetary policy surprises. This index is also employed in Section 4.2 of the manuscript, with construction details outlined in Appendix A.4.

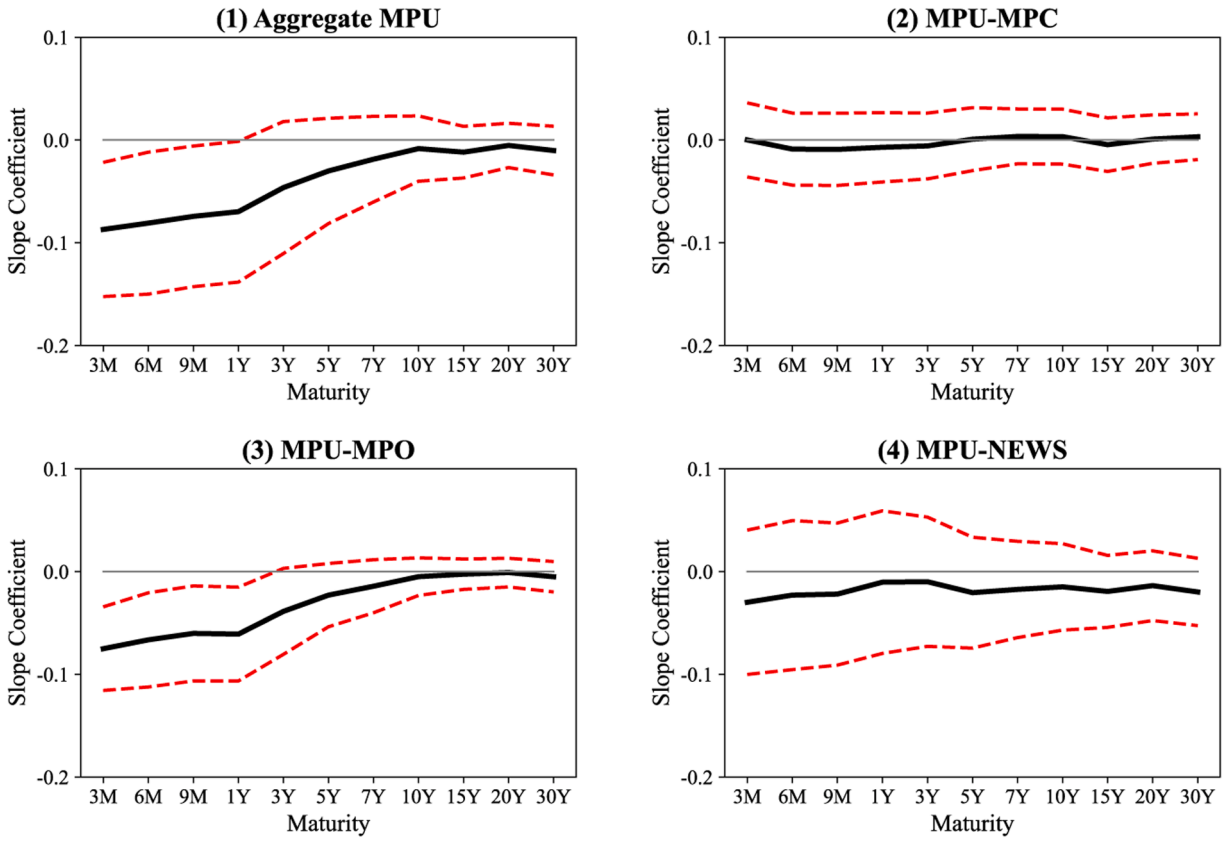


Fig. D.1. Contemporaneous responses of yield changes to monetary policy uncertainty.
Note: The figure shows the slope coefficients (black solid line) and the relevant 90% confidence bounds (red dashed line) from regressing the contemporaneous yield changes of different maturity bonds on either the aggregate or individual monetary policy uncertainty measure. To save space, "M" typically denotes months, whereas "Y" represents years in the labels of maturity. The communication-based MPU index (MPU-MPC, subfigure 2) is obtained by counting uncertainty-related keywords in PBC communication texts. The operation-based MPU index (MPU-MPO, subfigure 3) is obtained from the unforecastable components of a broad set of monetary policy variables following the method of Jurado et al. (2015). The news-based MPU index (MPU-NEWS, subfigure 4) is obtained from Huang and Luk (2020). The aggregate MPU (subfigure 1) index is the first principal component of the three individual measures. The sample is from March 2006 to December 2020.

Compared with different individual MPU measures, the operation-based index has the closest alignment with the aggregate measure: it correlates significantly and negatively with short-term yield changes but has less pronounced effects on mid- and long-term yields. This alignment likely arises from the operation-based MPU's direct construction using various policy tools, including interest rates, which enhances its conceptual and empirical relevance to fluctuations in government bond yields. However, we observe that for bonds with the same maturity, the slope coefficient of the aggregate MPU measure is larger than that of the operation-based MPU, indicating that the aggregate measure has a stronger impact on yields. In addition, unlike the aggregate and operation-based indices, the communication and news-based measures do not have statistically significant effects on yield changes, and their coefficients (in absolute terms) are comparatively small. This further underscores the advantages and necessity of aggregating information.

D.3.2 Response of yield volatility

For realized yield volatility, the estimated slope coefficients are presented in Fig. D.2. Not surprisingly, we observe that the impact of an increase in aggregate MPU measure on the level of yield volatility is positive and statistically significant. Unlike before, MPU affects not only short-term bonds but also long-term bonds. As anticipated, however, these impacts decrease monotonically with increasing term structure. These findings suggest that yield volatility increases with increasing uncertainty. Similar results are also observed in the U.S. (e.g., Leippold and Matthys, 2022). In terms of the individual measures, we find that the slope coefficients for both the operation-based and news-based indices are positive and statistically significant, especially for short-end bonds. In contrast, the coefficient for the communication-based index has a positive sign but is statistically insignificant. Compared with those coefficients in the four subfigures, the coefficient for the aggregate index is the largest across all maturities.

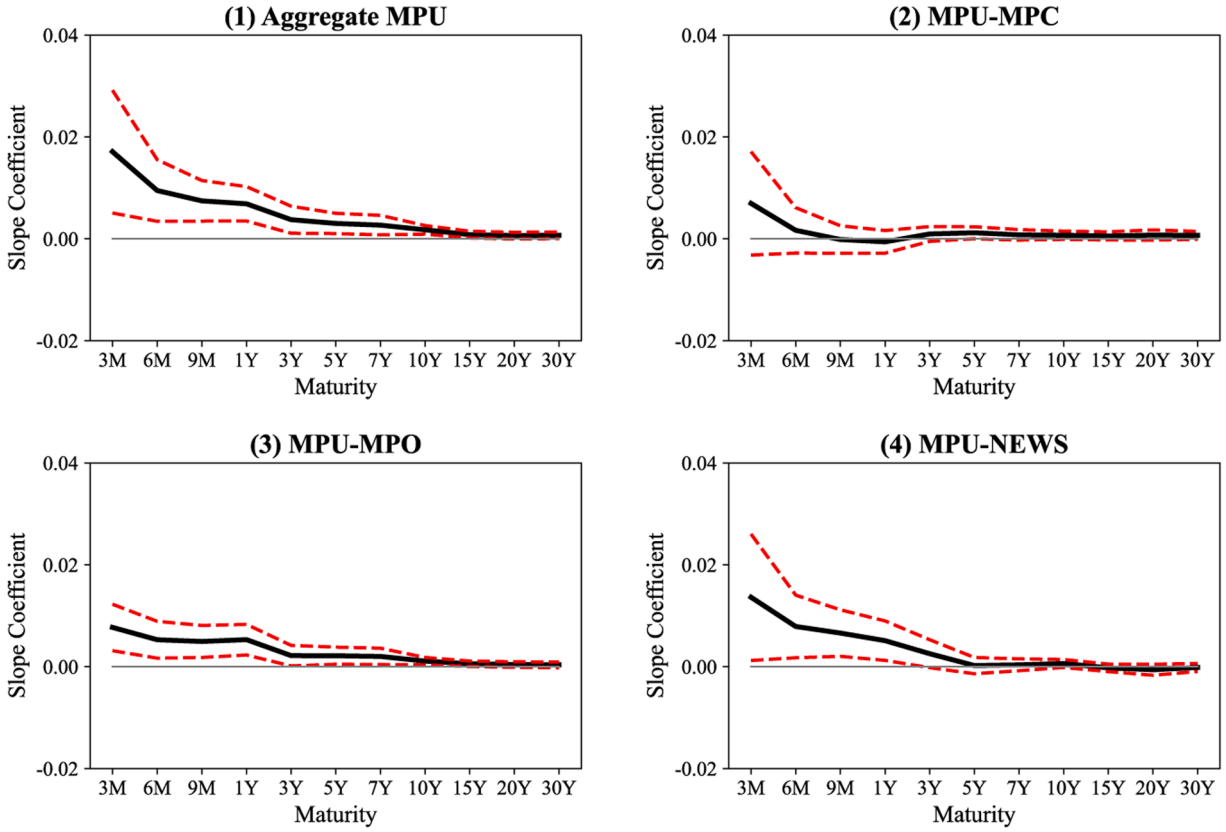


Fig. D.2. Contemporaneous responses of yield volatility to monetary policy uncertainty.
Note: The figure shows the slope coefficients (black solid line) and the relevant 90% confidence bounds (red dashed line) from regressing the contemporaneous yield volatility of different maturity bonds on either the aggregate or individual monetary policy uncertainty measure. The sample is from March 2006 to December 2020.

D.3.3 Response of yield curve factors

Finally, we evaluate the impact of monetary policy uncertainty on yield curve factors. The *t*-statistics of the MPU in Equation (D.2) are presented in Fig. D.3. We focus our main analysis on the aggregate MPU measure and find that the slope factor is most responsive to MPU shocks, aligning with Rudebusch and Wu (2008)'s argument that the slope factor is related to countercyclical monetary policy. This observation is also consistent with our results regarding the heterogeneous effects of the term structure on MPU: short-term bond yields respond much more significantly to MPU than long-term yields do. Therefore, changes in MPU primarily influence the slope of the yield curve, driven by the pronounced impact on short-term bond yields relative to long-term yields. The directionality of the estimated coefficients is also in agreement with our intuition. Specifically, the coefficient for the level factor has a negative sign, indicating that the overall yield curve tends to decrease with increasing MPU. The positive coefficient for the slope factor is attributed to the fact that when MPU increases, short-term bond yields decrease more, whereas long-term yields change minimally. This is in accordance with both the aggregate MPU measure and the slope factor being countercyclical (Rudebusch and Wu, 2008). Moreover, Neuhierl and Weber (2019) observe that the slope factor increases during hawkish periods, which corresponds to our finding that the aggregate MPU measure also increases when the monetary policy stance is hawkish. The positive coefficient of the curvature factor is also intuitive. As the MPU increases, the short-term yields decrease, whereas the other yields remain unchanged, which in turn increases the curvature. The results of the individual measures are in line with our previous findings. The performance pattern of the operation-based index closely resembles that of the aggregate MPU measure. However, we do not find strong evidence that the communication- and news-based indices have significant impacts on the yield curve factors.

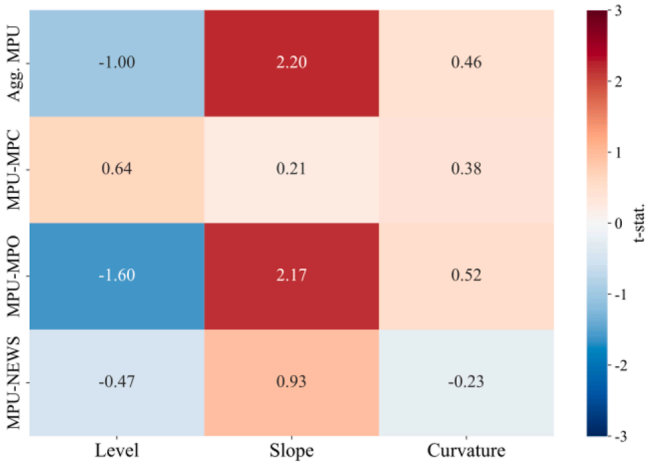


Fig. D.3. Response of yield curve factors to monetary policy uncertainty.
Note: The figure shows the *t*-statistics (shaded area) from regressing the contemporaneous yield curve factors of different maturity bonds on either the aggregate or individual monetary policy uncertainty measure. The sample is from March 2006 to December 2020.

Data availability

Data will be made available on request.

References

Allen, F., Qian, J., Shan, C., Zhu, J.L., 2024. Dissecting the long-term performance of the Chinese stock market. Journal of Finance 79 (2), 993–1054.
Amihud, Y., 2002. Illiquidity and stock returns: cross-section and time-series effects. Journal of Financial Markets 5 (1), 31–56.
Amstad, M., Sun, G., Xiong, W., 2020. The handbook of China's financial system. Princeton University Press.
Anderson, E.W., Ghysels, E., Juergens, J.L., 2009. The impact of risk and uncertainty on expected returns. Journal of Financial Economics 94 (2), 233–263.
Ang, A., Hodrick, R.J., Xing, Y., Zhang, X., 2006. The cross-section of volatility and expected returns. Journal of Finance 61 (1), 259–299.
Baker, M., Wurgler, J., 2006. Investor sentiment and the cross-section of stock returns. Journal of Finance 61 (4), 1645–1680.
Baker, S.R., Bloom, N., Davis, S.J., 2016. Measuring economic policy uncertainty. Quarterly Journal of Economics 131 (4), 1593–1636.
Bali, T., Engle, R., Murray, S., 2016. Empirical asset pricing: The cross section of stock returns.
Bali, T.G., Brown, S.J., Tang, Y., 2017. Is economic uncertainty priced in the cross-section of stock returns? Journal of Financial Economics 126 (3), 471–489.
Bauer, M.D., Lakdawala, A., Mueller, P., 2022. Market-based monetary policy uncertainty. Economic Journal 132 (644), 1290–1308.
Bekaert, G., Engstrom, E., 2017. Asset return dynamics under habits and bad environment–good environment fundamentals. Journal of Political Economy 125 (3), 713–760.
Bekaert, G., Hoerova, M., Lo Duca, M., 2013. Risk, uncertainty and monetary policy. Journal of Monetary Economics 60 (7), 771–788.
Bekaert, G., Ke, S., Wang, X., Zhang, X., 2022. The China-U.S. equity valuation gap. SSRN Electronic Journal 1–88.
Blinder, A., Krueger, A., 2004. What does the public know about economic policy, and how does it know it? NBER Working Paper (10787). No.
Bloom, N., 2009. The impact of uncertainty shocks. Econometrica 77 (3), 623–685.
Bossaerts, P., Ghirardato, P., Guarnaschelli, S., Zame, W.R., 2010. Ambiguity in asset markets: Theory and experiment. Review of Financial Studies 23 (4), 1325–1359.
Brogaard, J., Detzel, A., 2015. The asset-pricing implications of government economic policy uncertainty. Management Science 61 (1), 3–18.
Brunnermeier, M.K., Sockin, M., Xiong, W., 2021. China's model of managing the financial system. Review of Economic Studies rdab098.
Cao, H.H., Wang, T., Zhang, H.H., 2005. Model uncertainty, limited market participation, and asset prices. Review of Financial Studies 18 (4), 1219–1251.
Carhart, M.M., 1997. On persistence in mutual fund performance. Journal of Finance 52 (1), 57–82.
Carpenter, J.N., Lu, F., Whitelaw, R.F., 2021. The real value of China's stock market. Journal of Financial Economics 139 (3), 679–696.
Chang, C., Chen, K., Waggoner, D.F., Zha, T., 2016. Trends and cycles in China's macroeconomy. NBER Macroeconomics Annual 30, 1–84.
Chang, E.C., Luo, Y., Ren, J., 2014. Short-selling, margin-trading, and price efficiency: Evidence from the Chinese market. Journal of Banking & Finance 48, 411–424.
Chen, A.Y., Zimmermann, T., 2022. Open source cross-sectional asset pricing. Critical Finance Review 27 (2), 207–264.
Chen, H., Chow, K., Tillmann, P., 2017. The effectiveness of monetary policy in China: Evidence from a Qual VAR. China Economic Review 43, 216–231.
Chen, H., Tillmann, P., 2021. Monetary policy uncertainty in China. Journal of International Money and Finance 110, 102309.
Chen, J., Tang, G., Yao, J., Zhou, G., 2022. Investor attention and stock returns. Journal of Financial and Quantitative Analysis 57 (2), 455–484.
Chen, K., Higgins, P., Waggoner, D.F., Zha, T., 2016. China pro-growth monetary policy and its asymmetric transmission. Federal Reserve Bank of Atlanta Working Paper Series.
Chen, K., Ren, J., Zha, T., 2018. The nexus of monetary policy and shadow banking in China. American Economic Review 108 (12), 3891–3936.
Chen, T., Gao, Z., He, J., Jiang, W., Xiong, W., 2019. Daily price limits and destructive market behavior. Journal of Econometrics 208 (1), 249–264.
Cieslak, A., Hansen, S., Mcmahon, M., Xiao, S., 2021. Policymakers' uncertainty. Working Paper.
Dang, T.V., Li, W., Wang, Y., 2024. Government stock purchase undermines price informativeness: Evidence from China's "National Team". Journal of Financial and Quantitative Analysis 59 (5), 2340–2374.
Davis, S.J., Liu, D., Sheng, X.S., 2019. Economic policy uncertainty in China since 1949: The view from mainland newspapers. Working Paper.
De Pooter, M., Favara, G., Modugno, M., Wu, J., 2021. Monetary policy uncertainty and monetary policy surprises. Journal of International Money and Finance 112, 102323.
Dey, M.K., 2005. Turnover and return in global stock markets. Emerging Markets Review 6 (1), 45–67.
Drechsler, I., 2013. Uncertainty, time-varying fear, and asset prices. Journal of Finance 68 (5), 1843–1889.
Easley, D., O'Hara, M., 2009. Ambiguity and nonparticipation: The role of regulation. Review of Financial Studies 22 (5), 1817–1843.
Epstein, L.G., Schneider, M., 2010. Ambiguity and asset markets. Annual Review of Financial Economics 2, 315–346.
Fama, E.F., French, K.R., 1989. Business conditions and expected returns on stocks and bonds. Journal of Financial Economics 25 (1), 23–49.
Fama, E.F., French, K.R., 1992. The cross-section of expected stock returns. Journal of Finance 47 (2), 427–465.
Fama, E.F., French, K.R., 1993. Common risk factors in the returns on stocks and bonds. Journal of Financial Economics 33 (1), 3–56.
Fama, E.F., French, K.R., 2015. A five-factor asset pricing model. Journal of Financial Economics 116 (1), 1–22.
Fama, E.F., MacBeth, J.D., 1973. Risk, return, and equilibrium: Empirical tests. Journal of Political Economy 81 (3), 607–636.
Farago, A., Tédongap, R., 2018. Downside risks and the cross-section of asset returns. Journal of Financial Economics 129 (1), 69–86.
Fernández-Villaverde, J., Guerrón-Quintana, P., Rubio-Ramírez, J.F., Uribe, M., 2011. Risk matters: The real effects of volatility shocks. American Economic Review 101 (6), 2530–2561.
Frazzini, A., Pedersen, L.H., 2014. Betting against beta. Journal of Financial Economics 111 (1), 1–25.
Gábor-Tóth, E., Georgarakos, D., 2019. Economic policy uncertainty and stock market participation. SSRN Electronic Journal 1–50.
Han, H., Wang, X., 2023. Monetary policy uncertainty and corporate cash holdings: Evidence from China. Journal of Financial Stability 67, 101138.
Harvey, C.R., Siddique, A., 2000. Conditional skewness in asset pricing tests. Journal of Finance 55 (3), 1263–1295.
Hassan, T.A., Hollander, S., van Lent, L., Tahoun, A., 2019. Firm-level political risk: Measurement and effects. Quarterly Journal of Economics 134 (4), 2135–2202.
Higgins, P., Zha, T., Zhong, W., 2016. Forecasting China's economic growth and inflation. China Economic Review 41, 46–61.

Hirshleifer, D., Kewei, H., Teoh, S.H., Yinglei, Z., 2004. Do investors overvalue firms with bloated balance sheets? Journal of Accounting and Economics 38, 297–331.
Hodrick, R.J., Prescott, E.C., 1997. Postwar U.S. business cycles: An empirical investigation. Journal of Money, Credit and Banking 29 (1), 1–16.
Hou, K., Mo, H., Xue, C., Zhang, L., 2021. An augmented *q*-factor model with expected growth. Review of Finance 25 (1), 1–41.
Hou, K., Xue, C., Zhang, L., 2015. Digesting anomalies: An investment approach. Review of Financial Studies 28 (3), 650–705.
Huang, D., Li, J., Wang, L., 2021. Are disagreements agreeable? Evidence from information aggregation. Journal of Financial Economics 141 (1), 83–101.
Huang, Y., Ge, T., Wang, C., 2020. Monetary policy framework and transmission mechanisms. Handbook of China's Financial System 38–62.
Huang, Y., Luk, P., 2020. Measuring economic policy uncertainty in China. China Economic Review 59 (2), 1–18.
Husted, L., Rogers, J., Sun, B., 2020. Monetary policy uncertainty. Journal of Monetary Economics 115 (11), 20–36.
Jegadeesh, N., 1990. Evidence of predictable behavior of security returns. Journal of Finance 45 (3), 881–898.
Jurado, K., Ludvigson, S.C., Ng, S., 2015. Measuring uncertainty. American Economic Review 105 (3), 1177–1216.
Koivu, T., 2009. Has the Chinese economy become more sensitive to interest rates? Studying credit demand in China. China Economic Review 20 (3), 455–470.
Kurov, A., Stan, R., 2018. Monetary policy uncertainty and the market reaction to macroeconomic news. Journal of Banking & Finance 86, 127–142.
Lee, C.M.C., Qu, Y., Shen, T., 2023. Gate fees: The pervasive effect of IPO restrictions on Chinese equity markets. Review of Finance 27 (3), 809–849.
Lee, D.-H., Min, B.-K., Kim, T.S., 2019. Dispersion of beliefs, ambiguity, and the cross-section of stock returns. Journal of Empirical Finance 50, 43–56.
Leippold, M., Matthys, F., 2022. Economic policy uncertainty and the yield curve. Review of Finance 26 (4), 751–797.
Light, N., Maslov, D., Rytchkov, O., 2017. Aggregation of information about the cross section of stock returns: A latent variable approach. Review of Financial Studies 30 (4), 1339–1381.
Lin, J., Fan, J., Zhang, Y., Chen, L., 2023. Real-time macroeconomic projection using narrative central bank communication. Journal of Applied Econometrics 38 (2), 202–221.
Lin, M., Niu, L., 2021. Echo over the great wall: Spillover effects of QE announcements on Chinese yield curve. Journal of International Money and Finance 111, 102294.
Liu, J., Stambaugh, R.F., Yuan, Y., 2019. Size and value in China. Journal of Financial Economics 134 (1), 48–69.
Ludvigson, S.C., Ma, S., Ng, S., 2021. Uncertainty and business cycles: Exogenous impulse or endogenous response? American Economic Journal: Macroeconomics 13 (4), 369–410.
Ma, J., He, X., 2020. China's interest rate liberalization. In: Amstad, M, Sun, G &, Xiong, W (Eds.), Handbook of China's Financial System.
McCracken, M.W., Ng, S., 2016. FRED-MD: A monthly database for macroeconomic research. Journal of Business & Economic Statistics 34 (4), 574–589.
McMahon, M., Schipke, A., Li, X., 2018. China's monetary policy communication: Frameworks, impact, and recommendations. *IMF Working Paper*, No. 18/244.
Merton, R.C., 1973. An intertemporal capital asset pricing model. Econometrica 41 (5), 867–887.
Mitton, T., Vorkink, K., 2007. Equilibrium underdiversification and the preference for skewness. Review of Financial Studies 20 (4), 1255–1288.
Mueller, P., Tahbaz-Salehi, A., Vedolin, A., 2017. Exchange rates and monetary policy uncertainty. Journal of Finance 72 (3), 1213–1252.
Munday, T., Brookes, J., 2021. Mark my words: the transmission of central bank communication to the general public via the print media. ECB Working Paper. No. 944.
Neuhierl, A., Weber, M., 2019. Monetary policy communication, policy slope, and the stock market. Journal of Monetary Economics 108, 140–155.
Newey, W.K., West, K.D., 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. Econometrica 55 (3), 703–708.
Ozdagli, A., Velikov, M., 2020. Show me the money: The monetary policy risk premium. Journal of Financial Economics 135 (2), 320–339.
Ozoguz, A., 2009. Good times or bad times? Investors' uncertainty and stock returns. Review of Financial Studies 22 (11), 4377–4422.
Pástor, Ľ., Stambaugh, Robert F., 2003. Liquidity risk and expected stock returns. Journal of Political Economy 111 (3), 642–685.
Pástor, Ľ., Veronesi, P., 2012. Uncertainty about government policy and stock prices. Journal of Finance 67 (4), 1219–1264.
Pástor, Ľ., Veronesi, P., 2013. Political uncertainty and risk premia. Journal of Financial Economics 110 (3), 520–545.
Qiao, K., Dam, L., 2020. The overnight return puzzle and the "T+1" trading rule in Chinese stock markets. Journal of Financial Markets 50, 100534.
Rudebusch, G.D., Wu, T., 2008. A macro-finance model of the term structure, monetary policy and the economy. The Economic Journal 118 (530), 906–926.
Shanken, J., 1992. On the estimation of beta-pricing models. Review of Financial Studies 5 (1), 1–33.
Sihvonen, M., 2024. Yield curve momentum. Review of Finance 28 (3), 805–830.
Sun, R., 2015. What measures Chinese monetary policy? Journal of International Money and Finance 59, 263–286.
Sun, R., 2020. Monetary policy announcements and market interest rates' response: Evidence from China. Journal of Banking & Finance 113, 105766.
Ter Ellen, S., Larsen, V.H., Thorsrud, L.A., 2021. Narrative monetary policy surprises and the media. Journal of Money, Credit and Banking.
Tillmann, P., 2020. Monetary policy uncertainty and the response of the yield curve to policy shocks. Journal of Money, Credit and Banking 52 (4), 803–833.
Umar, Z., Yousaf, I., Aharon, D.Y., 2021. The relationship between yield curve components and equity sectorial indices: Evidence from China. Pacific-Basin Finance Journal 68, 101591.
Wang, B., Li, L., Hao, D., 2019. Monetary policy uncertainty, default risks and macroeconomic fluctuations. Econometric Research Journal 54 (3), 119–134 in Chinese.
Xiong, W.B., 2012. Measuring the monetary policy stance of the People's bank of china: An ordered probit analysis. China Economic Review 23 (3), 512–533.
Yi, G., 2021. China's interest rate system and market-based reform of interest rate. Journal of Financial Research (9), 1–11 in Chinese.